# Off on a Tangent

**Meditations on the derivative**

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# Introduction

These notes are intended as a guided meditation on the concept of the derivative.

## How to use this book

There are many exercises in these notes, because I find that I can only really understand math by working through exercises. So, I encourage you to work through the exercises.

If you're working through these notes and find one of the exercises challenging, that's great news! That means you'll have learned something when you solve it. Don't let the challenging exercises bog you down; keep reading onwards, but do come back to the exercises that gave you trouble when you have time.

You are advised to *not* spend time reading chapter 0 thoroughly before jumping into the main part of the text (ie, chapter 1 onwards). When ideas from chapter 0 are invoked in the main part, a reference to the relevant part of chapter 0 is included. My suggestion is to only look at chapter 0 when you run into a reference to it, chasing references back as needed.

Some sections are marked with a star ($\star$). This is intended to indicate that the section is more challenging, and perhaps not quite as important for the overall direction that these notes take. This is not to say they are not important; some very important facts are discussed in starred sections (eg, Taylor's theorem), and unstarred sections do sometimes reference starred sections. But, if you find yourself struggling and need help deciding what's most important to focus on, focus on the unstarred sections.

## Prerequisites

This text assumes basic familiarity with point-set topology (specifically, metric spaces, and with topological properties of $\mathbb{R}$) and with linear algebra. At a few points, there may be some ideas from point-set topology and linear algebra that you have not encountered before. A sort of "bare minimum" exposition of some of these ideas is included in chapter 0.

## Suggestions for improvement

This is a first draft, and there are bound to be many errors. Please be on the lookout for them! If you think you've found one, please share it with me. I'll also be very interested to hear about non-errors: sentences that are maybe correct but phrased confusingly, places where adding a picture could be helpful, etc.

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# 0 Preliminaries

The sections in this chapter cover some ideas relating to point-set topology and linear algebra that will be invoked in the main part of the text. Some sections are merely intended to establish notation; others cover topics that you may not have encountered before. I encourage skipping this chapter, and referring back to it only when you need to.

You should consult a dedicated linear algebra textbook to review the definitions of vector spaces, subspaces, linear independence, span, dimension, linear maps, matrices, matrix multiplication, determinants, and minors. You should also consult an analysis textbook to review the definitions of metric spaces, equivalence of metrics, open subsets, closed subsets, continuous maps, compactness, and connectedness.

## 0.1 Little-oh notation

Suppose $X$ is a metric space[1] and $x_0 \in X$ is a point.

**Definition 0.1.1.** A function $g : X \setminus \{x_0\} \to \mathbb{R}$ is *positive* if $g(x) > 0$ for all $x \in X \setminus \{x_0\}$. Similarly, $g$ is *non-negative* if $g(x) \geqslant 0$ for all $x \in X \setminus \{x_0\}$.

Suppose $g$ is a positive function. Definition 0.1.2 below formulates what it means for a non-negative function $f : X \setminus \{x_0\} \to \mathbb{R}$ to be "little-oh of $g$ as $x$ tends to $x_0$." Intuitively, this condition means that $f$ is *much smaller* than $g$ near $x_0$.

**Definition 0.1.2.** A function $f : X \setminus \{x_0\} \to \mathbb{R}$ is *little-oh of* $g$ *as* $x$ *tends to* $x_0$, written

$$f(x) = o(g(x)) \text{ as } x \to x_0$$

if, for every $\epsilon > 0$, there exists an open neighborhood $U$ of $x_0$ such that $f(x) \leqslant \epsilon g(x)$ for all

[1] Even more generally, $X$ could be a topological space here. See section 0.7.

$x \in U \setminus \{x_0\}$. Equivalently, this means that

$$\lim_{x \to x_0} \frac{|f(x)|}{g(x)} = 0.$$

When $x_0$ can be inferred from context, we write simply $f = o(g)$.

**Remark 0.1.3.** It's worth noting that the use of the symbol "=" above is mathematically abusive. The left-hand side of the "=" is a function and the right-hand side is a property of functions; of course, a function cannot literally be equal to a property. Rather, the "=" is being used here to mean that the function on the left-hand side *has* the property described on the right-hand side. As annoying as it is, this abuse of notation is fairly standard, so it's probably best to just get used to it.

**Exercise 0.1.4.** Prove that the two conditions in definition 0.1.2 are in fact equivalent.

**Exercise 0.1.5.** Let $X = \mathbb{R}$ and $x_0 = 0$. For each of the functions $f$ and $g$ described below, sketch graphs of $f$ and $g$ and then determine whether or not $f = o(g)$ as $x \to 0$.

(a) $f(x) = x$ and $g(x) = x^2$.

(b) $f(x) = x$ and $g(x) = |x|$.

(c) $f(x) = x^2$ and $g(x) = |x|$.

**Exercise 0.1.6.** For a fixed positive function $g : X \setminus \{x_0\} \to \mathbb{R}$, prove that the set of functions $f : X \setminus \{x_0\} \to \mathbb{R}$ which are $o(g)$ as $x \to x_0$ is a vector space under the natural operations. In other words, verify the following three facts.

(1) ("Zero is small") The zero function is $o(g(x))$.

(2) ("Scalar multiples of small are still small") If $c \in \mathbb{R}$ is a scalar and $f(x) = o(g(x))$, then

$$cf(x) = o(g(x)).$$

(3) ("Sum of smalls is still small") If $f_1(x) = o(g(x))$ and $f_2(x) = o(g(x))$, then

$$(f_1 + f_2)(x) = o(g(x)).$$

**Exercise 0.1.7** ("Smaller than small is still small"). Suppose $f_1, f_2 : X \backslash \{x_0\} \to \mathbb{R}$ are functions such that $|f_1(x)| \leqslant |f_2(x)|$ for all $x$ in a punctured neighborhood of $x_0$, and $f_2(x) = o(g(x))$ as $x \to x_0$. Then $f_1(x) = o(g(x))$ also.

## 0.2 Product metric

**Exercise 0.2.1.** Suppose $X$ and $Y$ are metric spaces with metrics $d_X$ and $d_Y$, respectively. Define a function $d : (X \times Y) \times (X \times Y) \to \mathbb{R}$ by

$$d((x_1, y_1), (x_2, y_2)) = \max\{d_X(x_1, x_2), d_Y(y_1, y_2)\}.$$

Show that $d$ is a metric on $X \times Y$.

**Exercise 0.2.2.** Suppose $X$ and $Y$ are metric spaces with metrics $d_X$ and $d_Y$, respectively. Define a function $d : (X \times Y) \times (X \times Y) \to \mathbb{R}$ by

$$d((x_1, y_1), (x_2, y_2)) = \sqrt{d_X(x_1, x_2)^2 + d_Y(y_1, y_2)^2}.$$

Show that $d$ is a metric on $X \times Y$, and that it is equivalent to the metric from section 0.2.

## 0.3 Norms on vector spaces

**Definition 0.3.1.** Let $V$ be a vector space. A *norm* $\| - \|$ on $V$ is a function $V \to \mathbb{R}$ satisfying the following two axioms.

(N1) $|v| \geqslant 0$.

(N2) $|v| = 0$ if and only if $v = 0$.

(N3) $|\lambda v| = |\lambda||v|$ for all $\lambda \in \mathbb{R}$ and $v \in V$.

(N4) $|v + w| \leqslant |v| + |w|$ for all $v, w \in V$.

### 0.3.A Norms induce metrics

**Exercise 0.3.2.** Suppose $\| - \|$ is a norm on a vector space $V$. Show that the function $d : V \times V \to \mathbb{R}$ given by $d(v, w) = |v - w|$ is a metric.

**Exercise 0.3.3.** Suppose $|-|$ is a norm on a vector space $V$. Show that the function $V \times V \to V$ given by $(v, w) \mapsto v + w$ is continuous, where $V \times V$ is regarded as a metric space via one of the metrics defined in section 0.2.

**Exercise 0.3.4.** Suppose $\|-\|$ is a norm on a vector space $V$. Show that the function $\mathbb{R} \times V \to V$ given by $(\lambda, v) \mapsto \lambda v$ is continuous, where $\mathbb{R} \times V$ is regarded as a metric space via one of the metrics defined in section 0.2.

### 0.3.B Equivalence of norms

**Definition 0.3.5.** Two norms $|-|$ and $|-|'$ on a vector space $V$ are *equivalent* if there exist nonzero constants $C_1$ and $C_2$ such that

$$C_1|v| \leqslant |v|' \leqslant C_2|v|$$

for all $v \in V$.

**Exercise 0.3.6.** If $V$ is a vector space, show that equivalence of norms is an equivalence relation on the set of all norms on $V$.

**Exercise 0.3.7.** Suppose $V$ is a vector space and $|-|$ and $|-|'$ are two equivalent norms on $V$. Show that the metrics $d$ and $d'$ corresponding to $|-|$ and $|-|'$ (cf. exercise 0.3.2) are equivalent.

### 0.3.C Norms on finite dimensional vector spaces

If $V$ is a finite dimensional vector space, we can construct a number of norms on $V$ by choosing a basis. Of these, two are of key importance.

**Example 0.3.8.** Suppose $V$ is a finite dimensional vector space, and $v_1, \ldots, v_n$ is a basis for $V$. The $L^2$ *norm* on $V$ with respect to this basis is defined by

$$|a_1 v_1 + \cdots + a_n v_n| = \sqrt{a_1^2 + \cdots + a_n^2},$$

where $a_1, \ldots, a_n \in \mathbb{R}$. Axioms (N1) through (N3) are straightforward to verify. For the

triangle inequality (N4), suppose $v = a_1v_1 + \cdots + a_nv_n$ and $w = b_1v_1 + \cdots + b_nv_n$. Then

$$\begin{aligned}|v+w|^2 &= (a_1+b_1)^2 + \cdots + (a_n+b_n)^2 \\ &= |v|^2 + 2(a_1b_1 + \cdots + a_nb_n) + |w|^2 \\ &\leqslant |v|^2 + 2|v||w| + |w|^2 \\ &= (|v|+|w|)^2,\end{aligned}$$

where we have used the Cauchy-Schwarz inequality[2] for the third step. Taking square roots yields (N4).

**Example 0.3.9.** Suppose $V$ is a finite dimensional vector space, and $v_1, \ldots, v_n$ is a basis for $V$. The $L^\infty$-*norm*, or the *max norm*, on $V$ with respect to this basis is defined by

$$|a_1v_1 + \cdots + a_nv_n|_{\max} = \max\{|a_1|, \ldots, |a_n|\}.$$

Axioms (N1) through (N3) are straightforward to verify. For the triangle inequality (N4), suppose $v = a_1v_1 + \cdots + a_nv_n$ and $w = b_1v_1 + \cdots + b_nv_n$. Then

$$\begin{aligned}|v+w|_{\max} &= \max\{|a_1+b_1| + \cdots + |a_n+b_n|\} \\ &\leqslant \max\{|a_1|+|b_1|, \cdots, |a_n|+|b_n|\} \\ &\leqslant \max\{|a_1|, \ldots, |a_n|\} + \max\{|b_1|, \ldots, |b_n|\} \\ &= |v|_{\max} + |w|_{\max}\end{aligned}$$

where we used the triangle inequality for real numbers for the second step.

The following states that these two norms are actually equivalent.

**Exercise 0.3.10.** Suppose $V$ is a finite dimensional vector space and $v_1, \ldots, v_n$ is a basis for $V$. Let $|-|$ denote the $L^2$-norm from example 0.3.8 and $|-|_{\max}$ the max norm from example 0.3.9. Show that, for any $v \in V$, we have

$$|v|_{\max} \leqslant |v| \leqslant \sqrt{n}|v|_{\max}.$$

[2]The *Cauchy-Schwarz inequality* states that

$$(a_1b_1 + \cdots + a_nb_n)^2 \leqslant (a_1^2 + \cdots + a_n^2)(b_1^2 + \cdots + b_n^2)$$

for all $a_1, \ldots, a_n, b_1, \ldots, b_n \in \mathbb{R}$. For a proof, see [PM91, theorem 6.1], for instance.

In fact, here is vast generalization of exercise 0.3.10.

**Theorem 0.3.11.** *If $V$ is a finite dimensional vector space, all norms on $V$ are equivalent.*

## 0.4 Euclidean space

For any non-negative integer $n$, we write $\mathbb{R}^n$ to denote the set of lists of $n$ real numbers. We will sometimes write its elements as horizontal lists of numbers separated by commas, as in

$$(h_1, \ldots, h_n),$$

and sometimes as vertical column of numbers, as in

$$\begin{bmatrix} h_1 \\ \vdots \\ h_n \end{bmatrix}.$$

The set $\mathbb{R}^n$ is also called $n$-*dimensional euclidean space*.

### 0.4.A Linear structure on $\mathbb{R}^n$

Given $v_1, v_2 \in \mathbb{R}^n$, we define their sum $v_1 + v_2$ by adding the entries coordinate-wise. Given $\lambda \in \mathbb{R}$ and $v \in \mathbb{R}$, we define $\lambda v$ by multiplying each entry of $v$ by $\lambda$. This endows $\mathbb{R}^n$ with the structure of a vector space.

For each $i = 1, \ldots, n$, we define the $i$*th standard basis vector*, denoted $e_i$, to be the list whose $i$th entry is 1 and all other entries are 0.

$$\begin{aligned} e_1 &= (1, 0, 0, \ldots, 0, 0) \\ e_2 &= (0, 1, 0, \ldots, 0, 0) \\ &\vdots \\ e_n &= (0, 0, 0, \ldots, 0, 1) \end{aligned}$$

Observe that

$$(h_1, \ldots, h_n) = h_1 e_1 + h_2 e_2 + \cdots + h_n e_n.$$

Thus the list $e_1, \ldots, e_n$ is a basis for $\mathbb{R}^n$.

For any $i = 1, \ldots, n$, we let $\pi_i : \mathbb{R}^n \to \mathbb{R}$ denote the $i$th *projection map*, given by

$$\pi_i(h_1, \ldots, h_n) = h_i.$$

This is a linear map.

### 0.4.B Norms on $\mathbb{R}^n$

The $L^2$ norm (cf. example 0.3.8) with respect to the standard basis $e_1, \ldots, e_n$ of $\mathbb{R}^n$ is the usual norm we use on $\mathbb{R}^n$. Explicitly, if $v = (h_1, \ldots, h_n) \in \mathbb{R}^n$, we define

$$|v| := \sqrt{h_1^2 + \cdots + h_n^2}.$$

This norm is sometimes also called the *euclidean norm* or the *standard norm* on $\mathbb{R}^n$. Unless explicitly specified otherwise, we regard $\mathbb{R}^n$ as a metric space with the metric induced by this norm (cf. exercise 0.3.2).

On the other hand, we could also endow $\mathbb{R}^n$ with the max norm (cf. example 0.3.9) with respect to the standard basis. We denote this norm by $|-|_{\max}$. Explicitly, if $v = (h_1, \ldots, h_n) \in \mathbb{R}^n$, we have

$$|v|_{\max} = \max\{|h_1|, \ldots, |h_n|\}.$$

### 0.4.C Sup norm on $\mathbb{R}^n$-valued functions

Let $X$ be a set and consider the set of functions $X \to \mathbb{R}^n$. This set is naturally a vector space under pointwise operations: if $f, g : X \to \mathbb{R}^n$, then

$$(f + g)(x) = f(x) + g(x)$$

and if $\lambda \in \mathbb{R}$, then

$$(\lambda f)(x) = \lambda f(x).$$

**Definition 0.4.1.** If $X$ is a set, we define the *sup norm* of a function $f : X \to \mathbb{R}$, denoted $\|f\|_{\sup}$, by

$$\|f\|_{\sup,X} = \sup_{x \in X} |f(x)|.$$

If $X$ can be inferred from context, we also write simply $\|f\|_{\sup}$. We say that $f$ is *bounded* if $\|f\|_{\sup} < \infty$.

**Exercise 0.4.2.** Show that $\| - \|_{\sup}$ is a norm on the vector space of bounded[3] functions $X \to \mathbb{R}^n$.

## 0.5 Matrices

The following is a whirlwind review of some facts about matrices, mostly intended to establish notation. The reader is expected to remember definitions of matrix multiplication, determinants, and minors. Few proofs are included in this section; for more details, you are encouraged to reference a dedicated linear algebra text.

**Definition 0.5.1.** We write $\mathcal{M}_{n\times m}$ for the set of all $n \times m$ matrices (ie, the matrices with $n$ rows and $m$ columns). This is naturally a vector space, where addition and scalar multiplication are defined entrywise.

**Definition 0.5.2.** Let $\mathrm{GL}_n$ denote the subset of $\mathcal{M}_{n\times n}$ consisting of invertible $n \times n$ matrices.

**Lemma 0.5.3.** *Let $A$ be a matrix. Then all of the following numbers are equal.*

*(1) The dimension of the span of the columns of $A$.*

*(2) The dimension of the span of the rows of $A$.*

*(3) The size of largest nonzero minor of $A$.*

*This integer is called the* rank *of $A$, and is denoted* $\mathrm{rank}(A)$.

### 0.5.A Matrix representations of linear maps

**Definition 0.5.4.** If $V$ and $W$ are both vector spaces, we write $\mathcal{L}(V, W)$ for the set of all linear maps $V \to W$. This is naturally a vector space.

**Definition 0.5.5.** If $V$ is a vector space, let $\mathrm{GL}(V)$ denote the set of invertible linear maps $V \to V$, regarded as a subset of $\mathcal{L}(V, V)$.

Throughout the rest of this section, we assume that $U$, $V$, and $W$ are finite dimensional vector spaces.

**Definition 0.5.6.** Suppose $B$ denotes a basis $v_1, \ldots, v_n$ for $V$. Then any $v \in V$ can be written uniquely as $a_1 v_1 + \cdots + a_n v_n$ for some scalars $a_1, \ldots, a_n \in \mathbb{R}$, and we define the *representation of $v$ with respect to* $B$, denoted $[v]_B$, to be the column vector with $n$ entries that records the scalars $a_1, \ldots, a_n$. In other words,

$$[v]_B = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}.$$

**Lemma 0.5.7.** *If $B$ denotes a basis $v_1, \ldots, v_n$ is a basis for $V$, then the function $V \to \mathbb{R}^n$ given by $v \mapsto [v]_B$ is an isomorphism, with inverse given by*

$$\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \mapsto a_1 v_1 + \cdots + a_n v_n.$$

**Definition 0.5.8.** Suppose $\ell : V \to W$ is a linear map. Let $B$ and $C$ denote bases $v_1, \ldots, v_m$ and $w_1, \ldots, w_n$ for $V$ and $W$, respectively. The *matrix representation of $\ell$ with respect to* $B$ *and* $C$, denoted $[\ell]_{B,C}$, is the $n \times m$ matrix whose $i$th column of $[\ell]_{B,C}$ is $[\ell(v_i)]_C$. In other words,

$$[\ell]_{B,C} = \begin{bmatrix} [\ell(v_1)]_C & \cdots & [\ell(v_m)]_C \end{bmatrix}.$$

**Lemma 0.5.9.** *Suppose that $V$ and $W$ are $m$- and $n$-dimensional vector spaces, respectively, and that and $B$ and $C$ are bases for $V$ and $W$, respectively. Then the function $\ell \mapsto [\ell]_{B,C}$ is an isomorphism*

$$\mathcal{L}(V, W) \longrightarrow \mathcal{M}_{n,m}.$$

**Lemma 0.5.10.** *If $\ell : V \to W$ is a linear map between finite dimensional vector spaces and $B$ and $C$ are bases for $V$ and $W$, respectively, then*

$$[\ell]_{B,C}[v]_B = [\ell(v)]_C$$

*for any $v \in V$.*

**Lemma 0.5.11.** *If $\ell : U \to V$ and $\ell' : V \to W$ are linear maps, and $A, B$ and $C$ are bases for $U, V$, and $W$, respectively, then*

$$[\ell' \circ \ell]_{A,C} = [\ell']_{B,C}[\ell]_{A,B}.$$

**Definition 0.5.12.** If $\ell : V \to W$ is a linear map, then the *rank* of $\ell$, denoted $\operatorname{rank}(\ell)$ is defined to be the dimension of the range of $\ell$.

**Lemma 0.5.13.** *If $\ell : V \to W$ is a linear map and $B$ and $C$ are bases for $V$ and $W$, respectively, then*

$$\operatorname{rank}(\ell) = \operatorname{rank}[\ell]_{B,C}.$$

**Definition 0.5.14.** If $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map, the *standard matrix representation* of $\ell$, denoted $[\ell]$, is the matrix of $\ell$ with respect to the standard bases on $\mathbb{R}^m$ and $\mathbb{R}^n$.

**Lemma 0.5.15.** *If $A$ is a $n \times m$ matrix and $\ell_A : \mathbb{R}^m \to \mathbb{R}^n$ is the linear map $\ell_A(v) = Av$, then*

$$[\ell_A] = A.$$

### 0.5.B Matrix norm

Observe that $\mathcal{M}_{n\times m}$ is a finite dimensional vector space. If $E_{j,i}$ denotes the matrix whose $(j,i)$-entry is 1 and all other entries are 0 for $i = 1, \dots, m$ and $j = 1, \dots, n$, then the list of all of these $mn$ matrices forms a basis for $\mathcal{M}_{n\times m}$. We can use this basis to define a $L^2$ and max norm on $\mathcal{M}_{n\times m}$, as in examples 0.3.8 and 0.3.9. Explicitly, if $A$ is a $n \times m$ matrix whose $(j,i)$-entry is $a_{j,i}$, we have the following.

$$\|A\| = \sqrt{\sum_{j,i} a_{j,i}^2}$$
$$\|A\|_{\max} = \max_{j,i} |a_{j,i}|$$

We know from exercise 0.3.10 that these two norms are equivalent. All of the following statements are independent of which norm on $\mathcal{M}_{n\times m}$ we have in mind.

**Lemma 0.5.16.** *The function* $\det : \mathcal{M}_{n\times n} \to \mathbb{R}$ *is continuous.*

*Proof.* Determinants can be computed using cofactor expansions, so $\det A$ is a polynomial in the entries of $A$. □

**Corollary 0.5.17.** $GL_n$ *is an open subset of* $\mathcal{M}_{n\times n}$.

*Proof.* Since $\det : \mathcal{M}_{n\times n} \to \mathbb{R}$ is continuous, and $\mathbb{R} \setminus \{0\}$ is an open subset of the codomain, we see that $GL_n = \det^{-1}(\mathbb{R} \setminus \{0\})$ must be open in $\mathcal{M}_{n\times n}$. □

**Lemma 0.5.18.** *The function* $\mathrm{GL}_n \to \mathrm{GL}_n$ *given by* $A \mapsto A^{-1}$ *is a homeomorphism.*

*Proof.* This function is its own inverse, so it is sufficient to show that $A \mapsto A^{-1}$ is continuous. But

$$A^{-1} = \det(A)^{-1} \operatorname{adj}(A),$$

where $\operatorname{adj}(A)$ denotes the adjugate matrix of $A$. We know that $\det(A)$ is a polynomial in the entries of $A$. Each entry of $\operatorname{adj}(A)$ is a minor of $A$ and is therefore also a polynomial in the entries of $A$. The lemma follows. □

The following generalizes corollary 0.5.17.

**Lemma 0.5.19.** *Suppose* $k \leqslant \min\{m, n\}$ *and let* $Z = \{A \in \mathcal{M}_{n\times m} : \operatorname{rank}(A) < k\}$. *Then* $Z$ *is a closed subset of* $\mathcal{M}_{n\times m}$.

*Proof.* Saying $\operatorname{rank}(A) < k$ is equivalent to insisting that all $k \times k$ minors of $A$ vanish. Each $k \times k$ minor is a polynomial in the entries of $A$, so the zero set of any particular $k \times k$ minor is a closed subset of $\mathcal{M}_{n\times m}$. We then take the intersection of the closed subsets corresponding to all possible $k \times k$ minors, and recall the fact that intersections of closed subsets must be closed. □

## 0.6 Operator norm

### 0.6.A Basics

**Definition 0.6.1.** Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. The *operator norm* of $\ell$, denoted $\|\ell\|$, is defined to be

$$\|\ell\| = \sup\{|\ell(v)| : |v| \leqslant 1\}.$$

The operator norm $\|\ell\|$ is *not* the same as the norm of the standard matrix representation $\|[\ell]\|$ as defined in section 0.5.B. The following gives an important reformulation of definition 0.6.1.

**Exercise 0.6.2.** Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Show that

$$\|\ell\| = \inf\{\lambda \in \mathbb{R} : |\ell(v)| \leqslant \lambda|v| \text{ for all } v \in \mathbb{R}^m\}.$$

In particular, this means that $|\ell(v)| \leqslant \|\ell\| \cdot |v|$ for all $v \in \mathbb{R}^m$.

Here are some fundamental properties of the operator norm.

**Exercise 0.6.3.** Show that the operator norm is a norm on $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ in the sense of definition 0.3.1.

**Lemma 0.6.4.** *Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a nonzero linear map. Then there exists $v_0 \in \mathbb{R}^m$ such that $|v_0| = 1$ and $|\ell(v_0)| = \|\ell\|$.*

*Proof.* The "unit disk"

$$D = \{x \in \mathbb{R}^m : |x| \leqslant 1\}$$

is a nonempty, connected, and compact subset of $\mathbb{R}^m$. The function $v \mapsto |\ell(v)|$ is also a continuous function $D \to \mathbb{R}$, so its image $|\ell(D)|$ must also be a nonempty, connected, and compact subset of $\mathbb{R}$. In other words, it must be a closed interval in $\mathbb{R}$. This means that there exists $v_0 \in D$ such that

$$|\ell(v_0)| = \sup |\ell(D)| = \|\ell\|.$$

Since $\ell \neq 0$, we know $\|\ell\| \neq 0$ from exercise 0.6.3, which in turn means that $v_0 \neq 0$. To prove that $|v_0| = 1$, assume for a contradiction that $|v_0| < 1$, and let $c = 1/|v_0| > 1$. Then $|cv_0| = |c||v_0| = 1$, so $cv_0 \in D$. But then

$$|\ell(cv_0)| = |c\ell(v_0)| = c|\ell(v_0)| > |\ell(v_0)| = \sup |\ell(D)|,$$

which is absurd. □

**Lemma 0.6.5.** *Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map and $|\ell(v)| = o(|v|)$ as $v \to 0$. Then $\ell = 0$.*

*Proof.* The fact that $|\ell(v)| = o(|v|)$ tells us that

$$\lim_{v \to 0} \frac{|\ell(v)|}{|v|} = 0. \tag{0.6.6}$$

Suppose for a contradiction that $\ell \neq 0$. Using lemma 0.6.4, choose $v_0 \in \mathbb{R}^m$ such that $|v_0| = 1$ and $|\ell(v_0)| = \|\ell\|$. For scalars $c$, observe that $cv_0 \to 0$ as $c \to 0$, which means that equation (0.6.6) implies that

$$\lim_{c \to 0} \frac{|\ell(cv_0)|}{|cv_0|} = 0.$$

But observe that, for nonzero $c$, we have

$$\frac{|\ell(cv_0)|}{|cv_0|} = \frac{|c\ell(v_0)|}{|cv_0|} = \|\ell\|$$

so

$$0 = \lim_{c \to 0} \frac{|\ell(cv_0)|}{|cv_0|} = \lim_{c \to 0} \|\ell\| = \|\ell\|.$$

This contradicts exercise 0.6.3. □

### 0.6.B Operator norm as a metric ⋆

Since the operator norm is a norm by exercise 0.6.3, we can regard $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ as a metric space by exercise 0.3.2. It turns out that many natural maps are continuous when we do this. For example, we have already seen that addition and scalar multiplication are continuous in exercises 0.3.3 and 0.3.4.

**Exercise 0.6.7** ("Evaluation is continuous")**.** Suppose $v_0 \in \mathbb{R}^m$ is a fixed vector. Then the "evaluate at $v_0$" function $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \to \mathbb{R}^n$ given by $\ell \mapsto \ell(v_0)$ is continuous.

*Possible hint.* Since the "evaluate at $v_0$" function is linear, it is sufficient to check continuity at 0.

**Exercise 0.6.8** (Submultiplicativity)**.** Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ and $\ell' : \mathbb{R}^n \to \mathbb{R}^p$ are both linear maps. Show that

$$\|\ell' \circ \ell\| \leqslant \|\ell'\| \cdot \|\ell\|.$$

Give an example to show that this inequality can be strict.

**Exercise 0.6.9** ("Composition is continuous")**.** Show that the map

$$\mathcal{L}(\mathbb{R}^n, \mathbb{R}^p) \times \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \longrightarrow \mathcal{L}(\mathbb{R}^m, \mathbb{R}^p)$$

given by $(\ell', \ell) \mapsto \ell' \circ \ell$ is continuous.

**Exercise 0.6.10.** Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is linear and $A = [\ell]$ is its standard matrix representation. Prove the following.

(a) $\|\ell\| \leqslant \|A\| \leqslant \sqrt{n}\|\ell\|$.

(b) $\|A\|_{\max} \leqslant \|\ell\| \leqslant \sqrt{mn}\|A\|_{\max}$.

**Exercise 0.6.11.** Show that the isomorphism $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \to \mathcal{M}_{n\times m}$ of lemma 0.5.9, given by $\ell \mapsto [\ell]$, is a homeomorphism.

Under the homeomorphism $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^n) \to \mathcal{M}_{n\times n}$ of exercise 0.6.11, the set $\mathrm{GL}(\mathbb{R}^n)$ of invertible linear maps $\mathbb{R}^n \to \mathbb{R}^n$ corresponds to the set $\mathrm{GL}_n$ of invertible $n \times n$ matrices; since $\mathrm{GL}_n$ is an open subset of $\mathcal{M}_{n\times n}$ by corollary 0.5.17, we conclude that $\mathrm{GL}(\mathbb{R}^n)$ is an open subset of $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$. Similarly, the following is a consequence of combining lemma 0.5.18 and exercise 0.6.11.

**Exercise 0.6.12.** Show that the map $\mathrm{GL}(\mathbb{R}^n) \to \mathrm{GL}(\mathbb{R}^n)$ given by $\ell \mapsto \ell^{-1}$ is a homeomorphism.

## 0.7 Topological spaces

This is a bare-bones introduction to topological spaces, intended for readers who have seen metric spaces but not topological spaces before.

### 0.7.A Basics

**Definition 0.7.1.** Let $X$ be a set. A *topology* on $X$ is a set $\tau_X$ of subsets of $X$, called *open subsets*, which satisfy the following axioms.

(T1) $\emptyset$ and $X$ are both open subsets.

(T2) The union of an arbitrary collection of open subsets is open.

(T3) The intersection of a finite collection of open subsets is open.

A pair $(X, \tau_X)$ consisting of a set $X$ and a topology $\tau$ on $X$ is called a *topological space*. Often, we write simply "$X$" in place of the pair $(X, \tau_X)$.

Topological spaces come up throughout mathematics, and in general can be very very bizarre. We will not delve into the general theory of topological spaces, and won't see any of these bizarre examples. Instead, you are encouraged to content yourself with the following examples.

**Example 0.7.2** (Metric spaces)**.** If $X$ is a metric space, the collection of subsets of $X$ that are open with respect to the metric defines a topology on $X$. Thus, every metric space can be

regarded as a topological space in a natural way. It's worth noticing that this process of regarding a metric space as a topological space "forgets information," since equivalent metrics on a set will define the same topology.

**Example 0.7.3** (Quotient spaces)**.** Suppose $Y$ is any topological space (for instance, one that comes from a metric space), and let $\sim$ be an equivalence relation on $Y$. For an element $y \in Y$, let $[y]$ denote its equivalence class, and let $X = Y/\sim$ denote the set of all equivalence classes. There is a natural function $\pi : Y \to X$ given by sending a point $y \in Y$ to its equivalence class $[y]$. We define a topology on $X$, called the *quotient topology*, by declaring a subset $U \subseteq X$ to be open if and only if its preimage

$$\pi^{-1}(U) = \{y \in Y : [y] \in U\}$$

is open in $Y$.

**Example 0.7.4** (Product spaces)**.** Suppose $X$ and $Y$ are topological spaces. We define a topology $\tau$ on the cartesian product $X \times Y$ by declaring a subset $U$ to be open if it is a union of sets of the form $V \times W$ where $U$ and $V$ are open subsets of $X$ and $Y$, respectively. This is called the *product topology* on $X \times Y$.

**Example 0.7.5.** Suppose $V$ is a finite dimensional vector space. Then all norms on $V$ are equivalent by theorem 0.3.11, so they all define the same topology on $V$ by example 0.7.2 and exercise 0.3.7. This topology is called the *canonical topology* on $V$. Unless explicitly specified otherwise, we will always regard finite dimensional vector spaces as topological spaces using the canonical topology.

We now make a few further definitions that readers are invited to compare against the definitions they are familiar with from their experience with metric spaces.

**Definition 0.7.6** (Closed subsets)**.** If $X$ is a topological space, a subset $F$ is *closed* if its complement $U$ is *open*.

**Definition 0.7.7** (Continuous functions)**.** If $X$ and $Y$ are topological spaces, a function $f : X \to Y$ is *continuous* if $f^{-1}(U)$ is an open subset of $X$ whenever $U$ is an open subset of $Y$.

**Example 0.7.8.** If $Y$ is a topological space, $\sim$ an equivalence relation on $Y$, and $X = Y/\sim$ the quotient space as in example 0.7.3 above, then the map $\pi : Y \to X$ which carries a point $y \in Y$ to its equivalence class is continuous. This follows immediately from the definition of the quotient topology.

**Exercise 0.7.9.** If $f : X \to Y$ and $g : Y \to Z$ are continuous functions between topological spaces, then $g \circ f$ is also continuous.

**Example 0.7.10.** If $V$ is a finite dimensional vector space, then the addition map $V \times V \to V$ and the scalar multiplication map $\mathbb{R} \times V \to V$ are continuous functions (using the product topology of example 0.7.4 on the domains). This statement is precisely the same as exercises 0.3.3 and 0.3.4.

**Example 0.7.11.** Suppose $\ell : V \to W$ is a linear map between two finite dimensional vector spaces. Then $\ell$ is automatically continuous.

**Definition 0.7.12** (Homeomorphisms)**.** Suppose $X$ and $Y$ are topological spaces and $f : X \to Y$ is a function. Then $f$ is a *homeomorphism* if it is bijective, continuous, and $f^{-1} : Y \to X$ is continuous. More generally, $f$ is a *homeomorphism onto its image* if is injective, continuous, and $f^{-1} : f(X) \to X$ is also continuous.

**Definition 0.7.13** (Subspace)**.** If $X$ is a topological space and $S$ is a subset, then the *subspace topology* on $S$ is

$$\tau_S := \{S \cap U : U \in \tau_X\}.$$

Further, $S$ equipped with the subspace topology is called a *subspace* of $X$.

**Exercise 0.7.14.** If $X$ is a topological space and $S$ is a subspace, prove that the inclusion map $i : S \to X$ is continuous.

### 0.7.B Hausdorff spaces $\star$

**Definition 0.7.15** (Hausdorff)**.** A topological space $X$ is *Hausdorff* if, for every pair of distinct points $x, y \in X$, there exist disjoint open subsets $U$ and $V$ containing $x$ and $y$, respectively.

**Example 0.7.16.** If $X$ is a metric space, the corresponding topological space as in example 0.7.2 is automatically Hausdorff.

However, quotient spaces need not be Hausdorff.

**Exercise 0.7.17.** Let $Y = \mathbb{R} \times \{\pm 1\}$ inside $\mathbb{R}^2$, and define an equivalence relation $\sim$ by declaring that $(a, 1) \sim (a, -1)$ for all $a \neq 0$. Show that $X = Y/\sim$ is not Hausdorff.

### 0.7.C Compact and σ-compact spaces ⋆

**Definition 0.7.18** (Open covers)**.** An *open cover* of a topological space $X$ is a collection $\mathcal{U}$ of open subsets of $X$ which cover $X$, in the sense that

$$\bigcup_{U \in \mathcal{U}} U = X.$$

A *subcover* of $\mathcal{U}$ is a subset $\mathcal{U}' \subseteq \mathcal{U}$ which is itself an open cover.

**Definition 0.7.19** (Compact)**.** A topological space $X$ is *compact* if every open subcover has a finite subcover.

**Definition 0.7.20** (σ-compact)**.** A topological space $X$ is σ-*compact* if it is a countable union of compact subspaces.

**Example 0.7.21.** $\mathbb{R}^n$ is σ-compact, since

$$\mathbb{R}^n = \bigcup_{k=1}^{\infty} [-k, k]^n$$

and each hypercube $[-k, k]^n$ is compact.

# 1 Single variable derivatives

The goal in this chapter is to formalize derivatives of functions whose input is a single real number and whose output is also a single real number.

## 1.1 Two definitions of the derivative

### 1.1.A Slope of the tangent line

Let $S$ be a subset of $\mathbb{R}$ and $a \in S$ an interior point. We would like to define the tangent line to the graph of a function $f : S \to \mathbb{R}$ at $a$. We know intuitively what "tangent line" should mean: a line that "just barely touches" the graph of $f$ at the point $(a, f(a))$. But it's not immediately clear how to formalize this intuition.

One approach to formalization is to notice that our intuitive idea of "tangent line" can be approximated by secant lines, and secant lines can be formalized without ambiguity. More precisely, for small values of $h$, the point $a + h$ is close to $a$ and still in $S$ since $a$ is an interior point of $S$. The slope of the secant line passing through the two points $(a, f(a))$ and $(a + h, f(a + h))$ is computed by the quantity

$$\frac{f(a+h) - f(a)}{h}.$$

This quantity is often called a *difference quotient*. See figure 1.1.1 for a geometric explanation of this difference quotient. By letting $h \to 0$, we get closer and closer to what we intuitively understand to be the slope of the tangent line.

**Definition 1.1.2.** A function $f : S \to \mathbb{R}$ is *differentiable* at an interior point $a \in S$ if

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists. If this limit does exist, its value is called the *derivative of* $f$ *at* $a$. We will usually denote this derivative as $f'(a)$, but sometimes also as one of the following, where $x$ is the

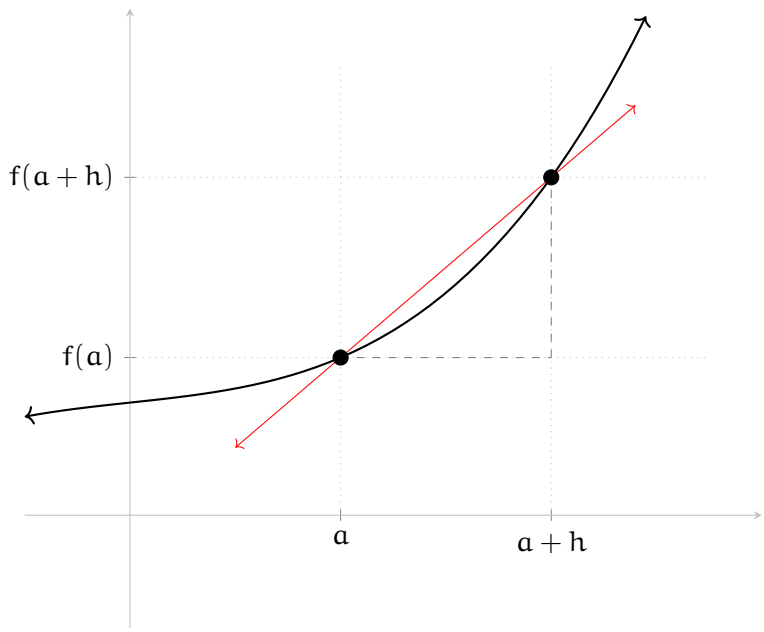


Figure 1.1.1: The graph of a function $f$ is depicted in black. The secant line passing through the two points $(a, f(a))$ and $(a+h, f(a+h))$ is depicted in red. The "run" between these points, is $h$. The "rise" between these points is $f(a+h) - f(a)$. Thus the slope of the secant line, computed as "rise over run," is precisely the difference quotient $(f(a+h) - f(a))/h$.

variable that is being used to denote the argument to $f$.

$$\left.\frac{df}{dx}\right|_{x=a} \qquad \left.\frac{d}{dx}f(x)\right|_{x=a} \qquad \frac{df}{dx}(a)$$

**Exercise 1.1.3.** Use the definition of the derivative to determine whether or not each of the following functions $f$ is differentiable at the given point $a$. If it is differentiable, calculate $f'(a)$. Check your answers by making sure they agree with geometric intuition and/or rules you learned during your introductory calculus course.

(a) $f(x) = |x|$ at $a = 0$.

(b) $f(x) = x$ at $a = 0$.

(c) $f(x) = x^2$ at $a = 0$.

(d) $f(x) = x^2$ at $a = 2$.

(e) $f(x) = \begin{cases} 1 & x \geqslant 0 \\ 0 & x < 0 \end{cases}$ at $a = 0$.

**Exercise 1.1.4** (Power rule for positive integer exponents)**.** Prove that, if $f(x) = x^n$ for a positive integer $n$, then $f'(a) = na^{n-1}$.

*Possible hint.* Recall the binomial theorem, which says that

$$(a+h)^n = \sum_{k=0}^{n} \binom{n}{k} a^k h^{n-k}.$$

**Exercise 1.1.5** (Differentiability implies continuity)**.** Prove that if $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in U$, then $f$ is also continuous at $a$.

**Definition 1.1.6.** We say that a function $f : S \to \mathbb{R}$ is *differentiable* if $f$ is continuous on $S$ and differentiable at every interior point of $S$. Letting $S^\circ$ denote the interior of $S$, the function $S^\circ \to \mathbb{R}$ defined by $a \mapsto f'(a)$ is called the *derivative* of $f$. We will usually denote this function by $f'$, but occasionally we will also use the notation

$$\frac{df}{dx},$$

where $x$ is the variable that is being used to denote the argument to $f$. Note that exercise 1.1.5 makes the continuity on $S^\circ$ automatic.

*Caution.* Other authors may define differentiability on a non-open subset slightly differently, so you should be cautious when reading other texts and you see assertions about a function on a non-open subset of $\mathbb{R}$ being differentiable. The definition we have given will be convenient for our purposes, but because of the potential for misunderstanding, we will apply it sparingly for non-open subsets. Specifically, we will only use this definition either when the domain is open, or when it is an interval.[1]

### 1.1.B Best linear approximation

Before proceeding further, we reinterpret the definition of the derivative in a way that will be better suited to generalization to the multivariable setting. Suppose a function $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$. The tangent line to the graph at the point $(a, f(a))$ is the graph of the function

$$t(x) = f'(a)(x - a) + f(a).$$

Since this function $t$ is "close" to $f$ near $a$, we should have that

$$f(a + h) - f(a) \approx t(a + h) - t(a) = f'(a)h$$

for small values of $h$. See figure 1.1.7.

Notice that the function $h \mapsto f'(a)h$ is a linear map $\mathbb{R} \to \mathbb{R}$ (in the sense of linear algebra). One way to interpret the following result is that the function $h \mapsto f'(a)h$ is the *only* linear map that is a good approximation to $h \mapsto f(a + h) - f(a)$. The precise statement of the result uses little-oh notation; if you have not seen this notation before, see definition 0.1.2.

**Proposition 1.1.8.** *A function $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$ if and only if there exists a linear map $\ell : \mathbb{R} \to \mathbb{R}$ such that*

$$|f(a + h) - f(a) - \ell(h)| = o(|h|) \text{ as } h \to 0.$$

*Moreover, if there exists such a linear map $\ell$, then $\ell$ is uniquely determined: it must be given by*

$$\ell(h) = f'(a)h$$

[1] By interval, we mean an uncountable connected subset of $\mathbb{R}$. This means that intervals can be open, closed, or half-open; and they can be bounded or unbounded.

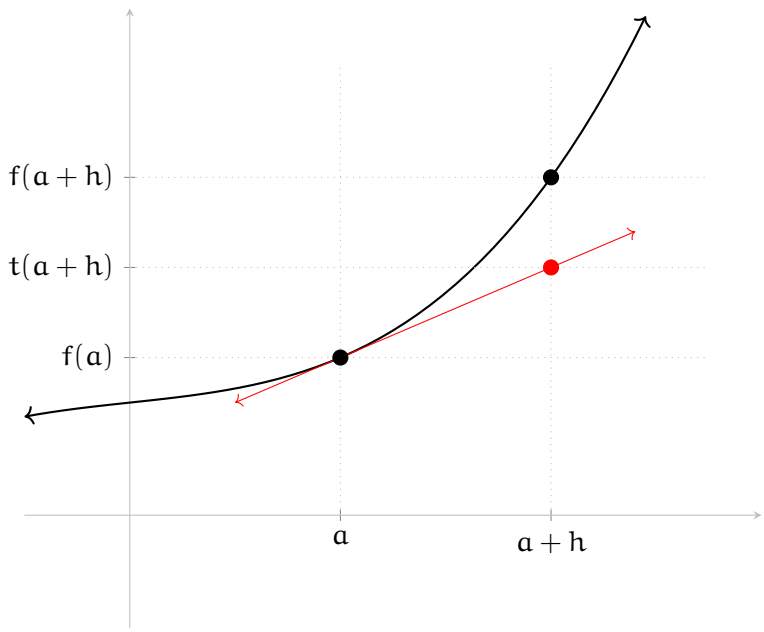


Figure 1.1.7: The graph of a function $f$ is depicted in black, and its tangent line $t$ at $a$ is depicted in red. Then the vertical distance $f(a+h) - f(a)$ is approximated by the vertical distance $t(a+h) - t(a)$. Since the tangent line has slope $f'(a)$, this "rise" can be computed as $f'(a)$ times the "run" (in other words, $f'(a)h$). As $h$ gets smaller, this approximation gets better.

*for all* $h \in \mathbb{R}$.

*Proof.* Let $\ell$ be a linear map $\ell(h) = ch$, where $c$ is some scalar. Observe that

$$\begin{aligned}\frac{|f(a+h) - f(a) - \ell(h)|}{|h|} &= \left|\frac{f(a+h) - f(a) - ch}{h}\right| \\ &= \left|\frac{f(a+h) - f(a)}{h} - c\right|.\end{aligned}$$

This leads to the following sequence of "if and only if" statements.

$$\begin{aligned}|f(a+h) - f(a) - \ell(h)| = o(|h|) &\iff \lim_{h\to 0} \frac{|f(a+h) - f(a) - \ell(h)|}{|h|} = 0 \\ &\iff \lim_{h\to 0} \left|\frac{f(a+h) - f(a)}{h} - c\right| = 0 \\ &\iff \lim_{h\to 0} \left(\frac{f(a+h) - f(a)}{h} - c\right) = 0 \\ &\iff \lim_{h\to 0} \frac{f(a+h) - f(a)}{h} = c\end{aligned}$$

The statement of the proposition follows. □

**Definition 1.1.9.** If $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in U$, we define $df_a : \mathbb{R} \to \mathbb{R}$ to be the linear map $df_a(h) = f'(a)h$.

In other words, $df_a$ is the only good linear approximation to the function

$$h \mapsto f(a+h) - f(a).$$

Being the *only good* linear approximation, it is also the *best* linear approximation, whence the name of this section.

The following is an important idea that we will come up frequently in proofs.

**Remark 1.1.10** ("Error in approximation" function)**.** Suppose $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$. For sufficiently small values of $h$, we can define the "error in approximation" function

$$r(h) = f(a+h) - f(a) - df_a(h).$$

See figure 1.1.11. Rearranging the definition of $r$, we have

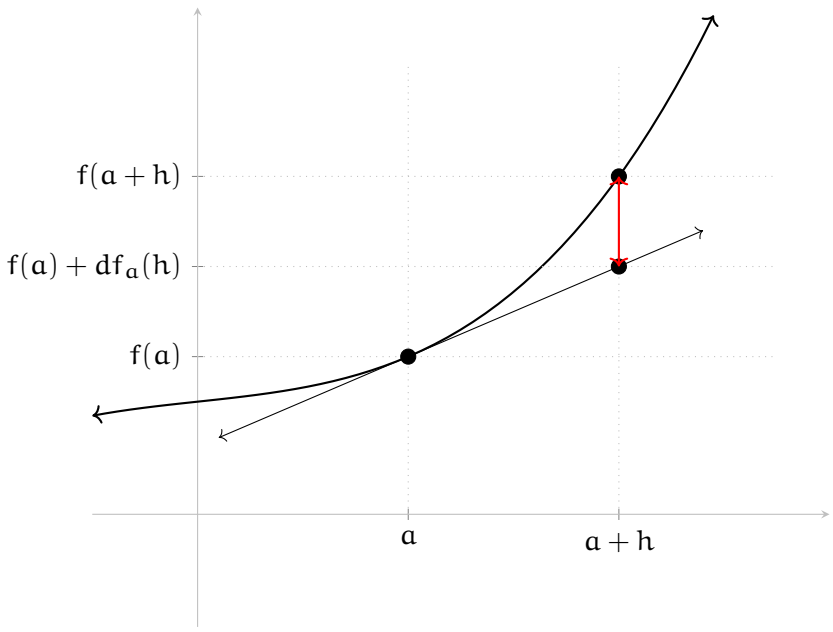


Figure 1.1.11: The thick black curve is the graph of a function $f$ and the thinner black line is its tangent line $f(a) + df_a$ are depicted in black. On a small input $h$, the "error in approximation" function $r$ outputs the vertical distance indicated in red.

$$f(a+h) = f(a) + df_a(h) + r(h).$$

This equation just says that $f(a+h)$ is the sum of the value of the linear approximation (which is $f(a) + df_a(h)$) and the error (which is $r(h)$).

Notice that $r(0) = 0$. Moreover, since $f$ is differentiable and therefore continuous at $a$ by exercise 1.1.5, we can see that $r$ is continuous at $0$. Finally, $r$ is "small." More precisely, we know that $|r(h)| = o(|h|)$ as $h \to 0$ by the discussion above, which means that

$$\lim_{h\to 0} \left| \frac{r(h)}{h} \right| = 0 \quad \Longrightarrow \quad \lim_{h\to 0} \frac{r(h)}{h} = 0.$$

As a first example of an application of this idea, here is a weak version of l'Hôpital's rule.

**Exercise 1.1.12** (L'Hôpital's rule, weak version)**.** Suppose $f, g : S \to \mathbb{R}$ are differentiable at an interior point $a \in S$, that $f(a) = g(a) = 0$, and that $g'(a) \neq 0$. Show that

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)}.$$

*Possible hint.* If $x = a + h$, we have

$$f(x) = f(a+h) = f(a) + df_a(h) + r(h) = f'(a)h + r(h),$$

where $r$ is the "error in approximation" function for $f$, as defined in remark 1.1.10.

## 1.2 Computing derivatives

The goal in this section is to prove a few of the basic rules of differentiation that one normally encounters in an introductory calculus course.

### 1.2.A Sum and scalar multiples rule

The following can be proved using either the definition of the derivative 1.1.2 or the equivalent "best linear approximation" characterization of proposition 1.1.8. My suggestion is that you try both methods for these exercises.

**Exercise 1.2.1** (Constant functions rule)**.** Prove that, if $f : S \to \mathbb{R}$ is *constant* (ie, there exists a real number $c$ such that $f(x) = c$ for all $x \in S$), then $f'(a) = 0$ for all interior points $a \in S$.

When you use proposition 1.1.8 to prove the following, you may find it useful to have done exercise 0.1.6.

**Exercise 1.2.2** (Sum rule)**.** Prove that, if $f, g : S \to \mathbb{R}$ are both differentiable at an interior point $a \in U$, then $f + g$ is also differentiable at $a$ and

$$(f + g)'(a) = f'(a) + g'(a).$$

**Exercise 1.2.3** (Scalar multiples rule)**.** Prove that, if $c$ is a constant and $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$, then $cf$ is also differentiable at $a$ and

$$(cf)'(a) = cf'(a).$$

### 1.2.B Product and quotient rules

**Proposition 1.2.4** (Product rule)**.** *Suppose* $f, g : U \to \mathbb{R}$ *are both differentiable at* $a \in U$*. Then* $fg$ *is also differentiable at* $a$ *and*

$$(fg)'(a) = g(a)f'(a) + f(a)g'(a).$$

*Proof.* The proof is a clever algebraic manipulation of difference quotients. The key trick is introducing a "cross term" of the form $f(a)g(a + h)$, and then cancelling it out by also introducing its negative. More precisely, for any sufficiently small real number $h$, we have

$$\begin{aligned}
\frac{(fg)(a+h) - (fg)(a)}{h} &= \frac{f(a+h)g(a+h) - f(a)g(a)}{h} \\
&= \frac{f(a+h)g(a+h) \color{blue}{- f(a)g(a+h) + f(a)g(a+h)} - f(a)g(a)}{h} \\
&= \frac{f(a+h)g(a+h) - f(a)g(a+h)}{h} + \frac{f(a)g(a+h) - f(a)g(a)}{h} \\
&= g(a+h) \cdot \frac{f(a+h) - f(a)}{h} + f(a) \cdot \frac{g(a+h) - g(a)}{h},
\end{aligned}$$

where the cross term and its negative is indicated in blue. Taking the limit as $h \to 0$ yields the result. □

**Exercise 1.2.5.** Check your understanding of the above proof of the product rule.

(a) Could you have instead introduced a cross term of the form $f(a+h)g(a)$? If so, rewrite the proof using this cross term. If not, explain why not.

(b) At what point is exercise 1.1.5 tacitly used in the above proof?

**Exercise 1.2.6.** Reprove the scalar multiples rule (exercise 1.2.3) using the product rule.

**Exercise 1.2.7** (Quotient rule)**.** Suppose $f, g : S \to \mathbb{R}$ are both differentiable at an interior point $a \in U$, and that $g(a) \neq 0$. Prove that $f/g$ is also differentiable at $a$ and

$$(f/g)'(a) = \frac{g(a)f'(a) - f(a)g'(a)}{g(a)^2}.$$

*Possible hint.* Observe that

$$\frac{(f/g)(a+h) - (f/g)(a)}{h} = \frac{\frac{f(a+h)}{g(a+h)} - \frac{f(a)}{g(a)}}{h}.$$

Multiply both the numerator and denominator by $g(a+h)g(a)$ to clear the denominators inside the numerator, and then introduce the cross term $f(a)g(a)$. Be sure to notice at what point you use exercise 1.1.5.

**Exercise 1.2.8** (Power rule for integer exponents)**.** Extend the result of exercise 1.1.4 to arbitrary integer exponents.

*Possible hint.* Use the quotient rule for negative integers. The exponent zero is a special but easy case; deal with it separately.

### 1.2.C Chain rule

**Proposition 1.2.9** (Chain rule)**.** *Suppose that $S$ and $T$ are both subsets of $\mathbb{R}$, that $f : S \to T$ is differentiable at an interior point $a$, that $f(a)$ is an interior point of $T$, and that $g : T \to \mathbb{R}$ is differentiable at $f(a)$. Then the composite $g \circ f : S \to \mathbb{R}$ is also differentiable at $a$, and*

$$(g \circ f)'(a) = g'(f(a))f'(a).$$

We will discuss two different (but similar) proofs of this important result. But first, here is an application. We will later prove a better version of this result (cf. theorem 1.3.18).

**Exercise 1.2.10** (Derivatives of inverses)**.** Suppose $f : S \to \mathbb{R}$ is injective, differentiable at an interior point $a \in S$, and that $b = f(a)$ is an interior point of $T$. Show that, if the inverse

$f^{-1} : f(S) \to \mathbb{R}$ is differentiable at $b$, then

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}.$$

**First proof**

The first proof uses the definition of the derivative definition 1.1.2, but there is a subtle technicality involved; so, let us discuss it informally first before diving into the formal proof.

The idea involved is again a clever rewriting of difference quotients. Again, we introduce a cross term, which in this is $f(a+h) - f(a)$. But this time we introduce this cross term "multiplicatively" rather than "additively." More precsiely, we have

$$\begin{aligned}\frac{(g \circ f)(a+h) - (g \circ f)(a)}{h} &= \frac{g(f(a+h)) - g(f(a))}{h} \\ &= \frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} \cdot \frac{f(a+h) - f(a)}{h}.\end{aligned}$$

The difference quotient on the right is precisely $f'(a)$. Thus, it would be sufficient to prove

$$\lim_{h \to 0} \frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} = g'(f(a)). \tag{1.2.11}$$

This formula looks an awful lot like the difference quotient that is used to define the derivative of $g$ at $f(a)$. In fact, since $f$ is continuous at $a$ by exercise 1.1.5, we have $f(a+h) \approx f(a) + h$ for small values of $h$, which means that

$$\frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} \approx \frac{g(f(a)+h) - g(f(a))}{f(a)+h-f(a)} = \frac{(g \circ f)(a+h) - (g \circ f)(a)}{h}$$

for small values of $h$. Taking the limit as $h \to 0$ on the right-hand side yields precisely $g'(f(a))$, so equation (1.2.11) seems like a reasonable conjecture.

The problem is that equation (1.2.11) isn't true. Issues arise when $f(a+h) = f(a)$ for arbitrarily small values of $h$, which leads to an infinite sequence of divisions by 0.

**Example 1.2.12.** Let

$$f(x) = \begin{cases} x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

and let $g(x) = x^2$. Observe that $g'(f(0)) = 0$. You will show later (in exercise 1.2.25) that $f$ is in fact differentiable at 0, so these functions do satisfy the hypothesis of the chain rule. However, the limit as $h \to 0$ of

$$\frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} = \frac{(h^2 \sin(1/h))^2}{h^2 \sin(1/h)}$$

does not exist. For example, the sequence $h_k = 1/k\pi$ converges to 0 as $k \to \infty$, but the expression above is undefined along this sequence.

We will get around the kind of issue that is described by the above example by carefully studying the expression

$$\frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)}. \tag{1.2.13}$$

Observe that, when $f(a+h) \neq f(a)$, then this expression is precisely the difference quotient that calculates the slope of the secant line passing through $(f(a), g(f(a))$ and $(f(a+h), g(f(a+h))$. When $f(a+h) = f(a)$, this secant line doesn't make sense; but what does make sense is the tangent line! So we'll remove the undefinedness of expression (1.2.13) by extending it so that it takes the value $g'(f(a))$ whenever $f(a+h) = f(a)$.

This leads us to our first proof of the chain rule.

*First proof of the chain rule.* Consider the function $\epsilon$ defined for small values of $h$ by $\epsilon(h) = f(a+h) - f(a)$. See figure 1.2.14. You will verify in exercise 2.3.5 below that $\epsilon$ is continuous at 0.

Next, define a function $\sigma$ by the following.

$$\sigma(k) = \begin{cases} \dfrac{g(f(a)+k) - g(f(a))}{k} & \text{if } k \neq 0 \\ g'(f(a)) & \text{if } k = 0 \end{cases}$$

In other words, $\sigma$ is the function that, on some small nonzero input $k$, outputs the slope of the secant line connecting the two points on the graph of $g$ where $x = f(a)$ and $x = f(a) + k$. See figure 1.2.15. It follows from the definition of the derivative of $g$ at $f(a)$ that the function $\sigma$ is continuous at 0.

Observe that

$$\frac{(g \circ f)(a+h) - (g \circ f)(a)}{h} = \sigma(\epsilon(h)) \cdot \frac{f(a+h) - f(a)}{h} \tag{1.2.16}$$

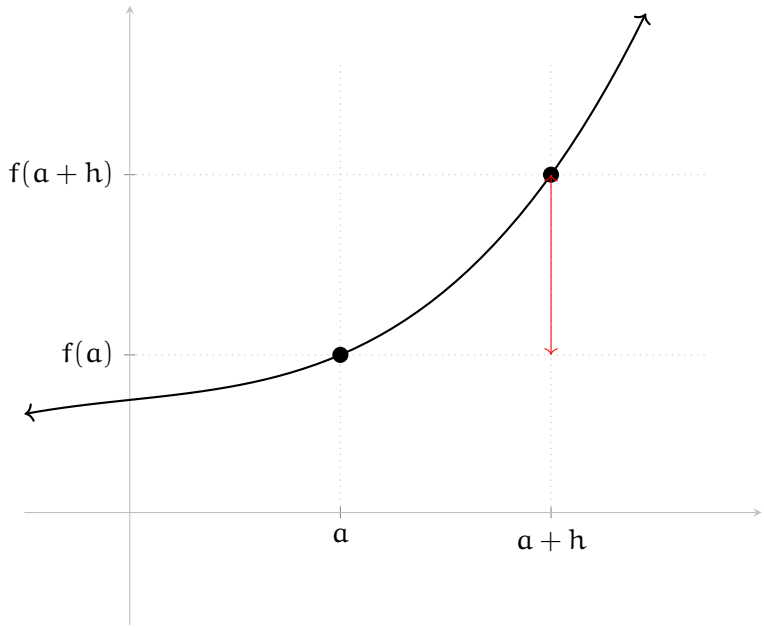


Figure 1.2.14: The graph of $f$ is depicted in black. The value of the function $t$ on a small nonzero input value $h$ is the vertical distance depicted in red.

for all small values of $h$, as you will verify in exercise 2.3.5 below. Thus

$$\begin{aligned}\lim_{h\to 0}\frac{(g\circ f)(a+h)-(g\circ f)(a)}{h} &= \lim_{h\to 0}\sigma(\epsilon(h))\cdot\frac{f(a+h)-f(a)}{h}\\ &\overset{(*)}{=} \sigma(\epsilon(0))f'(a)\\ &= g'(f(a))f'(a)\end{aligned}$$

where we have used the continuity of $\epsilon$ and $\sigma$ and the definition of the derivative of $f$ at $a$ for the step labeled $(*)$ above. This proves the chain rule. □

**Exercise 1.2.17.** Verify the following facts used in the proof above.

(a) Prove that $\epsilon$ is continuous at 0.

(b) Prove equation (1.2.16).

*Possible hint.* You might split up your proof of equation (1.2.16) into two cases: when $f(a+h) \neq f(a)$, and when $f(a+h) = f(a)$.

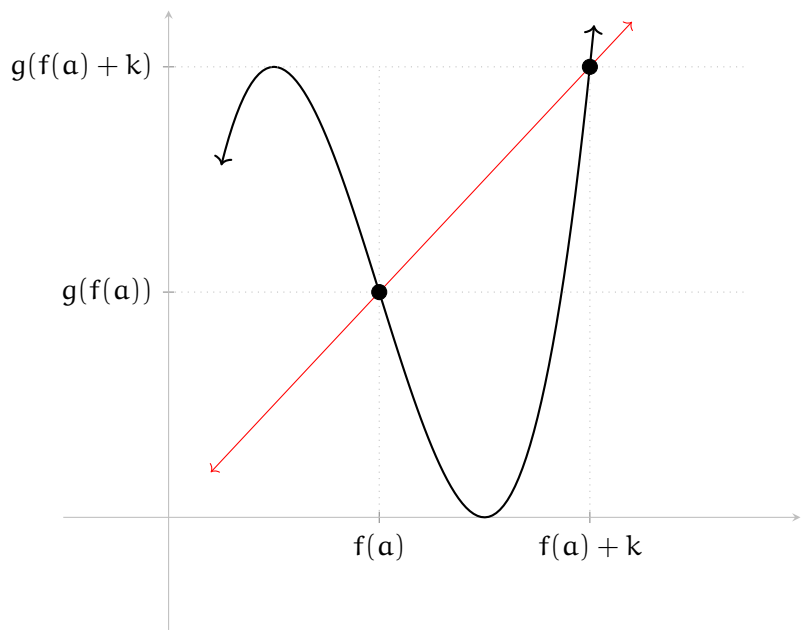


Figure 1.2.15: The graph of $g$ is depicted in black. The output of the function $\sigma$ on a small nonzero input value $k$ is the slope of the red secant line passing through the two points $(f(a), g(f(a))$ and $(f(a)+k, g(f(a)+k))$.

### Second proof

Our second proof of the chain rule will use the "best linear approximation" interpretation of derivatives provided by proposition 1.1.8.

*Second proof of the chain rule.* By proposition 1.1.8, it is sufficient to prove that

$$g(f(a+h)) - g(f(a)) - dg_{f(a)}(df_a(h)) = o(|h|) \text{ as } h \to 0.$$

Define the following "error in approximation" functions, as in remark 1.1.10.

$$\begin{aligned} r(h) &= f(a+h) - f(a) - df_a(h) \\ s(k) &= g(f(a)+k) - g(f(a)) - dg_{f(a)}(k) \end{aligned}$$

Since

$$f(a+h) = f(a) + df_a(h) + r(h)$$

by definition of $r$, we have

$$\begin{aligned} g(f(a+h)) &= g(f(a) + df_a(h) + r(h)) \\ &= g(f(a)) + dg_{f(a)}(df_a(h) + r(h)) + s(df_a(h) + r(h)) \\ &= g(f(a)) + dg_{f(a)}(df_a(h)) + dg_{f(a)}(r(h)) + s(df_a(h) + r(h)) \end{aligned}$$

where we use the definition of $s$ on input $k = df_a(h) + r(h)$ for the second step, and the fact that $dg_{f(a)}$ is linear for the third. Thus

$$g(f(a+h)) - g(f(a)) - dg_{f(a)}(df_a(h)) = dg_{f(a)}(r(h)) + s(df_a(h) + r(h)).$$

We therefore want to prove that the right-hand side is $o(|h|)$.

Since

$$dg_{f(a)}(r(h)) = g'(f(a)) \cdot r(h)$$

is just a scalar multiple of $r$, and $r = o(|h|)$, we see that $dg_{f(a)}(r) = o(|h|)$ also. So it is sufficient to prove that

$$s(df_a(h) + r(h)) = o(|h|).$$

We've now hit the technical part, so let's hit pause.

To see why this is technical, and for essentially the same reasons as the previous proof, notice that we're trying to prove that

$$\lim_{h \to 0} \frac{|s(df_a(h) + r(h)|)}{|h|} = 0.$$

To prove this, it's tempting to introduce a multiplicative cross-term as follows.

$$\frac{|s(df_a(h) + r(h)|)}{|h|} = \frac{|s(df_a(h) + r(h)|)}{|df_a(h) + r(h)|} \cdot \frac{|df_a(h) + r(h)|}{|h|}$$

But notice that $df_a(h) + r(h) = f(a + h) - f(a)$ by definition of $r$, so this cross term is problematic for the same reasons as before! If $f(a + h) = f(a)$ for $h$ arbitrary close to $0$, we have an infinite sequence of divisions by zeroes that causes problems.

We'll deal with this problem in essentially the same way as before: namely, we extend the definition of

$$\frac{|s(df_a(h) + r(h)|)}{|df_a(h) + r(h)|}$$

so that it is also defined when $f(a + h) = f(a)$.

*Second proof of the chain rule, continued.* Define $\eta(k) = s(k)/k$ for nonzero $k$. If we set $\eta(0) = 0$, the fact that $s(k) = o(|k|)$ implies that $\eta$ is continuous at $0$. Then

$$\begin{aligned} \frac{|s(df_a(h) + r(h))|}{|h|} &= \frac{|\eta(df_a(h) + r(h))| \cdot |df_a(h) + r(h)|}{|h|} \\ &\leqslant |\eta(df_a(h) + r(h))| \cdot \left( |f'(a)| + \frac{|r(h)|}{|h|} \right). \end{aligned}$$

Observe that $r$ and therefore $df_a + r$ is continuous at $0$ (cf. remark 1.1.10). Moreover, we have just seen that $\eta$ is continuous at $(df_a + r)(0) = 0$. Thus the composite $\eta \circ (df_a + r)$ is also continuous at $0$. Thus

$$\lim_{h \to 0} |\eta(df_a(h) + r(h))| \cdot \left( |f'(a)| + \frac{|r(h)|}{|h|} \right) = 0,$$

where we have used the fact that $|r(h)| = o(|h|)$ to see that the parenthetical expression has a finite limit (namely, $|f'(a)|$). So, by the squeeze theorem, we can therefore conclude that $|s(df_a(h) + r(h))| = o(|h|)$. □

Notice that in the first proof, the trick to overcoming the technicality was defining the

function $\sigma$ at $k = 0$. In the second proof, the trick to overcoming the same technicality was defining $\eta$ at $k = 0$. These two tricks are really the same trick, as shown by the following.

**Exercise 1.2.18.** Prove that the function $\eta$ from the second proof and the function $\sigma$ from the first proof are related by the equation

$$\sigma(k) = g'(f(a)) + \eta(k).$$

### 1.2.D Interior extremum theorem

We first recall the following definitions.

**Definition 1.2.19.** Supppose $X$ is a metric space and $f : X \to \mathbb{R}$ is a function. Suppose $a \in X$.

- $a$ is a *local maximum* of $f$ if there exists a neighborhood $U$ of $a$ in $X$ such that $f(x) \leqslant f(a)$ for all $x \in U$.
- $a$ is a *local minimum* of $f$ if there exists a neighborhood $U$ of $a$ in $X$ such that $f(x) \geqslant f(a)$ for all $x \in U$.
- $a$ is a *local extremum* if it is either a local maximum or a local minimum.
- $a$ is an *absolute maximum*, or just *maximum*, of $f$ if $f(x) \leqslant f(a)$ for all $x \in X$.
- $a$ is an *absolute minimum*, or just *minimum*, of $f$ if $f(x) \geqslant f(a)$ for all $x \in X$.
- $a$ is a *absolute extremum* or just *extremum*, if it is either an absolute maximum or an absolute minimum.

The following is a key property of derivatives that you probably remember from calculus.

**Exercise 1.2.20** (Interior extremum theorem)**.** Suppose an interior point $a \in S$ is a local extremum of a function $f : S \to \mathbb{R}$. If $f$ is differentiable at $a$, prove that $f'(a) = 0$.

*Possible hint.* Consider the following "slope of secant" function (which we've seen before, during the first proof of the chain rule in section 1.2.C).

$$\sigma(h) = \begin{cases} \dfrac{f(a+h) - f(a)}{h} & \text{if } h \neq 0 \\ f'(a) & \text{if } h = 0 \end{cases}$$

Since $f$ is differentiable at $x = a$, we know that $\sigma$ is continuous at $h = 0$. If $a$ is a local minimum, what can you say about the values of $\sigma$ for positive values of $h$? Negative values of $h$? Then consider the case when $a$ is a local minimum.

But remember, the converse to the interior extremum theorem 1.2.20 is *not* true.

**Exercise 1.2.21.** Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ where $f'(0) = 0$, but $f$ does not have a local extremum at 0.

### 1.2.E Non-rational functions

Using the rules we've developed so far, we can differentiate any rational function (ie, a quotient of a polynomial by a polynomial). In this section, we will simply state some facts about differentiation of other types of functions one frequently encounters. We will prove most of these a bit later; we are stating them now so that we can discuss a richer repertoire of examples as we go along.

First off, here is the most general statement of the power rule.

**Proposition 1.2.22** (Power rule for real exponents)**.** *If $r$ is any real number and $f : (0, \infty) \to \mathbb{R}$ is the function $f(x) = x^r$, then $f$ is differentiable and $f'(x) = rx^{r-1}$.*

We will prove this for *rational* exponents in exercise 1.3.19. We won't prove it in full generality, because that would require a long tangential discussion of what $x^r$ even means for irrational $r$. Actually, once we have a sufficiently rigorous definition of $x^r$ for irrational $r$, it turns out that proposition 1.2.22 is actually a relatively straightforward *consequence* of exercise 1.3.19.

Next up, we have the exponential, sine, and cosine functions. These are usually defined by their power series, as follows.

$$\begin{aligned} \exp(x) &= \sum_{n=0}^{\infty} \frac{x^n}{n!} \\ \cos(x) &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} \\ \sin(x) &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \end{aligned} \tag{1.2.23}$$

We'll prove that power series can be differentiated term-by-term in theorem 1.3.25. This allows us to prove the following standard facts.

(1) If $f(x) = \exp(x)$, then $f'(x) = \exp(x)$.

(2) If $f(x) = \sin(x)$, then $f'(x) = \cos(x)$.

(3) If $f(x) = \cos(x)$, then $f'(x) = -\sin(x)$.

**Remark 1.2.24.** Defining the exponential, sine, and cosine functions using power series is mathematically fine, but there are numerous facts about these functions that are not clear from the power series, and one needs to prove these facts separately. For example, it's useful knowing that, if $e := \exp(1)$, then $\exp(x) = e^x$. This fact is not immediately clear from the power series definition; it requires some proof, and we won't prove it here, but you should feel free to use it. It's obviously also useful to know that sines and cosines have something to do with trigonometry. Again, this is not clear from the power series definition; it requires some proof, and we won't prove it, but you should feel free to use it.

Let's now discuss some examples using these functions. Specifically, let's think about functions involving $\sin(1/x)$. As we will see, these functions exhibit a wild variety of pathological behavior.

**Exercise 1.2.25.** (a) Prove that the function

$$f(x) = \begin{cases} \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is discontinuous at 0.

*Possible hint.* Show the following.

$$\limsup_{x \to 0^{\pm}} \sin(1/x) = 1 \qquad \liminf_{x \to 0^{\pm}} \sin(1/x) = -1$$

(b) Prove that the function

$$f(x) = \begin{cases} x \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is continuous, but that it is not differentiable at 0.

(c) Prove that the function

$$f(x) = \begin{cases} x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is differentiable, but that the derivative $f'$ is not continuous at 0.

The following example, again involving $\sin(1/x)$, illustrates a caveat to the interior extremum theorem exercise 1.2.20: even if $f$ has a local extremum at a point, it need not be that the derivative "changes sign" at that point.

**Exercise 1.2.26.** Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = \begin{cases} x^4 \left(2 + \sin(1/x)\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

(a) Show that $f$ has an absolute minimum at 0.

(b) Show that $f$ is differentiable and calculate the derivative $f'$.

(c) Show that, on every open interval of the form $(a, 0)$ or $(0, b)$, the derivative $f'$ takes on both positive and negative values.

## 1.3 Differentiable functions

In this section, we'll discuss functions which are differentiable not just at a point, but on an entire interval.

### 1.3.A Mean value theorem

The mean value theorem 1.3.2 is an extremely important foundational result; it comes up incredibly often during proofs. In this section, we'll discuss the theorem and a few of its consequences; we will see many more of its consequences later on.

Rolle's theorem 1.3.1 is technically a special case of the mean value theorem, but the general case can be derived from this special case.

**Exercise 1.3.1** (Rolle's theorem)**.** Suppose $a < b$ and let $f : [a, b] \to \mathbb{R}$ be a differentiable function. If $f(a) = f(b) = 0$, prove that there exists $c \in (a, b)$ such that $f'(c) = 0$.

*Possible hint.* Any function must fall into at least one of the following three cases: (i) $f$ is constantly zero, (ii) $f$ takes on positive values somewhere on $[a, b]$, and (iii) $f$ takes on negative values somewhere on $[a, b]$. Deal with the first case using exercise 1.2.1, and the latter two cases using the extreme value theorem[2] together with exercise 1.2.20.

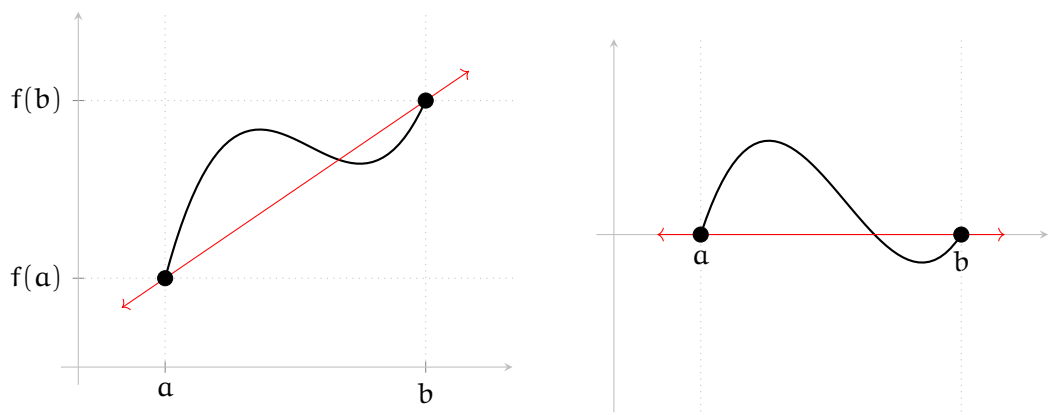


Figure 1.3.3: On the left is depicted the graph of a function $f$ and the secant line $\ell$. Subtracting $\ell$ from the picture on the left yields the picture on the right.

**Exercise 1.3.2** (Mean value theorem)**.** Suppose $a < b$ and let $f : [a, b] \to \mathbb{R}$ be a differentiable function. Prove that there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

*Possible hint.* Let $\ell$ be the secant line passing through $(a, f(a))$ and $(b, f(b))$. Then consider the function $g = f - \ell$. See figure 1.3.3. Then $g(a) = g(b) = 0$, so we can apply Rolle's theorem 1.3.1 to $g$.

We can use the mean value theorem to prove the following "if and only if upgrade" of exercise 1.2.1.

**Proposition 1.3.4.** *Suppose* $I$ *is an interval and that* $f : I \to \mathbb{R}$ *is differentiable. Then* $f' = 0$ *if and only if* $f$ *is constant.*

*Proof.* We showed in exercise 1.2.1 that $f$ being constant implies $f' = 0$. Conversely, suppose we have $a < b$ in $I$. We want to show that $f(a) = f(b)$. Since $f$ is differentiable on $I$, we can apply the mean value theorem on the smaller closed subinterval $[a, b]$. In other words, there exists a point $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$

[2]The *extreme value theorem* says that if $K$ is a compact metric space and $f : K \to \mathbb{R}$ is continuous, then $f$ achieves its global maximum and its global minimum somewhere on $K$ (cf. [PM91, theorems 3.17 and 6.30]).

But $f'(c) = 0$ by assumption, so $f(a) = f(b)$. $\square$

Here is another consequence, which says roughly that functions with bounded derivative cannot grow too fast.

**Exercise 1.3.5.** Suppose $I$ is an open interval and $f : I \to \mathbb{R}$ is a differentiable function such that there exists a number $M$ such that $|f'(x)| \leqslant M$ for all $x \in I$. Show that $|f(b) - f(a)| \leqslant M|b - a|$ for all $a, b \in I$.

### 1.3.B Monotonicity

Recall the following definition.

**Definition 1.3.6.** Suppose $f : S \to \mathbb{R}$ is a function and $x, y \in S$.

- $f$ is *increasing* if $x < y$ implies $f(x) \leqslant f(y)$.
- $f$ is *decreasing* if $x < y$ implies $f(x) \geqslant f(y)$.
- $f$ is *strictly increasing* if $x < y$ implies $f(x) < f(y)$.
- $f$ is *strictly decreasing* if $x < y$ implies $f(x) > f(y)$.

Further, $f$ is *monotone* if it is either increasing or decreasing, and *strictly monotone* if it is either strictly increasing or strictly decreasing.

Here is an important property of strictly monotone continuous functions.

**Exercise 1.3.7.** Prove that, if $I$ is an open interval and $f : I \to \mathbb{R}$ is continuous and strictly monotone, then $f(I)$ is also an open interval. Conclude that $f$ is an open map (ie, that the image under $f$ of any open subset of the domain is open).

*Possible hint.* Suppose $f$ is strictly increasing. If $I = (a, b)$, define $c = \lim_{x \to a^+} f(x)$ and $d = \lim_{x \to b^-} f(x)$ and use the intermediate value theorem to prove that $f(I) = (c, d)$. Then deal with the case when $f$ is strictly decreasing.

The proofs of the following standard relationships between derivatives and monotonicity are similar to the proof of proposition 1.3.4.

**Exercise 1.3.8** (Derivatives and monotonicity)**.** Suppose $I$ is an interval and that $f : I \to \mathbb{R}$ is differentiable. Prove the following.

(a) $f' \geqslant 0$ if and only if $f$ is increasing.

(b) $f' \leqslant 0$ if and only if $f$ is decreasing.

**Exercise 1.3.9** (Derivatives and strict monotonicity)**.** Suppose $I$ is an interval and that $f : I \to \mathbb{R}$ is differentiable. Prove the following.

(a) If $f' > 0$, then $f$ is strictly increasing.

(b) If $f' < 0$, then $f$ is strictly decreasing.

It's worth noticing that exercise 1.3.9 cannot be upgraded to if and only if statements.

**Exercise 1.3.10.** Give an example of a strictly increasing differentiable function $f : \mathbb{R} \to \mathbb{R}$ for which there exists some $a \in \mathbb{R}$ such that $f'(a) = 0$.

And again, there are pathologies involving monotonicity than can be exhibited by functions involving $\sin(1/x)$. Here is an example of a function whose derivative is positive at a single point, but it is not monotone near that point.

**Exercise 1.3.11.** Let $f : \mathbb{R} \to \mathbb{R}$ denote the function

$$f(x) = \begin{cases} x + 2x^2 \sin(1/x) & \text{if } x > 0 \\ 0 & \text{if } x = 0. \end{cases}$$

(a) Show that $f$ is differentiable and calculate $f'$. In particular, show that $f'(0) = 1$.

(b) Show that $f$ is not monotone on any open interval containing 0.

### 1.3.C Darboux's theorem ⋆

Derivatives are often "close" to being continuous, but in a weird way. They cannot have "simple" discontinuities, which means that when they are discontinuous at all, they are discontinuous in wild ways. Darboux's theorem is one way in which this principle manifests itself. Before getting to the precise statement, we make the following definition.

**Definition 1.3.12.** Suppose $I$ is an interval and $f : I \to \mathbb{R}$ is a function. Then $f$ *has the intermediate value property* if, for every $a < b$ in $I$ and every $y \in \mathbb{R}$ strictly between $f(a)$ and $f(b)$, there exists $x \in (a, b)$ such that $f(x) = y$.[3]

[3]Functions which have the intermediate value property are sometimes also called *Darboux functions*.

Using this vocabulary, the intermediate value theorem asserts that every continuous function has the intermediate value property [PM91, theorem 3.3]. Simple examples of discontinuous functions don't have the intermediate value property (cf. exercise 1.3.13 below). But, it turns out that some fairly bizarre discontinuous functions do have the intermediate value property (cf. exercise 1.3.14 below).

**Exercise 1.3.13.** Let $I$ be an interval and $f : I \to \mathbb{R}$ a function.

(a) Recall that $f$ has a *removable discontinuity* at a point $a \in I$ if

$$\lim_{x \to a} f(x)$$

exists, but the value of this limit does not equal $f(a)$. Show that, if $f$ has a removable discontinuity, then $f$ does *not* have the intermediate value property.

(b) Recall that $f$ has a *jump discontinuity* at a point $a \in I$ if the two one-sided limits

$$\lim_{x \to a^-} f(x) \text{ and } \lim_{x \to a^+} f(x)$$

both exist, but their values are not equal to each other. Show that, if $f$ has a jump discontinuity, then $f$ does *not* have the intermediate value property.

**Exercise 1.3.14.** Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

We know from exercise 1.2.25 that this function is discontinuous at $0$. Show that it has the intermediate value property.

*Possible hint.* First show that for any interval $I$ containing $0$, we have $f(I) = [-1, 1]$. Then explain how this implies the intermediate value property.

Now we know that the derivative $f'$ of a differentiable function $f$ need not be continuous (cf. exercise 1.2.25). Darboux's theorem asserts that, even though $f'$ might not be continuous, it must still have the intermediate value property! This means, for example, that derivatives cannot have removable or jump discontinuities (because, as we saw in exercise 1.3.13, functions with these kinds of discontinuities do not have the intermediate value property).

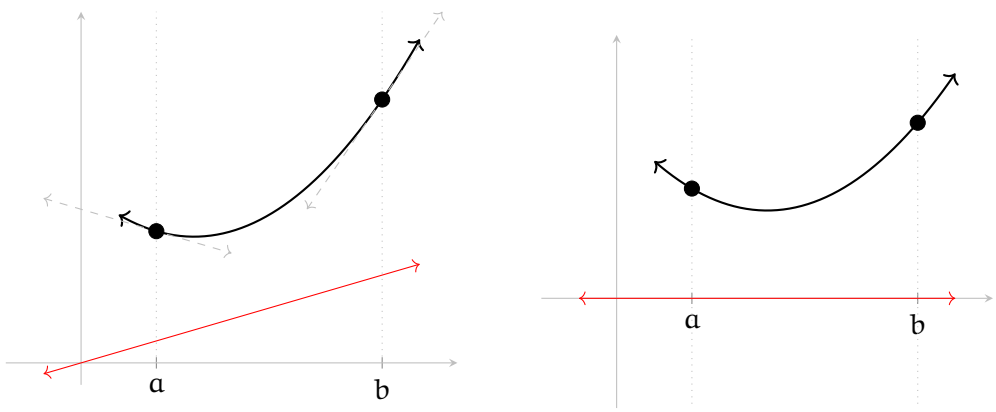


Figure 1.3.16: On the left is depicted the graph of a function $f$, the two tangent lines at $a$ and $b$, and the graph of the line $\ell(x) = mx$ for some $m$ in between $f'(a)$ and $f'(b)$. Subtracting $\ell$ from this picture yields the picture on the right.

**Theorem 1.3.15** (Darboux)**.** *Suppose* $I$ *is an open interval and* $f : I \to \mathbb{R}$ *is differentiable. Then* $f'$ *has the intermediate value property.*

*Proof.* Suppose $a < b$ are two elements of $I$ and $m$ is a real number strictly between $f'(a)$ and $f'(b)$. We want to show that there exists $c \in (a, b)$ such that $f'(c) = m$. This statement somewhat resembles the statement of the mean value theorem 1.3.2, and indeed, our proof will also somewhat resemble the proof we gave of the mean value theorem.

Consider the differentiable function $g : I \to \mathbb{R}$ defined by $g(x) = f(x) - mx$. See figure 1.3.16. Observe that $f'(c) = m$ if and only if $g'(c) = 0$. Thus we would like to show that there exists $c \in (a, b)$ such that $g'(c) = 0$.

There are two cases two consider: either (i) $f'(a) < m < f'(b)$ or (ii) $f'(a) > m > f'(b)$. Suppose we are in case (i). By the extreme value theorem, $g$ attains its minimum on the compact interval $[a, b]$ at some point $c \in [a, b]$. We claim that $c$ must be an interior point of $[a, b]$, and then interior extremum theorem 1.2.20 will imply that $g'(c) = 0$.

If $c = a$ were the minimum of $g$ on $[a, b]$, then we would have

$$\frac{g(a+h) - g(a)}{h} > 0$$

for all sufficiently small positive values of $h$. This would mean that $g'(a) \geqslant 0$, but we know that $g'(a) = f'(a) - m < 0$, which is a contradiction. Similarly, if $c = b$ were the minimum

of $g$ on $[a, b]$, then we would have

$$\frac{g(b+h) - g(b)}{h} < 0$$

for all sufficiently small negative values of $h$. This would imply that $g'(b) \leqslant 0$, but we know that $g'(b) = f'(b) - m > 0$.

In case (ii), we instead let $c$ be the *maximum* of $g$ on $[a, b]$, and argue similarly to show that $c$ must be an interior point. □

Here is an example application.

**Exercise 1.3.17.** Suppose $I$ is an open interval and $f : I \to \mathbb{R}$ is a differentiable function such that $f'(x) \neq 0$ for all $x \in I$. Show that $f$ is strictly monotone.

### 1.3.D Inverse function theorem

**Theorem 1.3.18.** *Suppose that $I$ is an open interval, that $f : I \to \mathbb{R}$ is differentiable, and that $f'(x) \neq 0$ for all $x \in I$. Then $f$ is open, the inverse function $f^{-1} : f(I) \to \mathbb{R}$ is differentiable, and*

$$f^{-1}(y) = \frac{1}{f'(f^{-1}(y))}$$

*for all $y \in f(I)$.*

*Proof.* Exercise 1.3.17 tells us that $f$ is strictly monotone, so exercise 1.3.7 tells us that $f$ is an open map and that $f(I)$ is an open interval. In particular, this means that $f^{-1}$ is continuous. To see that it is differentiable, fix $y \in f(I)$ and let $x = f^{-1}(y)$. For each $k$, define

$$h(k) = f^{-1}(y+k) - f^{-1}(y).$$

Rearranging, this is equivalent to all of the following.

$$\begin{aligned} f^{-1}(y+k) &= f^{-1}(y) + h(k) \\ f^{-1}(y+k) &= x + h(k) \\ y + k &= f(x + h(k)) \\ k &= f(x + h(k)) - y \\ k &= f(x + h(k)) - f(x) \end{aligned}$$

Thus

$$\frac{f^{-1}(y+k)-f^{-1}(y)}{k} = \frac{h(k)}{k} = \frac{h(k)}{f(x+h(k))-f(x)} = \frac{1}{\left(\dfrac{f(x+h(k))-f(x)}{h(k)}\right)}$$

Since $f^{-1}$ is continuous at $y$, we have $\lim_{k\to 0} h(k) = 0$. Thus the denominator on the far right above tends to $f'(x) = f'(f^{-1}(y))$ as $k \to 0$. Since the limit of the denominator is nonzero by assumption, the result follows from the quotient rule for limits. □

**Exercise 1.3.19** (Power rule for fractional exponents)**.** Extend the result of exercise 1.2.8 to arbitrary rational exponents (ie, to exponents that can be written in the form $m/n$ where $m$ and $n$ are both integers).

**Exercise 1.3.20.** If $f(x) = \ln(x)$, prove that $f'(x) = 1/x$. You can use the facts that $\ln : (0, \infty) \to \mathbb{R}$ is the inverse of the exponential function, and that the exponential function is its own derivative.

### 1.3.E Uniform limits ⋆

Derivatives interact somewhat strangely with limits. Here are some examples to illustrate that the limit of the derivatives is necessarily the derivative of the limit.

**Exercise 1.3.21.** For any positive integer $n$, let $f_n : \mathbb{R} \to \mathbb{R}$ be the function

$$f_n(x) = \frac{\sin(nx)}{n}.$$

(a) Show that $f_n$ is differentiable.

(b) Show that the $f_n$ converge uniformly to 0.

(c) Show that the derivatives $f_n'$ do *not* converge pointwise.

**Exercise 1.3.22.** For any positive integer $n$, let $f_n : \mathbb{R} \to \mathbb{R}$ be the function

$$f_n(x) = \sqrt{x^2 + \frac{1}{n}}.$$

(a) Show that $f_n$ is differentiable.

(b) Show that the $f_n$ converge uniformly to the absolute value function $f(x) = |x|$.

(c) Show that the derivatives $f_n'$ converge pointwise but *not* uniformly.

In fact, to get well-behaved interacted between limits and derivatives, what we really need is uniform convergence *of the derivatives*.

**Theorem 1.3.23.** *Suppose* $U$ *is an open subset and* $f_n : U \to \mathbb{R}$ *is a differentiable function for each* $n = 0, \dots$ *such that* $\lim f_n = f$ *pointwise. If the derivatives* $f_n'$ *converge uniformly on compact subsets of* $U$*, then* $f$ *is differentiable and* $\lim f_n' = f'$.

*Proof.* Set $g = \lim f_n'$. Fix $a \in U$ and consider the following "slope of secant" functions (as in the first proof of the chain rule in section 1.2.C).

$$\sigma_n(h) = \begin{cases} \dfrac{f_n(a+h) - f_n(a)}{h} & \text{if } h \neq 0 \\ f_n'(a) & \text{if } h = 0 \end{cases}$$

$$\sigma(h) = \begin{cases} \dfrac{f(a+h) - f(a)}{h} & \text{if } h \neq 0 \\ g(a) & \text{if } h = 0 \end{cases}$$

Continuity of $\sigma_n$ and $\sigma$ away from $h = 0$ is clear. At $h = 0$, we know that $\sigma_n$ is continuous because $f_n$ is differentiable at $a$. Note moreover that $\sigma_n$ converges pointwise to $\sigma$.

In fact, $\sigma_n$ converges *uniformly* to $\sigma$ on a compact interval $I$ containing $0$. To see this, observe that for any $m, n$, and $h \in I \setminus \{0\}$, we have

$$\sigma_m(h) - \sigma_n(h) = \frac{(f_m(a+h) - f_n(a+h)) - (f_m(a) - f_n(a))}{h} = f_m'(\xi) - f_n'(\xi),$$

where for the last step we have applied the mean value theorem to the function $f_m - f_n$ on the closed interval between $a$ and $a + h$ to find $\xi$ strictly between $a$ and $a + h$. Moreover, we have $\sigma_m(0) - \sigma_n(0) = f_m'(a) - f_n'(a)$. In other words, for every $h \in I$, there exists $\xi \in a + I$ such that $\sigma_m(h) - \sigma_n(h) \leqslant f_m'(\xi) - f_n'(\xi)$. Thus

$$\|\sigma_m - \sigma_n\|_{\sup, I} \leqslant \|f_m' - f_n'\|_{\sup, a+I}.$$

Since the derivatives converge uniformly on the compact subset $a + I$, they are uniformly Cauchy. It follows from the above inequality that the $\sigma_n$ are also uniformly Cauchy, and therefore uniformly convergent. But $\sigma$ is the pointwise limit of the $\sigma_n$, so it must be the uniform limit.

Since the $\sigma_n$ are continuous on $I$, and uniform limits of continuous functions are also continuous, we conclude that $\sigma$ is also continuous. This proves that $f$ is differentiable at $a$ and $f'(a) = g(a) = \lim f_n'(a)$. $\square$

**Remark 1.3.24.** The hypotheses of this theorem can be made weaker by only assuming that the functions $f_n$ converge *just at a single point* rather than pointwise on all of $U$, but one still needs to assume uniform convergence of the derivatives on compact subsets [Rud76, theorem 7.17]. The idea of the proof of this generalization is similar to the proof above, but the flow of the argument gets obscured by more technical details.

This theorem lets us prove that we can differentiate power series term-by-term.

**Theorem 1.3.25.** *Suppose the power series* $\sum_{n=0}^{\infty} a_n x^n$ *has a nonzero radius of convergence* $R$. *Then the function* $f : (-R, R) \to \mathbb{R}$ *given by*

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

*is differentiable and*

$$f'(x) = \sum_{n=0}^{\infty} n a_n x^{n-1}.$$

*Moreover, the radius of convergence of the power series* $\sum_{n=0}^{\infty} n a_n x^{n-1}$ *is also* $R$.

*Proof.* Let

$$f_n(x) = a_0 + a_1 x + \cdots + a_n x^n$$

be the sequence of the partial sums of the power series. Then clearly $\lim f_n = f$ pointwise. The conclusion of theorem 1.3.25 would therefore follow by applying theorem 1.3.23 to the sequence $f_n$, provided we first show that $f_n'$ converges uniformly on compact subsets of $(-R, R)$. To prove this, it is sufficient to show that the radius of convergence of the power series

$$\sum_{n=0} n a_n x^{n-1}$$

is also $R$. Let $R'$ denote the radius of convergence of this power series; we want to show that $R' = R$. Observe that

$$\frac{1}{R'} = \limsup_{n\to\infty} (n a_n)^{1/(n-1)} = \limsup_{n\to\infty} ((n a_n)^{1/n})^{n/(n-1)}.$$

We know that $\lim n^{1/n} = 1$ and that $\lim n/(n-1) = 1$. We also know that

$$\frac{1}{R} = \limsup_{n\to\infty} a_n^{1/n}.$$

Putting all of this together, we conclude that $R = R'$. □

**Exercise 1.3.26.** Verify that, if the exponential, sine, and cosine functions are defined by their power series as in equation (1.2.23), then their derivatives are calculated by the formulas we stated in section 1.2.E.

## 1.4 Levels of differentiability

### 1.4.A Continuous differentiability

We saw in section 1.3.C that the derivative of a differentiable function is always "close" to being continuous, though in a slightly bizarre way. Insisting that the derivative actually be continuous is often useful for ruling out pathological behavior.

**Definition 1.4.1.** Suppose $U$ is an open subset of $\mathbb{R}$ and $f : U \to \mathbb{R}$ is differentiable. If the derivative $f' : U \to \mathbb{R}$ is also continuous, then $f$ is said to be *continuously differentiable*.

For example, we saw in exercise 1.3.11 that there is a differentiable function whose derivative is positive at a point, but which is not monotone on any interval containing that point. This is rather bizarre behavior, and it cannot happen with continuously differentiable functions.

**Exercise 1.4.2.** Suppose $f : U \to \mathbb{R}$ is continuously differentiable, and $a \in U$ is a point such that $f'(a) \neq 0$. Show that there exists an open interval containing $a$ on which $f$ is strictly monotone.

To appreciate the following fact, it's worth noting the following: for any open interval $I$, there exist differentiable functions $f : I \to \mathbb{R}$ such that $f'(x) = 0$ for all $x \in I \cap \mathbb{Q}$, but $f$ is *not* constant (and $f'$ does *not* constantly vanish)! These extremely bizarre functions were first found by Pompeiu [Pom07]. This pathological behavior is also ruled out by insisting that the derivative be continuous.

**Exercise 1.4.3.** Suppose $f : U \to \mathbb{R}$ is continuously differentiable.

(a) The *critical locus* of $f$ is the set

$$Z = \{x \in U : f'(x) = 0\}.$$

Show that $Z$ is a closed subset of $U$.

(b) Conclude that, if $f'(x) = 0$ for all $x \in U \cap \mathbb{Q}$, then $f'(x) = 0$ for all $x \in U$.

Unfortunately, insisting that the derivative of a function be continuous does not rule out all possible bizarre behavior. For example, in exercise 1.2.26, we saw a differentiable function $f$ with an absolute minimum at a point, but the derivative doesn't just simply "change sign" at that point. In fact, it turns out that function is also continuously differentiable (cf. exercise 1.4.4), so merely insisting on continuous differentiability does not quite manage to rule out that kind of pathological behavior. However, as it turns out, the derivative of $f'$ (called the "second derivative" of $f$, and denoted $f''$), turns out *not* to be continuous.

**Exercise 1.4.4.** Let $f : \mathbb{R} \to \mathbb{R}$ be the function from exercise 1.2.26. Show that $f'$ is differentiable, but that $f''$ is not continuous.

This leads us to thinking about iterated derivatives.

### 1.4.B $C^k$ functions

**Definition 1.4.5.** Let $U$ be an open subset of $\mathbb{R}$ and $f : U \to \mathbb{R}$ a function. We then define the following.

- $f$ is said to be $C^0$ if it is continuous.
- $f$ is said to be $C^1$ if it is continuously differentiable.
- $f$ is said to be $C^2$ if it is differentiable and $f'$ is $C^1$. In this case, the derivative $(f')'$ of $f'$ is called the *second derivative* of $f$, and is denoted either $f''$ or $f^{(2)}$.
- $f$ is said to be $C^3$ if it is differentiable and $f'$ is $C^2$. In this case, the second derivative $(f')''$ of $f'$ is called the *third derivative* of $f$ and is denoted $f^{(3)}$.
- $f$ is said to be $C^4$ if it is differentiable and $f'$ is $C^3$. In this case, the third derivative $(f')^{(3)}$ of $f'$ is called the *fourth derivative* of $f$ and is denoted $f^{(4)}$.

Inductively, for any positive integer $k$, $f$ is said to be $C^k$ if it is differentiable and $f'$ is $C^{k-1}$. In this case, the $(k-1)$st derivative $(f')^{(k-1)}$ of $f'$ is called the $k$*th derivative* of $f$ and is denoted $f^{(k)}$.

First up, let's prove that all reasonable ways of combining functions in a particular differentiability class stays within that differentiability class.

**Exercise 1.4.6.** Prove that the set of all $k$ times differentiable functions $U \to \mathbb{R}$ is a vector space.

**Exercise 1.4.7.** Suppose that $f, g : U \to \mathbb{R}$ are $C^k$ functions. Prove that the product $fg$ is also $C^k$, and that

$$(fg)^{(k)} = \sum_{i=0}^{k} \binom{k}{i} f^{(i)} g^{(k-i)}.$$

**Exercise 1.4.8.** Show that the composite of two $C^k$ functions is also $C^k$.

*Possible hint.* Induct on $k$. The base case $k = 0$ is clear. For the inductive step, use the chain rule 1.2.9 and exercise 1.4.7.

**Exercise 1.4.9.** Suppose $f : U \to \mathbb{R}$ is $C^k$ and $f(x) \neq 0$ for all $x \in U$. Show that $1/f$ is also $C^k$.

*Possible hint.* You might first show that the function $g(x) = 1/x$ is $C^k$ for all $k$. Then $1/f = g \circ f$, so you can apply exercise 1.4.8.

**Lemma 1.4.10.** *Suppose* $I$ *is an open interval,* $f : I \to \mathbb{R}$ *is* $C^k$ *injection for some* $k \geqslant 1$*, and that* $f'(x) \neq 0$ *for all* $x \in I$*. Then* $f^{-1} : f(I) \to U$ *is also* $C^k$*.*

*Proof.* Since $f'$ is nonzero on $I$, we know from theorem 1.3.18 that $f^{-1}$ is differentiable and that

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}.$$

In other words, $(f^{-1})'$ is the composite of three functions: $f^{-1}$, $f'$, and the function $x \mapsto 1/x$.

We will inductively prove that $f^{-1}$ is $C^r$ for all $0 \leqslant r \leqslant k$. Certainly $f^{-1}$ is $C^0$ (ie, continuous), since it is even differentiable, as we saw above. Inductively, suppose $f^{-1}$ is $C^r$ for some $0 \leqslant r < k$. We know that $f$ is $C^k$ and therefore $C^r$, and also that $x \mapsto 1/x$ is $C^r$ (cf. exercise 1.4.9), so $(f^{-1})'$ is the composite of three $C^r$ functions. Thus exercise 1.4.8 tells us that $(f^{-1})'$ is $C^r$, which shows that $f^{-1}$ is $C^{r+1}$. This completes the induction. $\square$

Observe that we have an infinite chain of implications

$$\cdots \implies C^3 \implies C^2 \implies C^1 \implies C^0.$$

The following exercise shows that all of these implications are strict; in other words, for every $k$, there exists a $C^k$ function which is not $C^{k+1}$.

**Exercise 1.4.11.** For every non-negative integer $k$, let

$$f_k(x) = x^k|x|.$$

Prove that $f_k$ is $C^k$ but not $C^{k+1}$.

*Possible hint.* First prove that $f'_{k+1} = (k+2)f_k$.

### 1.4.C Taylor's theorem ⋆

**Theorem 1.4.12** (Taylor)**.** *Suppose $f : U \to \mathbb{R}$ is $C^k$ for some positive integer $k$ and $a \in U$. Then there exists a unique polynomial $p_k$ at most $k$, called the* degree $k$ Taylor polynomial of $f$ at $a$, *such that $|f(a+h) - p_k(h)| = o(|h|^k)$ as $h \to 0$. Moreover, we have*

$$p_k(h) = f(a) + f'(a)h + \frac{f''(a)}{2!}h^2 + \frac{f^{(3)}(a)}{3!}h^3 + \cdots + \frac{f^{(k)}(a)}{k!}h^k. \tag{1.4.13}$$

*Proof.* Let $p_k(h)$ be defined as equation (1.4.13) and set $r(h) = f(a+h) - p_k(h)$. Observe that $r$ is $C^k$ by exercise 1.4.6. By direct calculation, we have $r^{(i)}(0) = 0$ for all $i = 0, 1, \ldots, k$.

Suppose $h > 0$. By iterating the mean value theorem, we have

$$\begin{aligned} r(h) &= r(h) - r(0) \\ &= r'(h_1)h = (r'(h_1) - r'(0))h \\ &= r''(h_2)h_1h = (r''(h_2) - r''(0))h_1h \\ &\vdots \\ &= r^{(k-1)}(h_{k-1})h_{k-2}h_{k-3}\cdots h_2h_1h \end{aligned}$$

where $0 < h_{k-1} < \cdots < h_2 < h_1 < h$. This means that

$$\begin{aligned}
\frac{|r(h)|}{|h|^k} &= \left|\frac{r^{(k-1)}(h_{k-1})h_{k-2}\cdots h_2 h_1 h}{h^k}\right| \\
&\leqslant \left|\frac{r^{(k-1)}(h_{k-1})h^{k-1}}{h^k}\right| \\
&= \left|\frac{r^{(k-1)}(h_{k-1})}{h}\right| \\
&\leqslant \left|\frac{r^{(k-1)}(h_{k-1})}{h_{k-1}}\right| \\
&= \left|\frac{r^{(k-1)}(h_{k-1}) - r^{(k-1)}(0)}{h_{k-1}}\right|.
\end{aligned}$$

Now as $h \to 0^+$, we also have that $h_{k-1} \to 0$, so this final expression tends to $r^{(k)}(0) = 0$. By the squeeze theorem, we conclude that

$$\lim_{h\to 0^+} \frac{|r(h)|}{|h|^k} = 0.$$

One argues similarly with $h < 0$ to prove that

$$\lim_{h\to 0^-} \frac{|r(h)|}{|h|^k} = 0.$$

In this case, we have $h < h_1 < h_2 < \cdots < h_{k-1} < 0$. This proves that $|r(h)| = o(|h|^k)$.

To prove that $p_k$ is unique, suppose $q$ is any polynomial of degree at most $k$ such that $|f(a+h) - q(h)| = o(|h|^k)$. Then

$$|p_k(h) - q(h)| = |p_k(h) - f(a+h) + f(a+h) - q(h)| \leqslant |f(a+h) - p_k(h)| + |f(a+h) - q(h)|$$

so exercises 0.1.6 and 0.1.7 imply that $|p_k(h) - q(h)| = o(|h|^k)$. But $p_k - q$ is a polynomial of degree at most $k$, and a nonzero polynomial of degree at most $k$ cannot be $o(|h|^k)$ as $h \to 0$. So we must have $p_k = q$. □

**Remark 1.4.14.** It's worth remarking that Taylor's theorem does not use the fact that $f^{(k)}$ is continuous; it merely requires that $f^{(k)}$ exists, ie, that $f$ is $k$ *times differentiable*. This is evident in the proof above.

We can now return to our discussion of the pathology exhibited by exercises 1.2.26

and 1.4.4, and prove that such pathologies cannot occur under some hypotheses. We prepare for this discussion with the following lemma.

**Lemma 1.4.15.** *Suppose* $f : U \to \mathbb{R}$ *is* $C^k$ *for some* $k \geqslant 1$*, that*

$$f(a) = f'(a) = f^{(2)}(a) = \cdots = f^{(k-1)}(a) = 0,$$

*and* $f^{(k)}(a) > 0$*. Then there exists an* $\epsilon > 0$ *such that* $f$ *is positive on* $(a, a+\epsilon)$*, and on* $(a-\epsilon, a)$*,* $f$ *is positive if* $k$ *is even and negative if* $k$ *is odd. If* $f^{(k)}(a) < 0$*, then all instances of "positive" in the conclusion get replaced by "negative" and vice versa.*

*Proof.* Let us suppose that $f^{(k)}(a) > 0$. By Taylor's theorem 1.4.12, we see that

$$f(a+h) = \frac{f^{(k)}(a)}{k!}h^k + r(h)$$

where $|r(h)| = o(|h|^k)$ as $h \to 0$. Dividing through by $h^k$, we see that

$$\frac{f(a+h)}{h^k} = \frac{f^{(k)}(a)}{k!} + \frac{r(h)}{h^k}.$$

Since $f^{(k)}(a) > 0$, we also have $f^{(k)}(a)/k! > 0$. Since $\lim_{h\to 0} |r(h)/h^k| = 0$, there exists $\epsilon > 0$ such that $|r(h)/h^k| < \epsilon$ for all $|h| < \epsilon$. Then, if $0 < |h| < \epsilon$, we have

$$\frac{f(a+h)}{h^k} = \frac{f^{(k)}(a)}{k!} + \frac{r(h)}{h^k} > 0.$$

Now note that if $k$ is even, then $h^k$ is always positive, so the lemma follows. On the other hand, if $k$ is odd, then $h^k$ is positive for positive $h$ and negative for negative $h$, and again the lemma follows. The proof when $f^{(k)}(a) < 0$ is similar. □

**Exercise 1.4.16.** Suppose $f$ is $C^k$ for some $k \geqslant 1$, that

$$f'(a) = f^{(2)}(a) = \cdots = f^{(k-1)}(a) = 0$$

and $f^{(k)}(a) \neq 0$. Then there exists an $\epsilon > 0$ such that $f$ is strictly monotone on $(a-\epsilon, a)$ and on $(a, a+\epsilon)$.

*Possible hint.* Apply lemma 1.4.15 to $f'$. Then use exercise 1.3.9.

**Example 1.4.17.** Let $f : \mathbb{R} \to \mathbb{R}$ be the function of exercises 1.2.26 and 1.4.4. Then $f$ is twice differentiable and $f'(0) = f''(0) = 0$, but it is not thrice differentiable at 0. In other words, it doesn't eventually have a nonzero derivative at 0, so we're not able to apply exercise 1.4.16.

### 1.4.D Smooth functions

Insisting that a function be $C^k$ for larger and larger values of $k$ rules out more and more pathological behavior. This suggests the following.

**Definition 1.4.18.** A function $f : U \to \mathbb{R}$ is said to be $C^\infty$, or *infinitely differentiable*, or *smooth*, if it is $C^k$ for all $k$.

It's a little annoying that there are three different words that all mean the same thing, but that's just how it is; all three are commonly used in the mathematical literature, so it's best to get used to all of them. We'll usually use the word "smooth."

It follows immediately from exercises 1.4.6 to 1.4.9 and lemma 1.4.10 that sums, scalar multiples, products, reciprocals, composites, and inverses of smooth functions are also smooth.

The class of smooth functions is very well-behaved. On the one hand, as we have already seen, smoothness rules out lots of pathological behavior. For example, functions like exercise 1.3.11 are not smooth, and in fact, if the derivative of a smooth function $f$ is nonzero at a single point, then $f$ must be strictly monotone in a neighborhood of that point (cf. exercise 1.4.2). We've also seen that functions like the one from exercise 1.2.26 and example 1.4.17 are not smooth, and that if $f$ is smooth and $a$ is a local extremum and eventually $f$ has a nonzero derivative at $a$, then in fact the derivative of $f$ must "change sign" at $a$ (cf. exercise 1.4.16).

On the other hand, the class of smooth functions is not overly restrictive. There is a vast array of smooth functions, and we can tailor smooth functions to almost arbitrary specifications.

#### Diffeomorphisms

Our first example of this principle is the following smooth bijection $(-1, 1) \to \mathbb{R}$ whose inverse is also smooth. A smooth bijection with a smooth inverse is also called a "diffeomorphism." Roughly, the fact that there exists diffeomorphism $(-1, 1) \to \mathbb{R}$ "smoothly stretch out" $(-1, 1)$ to all of $\mathbb{R}$, and also "smoothly shrink" all of $\mathbb{R}$ down to $(-1, 1)$.

**Example 1.4.19.** Consider the function $f : (-1, 1) \to \mathbb{R}$ defined by

$$f(x) = \frac{x}{1 - x^2}.$$

See figure 1.4.20. Since this is a quotient of two smooth functions and the denominator is nonzero on $(-1, 1)$, this function is also smooth. We can calculate that

$$f'(x) = \frac{1 + x^2}{(1 - x^2)^2}$$

and we can see from this formula that the derivative is always strictly positive, so $f$ is strictly increasing (cf. exercise 1.3.9). In particular, it is injective. Moreover, we have

$$\lim_{x \to -1^+} f(x) = -\infty \quad \text{and} \quad \lim_{x \to 1^-} f(x) = +\infty$$

so the intermediate value theorem guarantees that $f$ is bijective. Thus the inverse function $f^{-1} : \mathbb{R} \to (-1, 1)$ exists. Finally, since $f'$ is always nonzero, it follows from lemma 1.4.10 that $f^{-1}$ must be smooth. It is possible to verify that

$$f^{-1}(x) = \frac{-1 + \sqrt{4x^2 + 1}}{2x}$$

by check that composing this formula with $f$ yields the identity, but, thanks to lemma 1.4.10, we don't actually need this formula for $f^{-1}$ at all to know that $f^{-1}$ is smooth.

Of course, there's nothing special about the open interval $(-1, 1)$. In fact, we can construct a diffeomorphism between any pair of open intervals!

**Exercise 1.4.21.** (a) For real numbers $a < b$, construct a diffeomorphism $f : (a, b) \to \mathbb{R}$.

(b) Show that the exponential function is a diffeomorphism $\mathbb{R} \to (0, \infty)$.

(c) Construct a diffeomorphism $\mathbb{R} \to (a, \infty)$ for any real number $a$.

(d) Construct a diffeomorphism $\mathbb{R} \to (-\infty, a)$ for any real number $a$.

(e) If $I$ and $I'$ are both open intervals, show that there exists a diffeomorphism $f : I \to I'$.

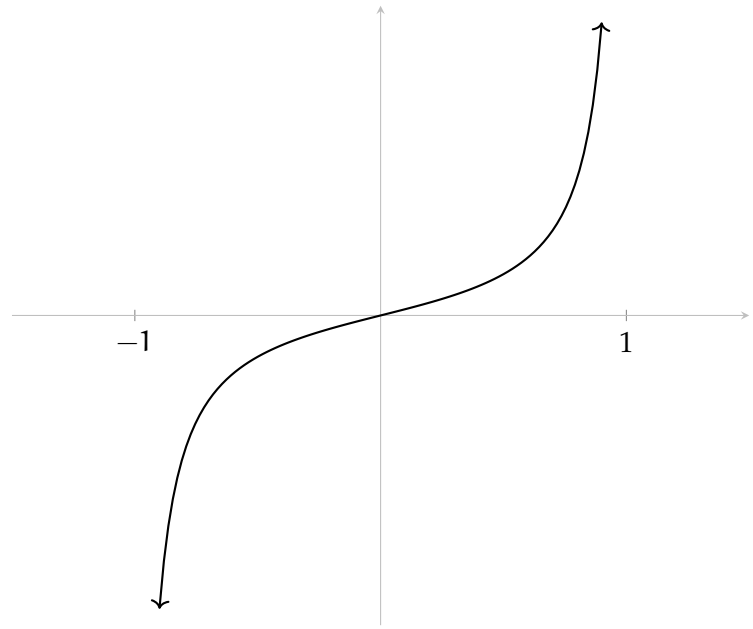


Figure 1.4.20: The graph of a diffeomorphism $(-1, 1) \to \mathbb{R}$.

#### Infinitely flat functions

We next discuss an example of a smooth function which gets "infinitely flat" at 0, but is not constantly equal to 0. But first, a preliminary remark.

**Remark 1.4.22.** Here is a fact that you might recognize:

$$\lim_{u \to \infty} \frac{u^m}{e^u} = 0 \tag{1.4.23}$$

for any non-negative integer $m$. We'll use this fact, and also the related fact that

$$\lim_{x \to 0^+} \frac{e^{-1/x}}{x^m} = 0$$

for any non-negative integer $m$, which follows immediately from equation (1.4.23) by making the substitution $u = 1/x$.

You probably remember proving equation (1.4.23) using l'Hôpital's rule in your calculus class. We haven't proved l'Hôpital's rule (except for the weak version in exercise 1.1.12, which you can check is not enough to prove equation (1.4.23)), but you can find proofs of

l'Hôpital's rule in many places (eg, [PM91, theorem 4.15], [Rud76, theorem 5.13], or even Wikipedia). The proof is a clever application of the mean value theorem 1.3.2.

An alternative proof of equation (1.4.23) uses the power series definition of the exponential function. The idea is to notice that

$$\frac{e^u}{u^m} = \frac{1}{u^m} + \cdots + \frac{1}{(m-1)!u} + \sum_{k=0}^{\infty} \frac{u^k}{(m+k)!}$$

As $u \to \infty$, the first few summands all tend to 0, and the series at the end tends to $\infty$. Equation (1.4.23) follows from this; details are omitted, but you should be able to work them out yourself if you're interested.

**Exercise 1.4.24.** Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined as follows.

$$f(x) = \begin{cases} e^{-1/x} & \text{if } x > 0 \\ 0 & \text{if } x \leqslant 0 \end{cases}$$

See figure 1.4.25.

(a) Use remark 1.4.22 to show, using the definition of the derivative definition 1.1.2, that $f'(0) = 0$.

(b) Check that

$$f'(x) = \begin{cases} \dfrac{e^{-1/x}}{x^2} & \text{if } x > 0 \\ 0 & \text{if } x \leqslant 0. \end{cases}$$

You've already computed $f'(0)$. You can compute formulas for the derivative $f'(x)$ for $x > 0$ and for $x < 0$ using the usual rules for differentiation.

(c) Use remark 1.4.22 to show that $f''(0) = 0$, and then check that

$$f''(x) = \begin{cases} p_2(x) \cdot \dfrac{e^{-1/x}}{x^4} & \text{if } x > 0 \\ 0 & \text{if } x \leqslant 0 \end{cases}$$

where $p_2$ is the function $p_2(x) = 1 - 2x$.

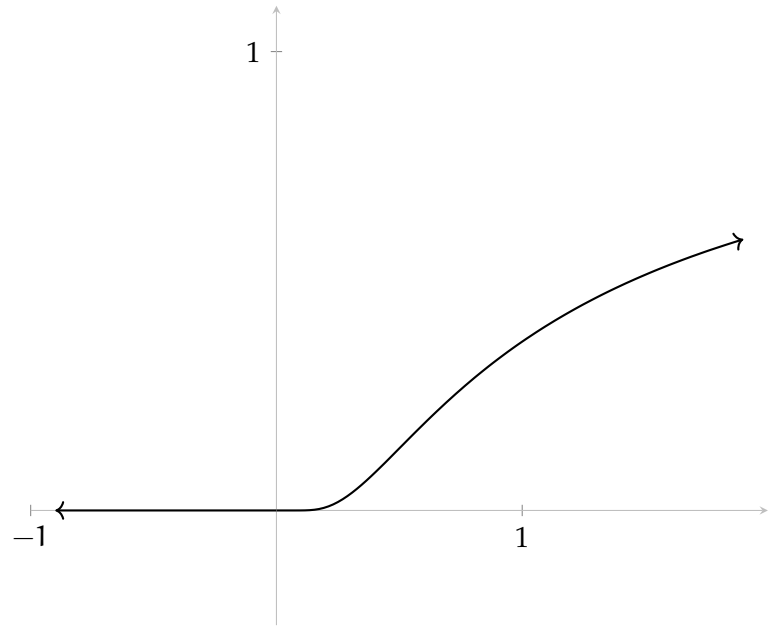


Figure 1.4.25: The graph of a function that is "infinitely flat" at 0.

(d) Inductively, prove that

$$f^{(k)}(x) = \begin{cases} p_k(x) \cdot \dfrac{e^{-1/x}}{x^{2k}} & \text{if } x > 0 \\ 0 & \text{if } x \leqslant 0 \end{cases}$$

where $p_k$ is a polynomial function of degree $k-1$ such that $p_k(0) = 1$.

Thus $f$ is an example of a smooth function that is "infinitely flat" at $x = 0$, in the sense that $f^{(k)}(x) = 0$ for all $k$, but is not constantly equal to 0.

### Bump functions

Next up, we have "bump functions." Recall that the "support" of a function $f : U \to \mathbb{R}$ is the closure of the set $\{x \in U : f(x) \neq 0\}$. A *bump function* is a smooth function with compact support.

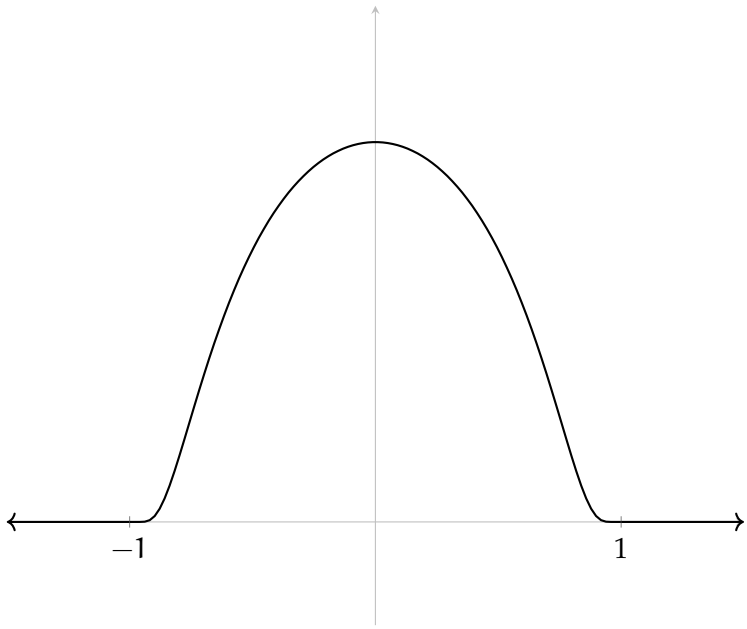


Figure 1.4.27: The graph of the function $\psi$ from example 1.4.26.

**Example 1.4.26.** Consider the function $\psi : \mathbb{R} \to \mathbb{R}$ given by the following.

$$\psi(x) = \begin{cases} e^{-1/(1-x^2)} & \text{if } x \in (-1, 1) \\ 0 & \text{otherwise.} \end{cases}$$

See figure 1.4.27 for its graph. It is clear from the definition of $\psi$ that the support of $\psi$ is the closed interval $[-1, 1]$. Moreover, observe that, if $f$ is the "infinitely flat" function from exercise 1.4.24 and $p : \mathbb{R} \to \mathbb{R}$ is the polynomial $p(x) = 1 - x^2$, then $\psi = f \circ p$. Thus $\psi$ is smooth.

**Exercise 1.4.28.** For any $a < b$ in $\mathbb{R}$, construct a bump function whose support is $[a, b]$.

**Exercise 1.4.29.** Let $F$ be a closed subset of $\mathbb{R}$. Construct a smooth function which is zero on $F$ and nonzero outside of $F$.

*Possible hint.* Recall that the complement of $F$ is a countable disjoint union of open intervals (cf. [PM91, theorem 6.17]).

### Bridge functions

Next up, we'll discuss an example of a "bridge" function. It will be constant on $(-\infty, -1]$, and also constant on $[1, \infty)$, but the values on these two intervals is different; and on $(-1, 1)$, the function "smoothly bridges the gap" between the two values.

In discussing this example, we will invoke the fundamental theorem of calculus, even though we have not proved it (or even defined integrals, for that matter). If you're unhappy with using things you haven't proved, you can find rigorous discussions of integration in many places (eg, [PM91, chapter 5], [Rud76, chapter 6], etc). Alternatively, you might also be interested in [GO03, chapter 3, example 12], which gives an example of a "bridging function" that involves no integrals.

**Example 1.4.30.** Let $\psi$ be the bump function from example 1.4.26, and then consider

$$\eta(x) = \int_{-1}^{x} \psi(t)\, dt.$$

Then

$$\eta'(x) = \frac{d}{dx} \int_{-1}^{x} \psi(t)\, dt = \psi(x)$$

by the fundamental theorem of calculus. Thus $\eta$ is smooth, since $\eta' = \psi$ is smooth. Note moreover that $\eta$ is increasing, since $\eta' = \psi \geqslant 0$. Finally, it is clear from the geometric interpretation of integrals as "area under the curve" that $\eta$ is constantly $0$ for all $x \leqslant -1$ and that it is constantly $\eta(1)$ for all $x \geqslant 1$.

**Exercise 1.4.31.** For $a < b$ in $\mathbb{R}$, let $I = [a, b]$ and let $U$ be an open subset containing $I$. Prove that there exists a bump function $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = 1$ for all $x \in K$ and $f(x) = 0$ for all $x \notin U$.

*Possible hint.* The characteristic function

$$\chi_I(x) = \begin{cases} 1 & \text{if } x \in I \\ 0 & \text{if } x \notin I \end{cases}$$

is close to having all of the properties we want, but it's not smooth; it's clearly not even continuous. "Bridge" the discontinuities.

# 2 Multivariable derivatives

In this chapter, we will study derivatives of functions $S \to \mathbb{R}^n$ where $S$ is a subset of $\mathbb{R}^m$. As it turns out, the most important case is when $n = 1$. In other words, the "multi" in the name of this chapter (as opposed to the "single" in the name of the previous chapter) is really referring to the fact that we might have $m > 1$ (rather than to the fact that we might have $n > 1$). Moreover, the single variable case is actually quite important for the multivariable case; we'll often use results from chapter 1 to prove their multivariable counterparts.

We'll begin by analyzing a particular example to build up some geometric intuition in section 2.1, and then proceed with the abstract discussion of multivariable derivatives.

## 2.1 Example

Recall that we started off our discussion of single variable derivatives starting with a very geometric idea of tangent lines to graphs. For large values of $m$ (and $n$), graphs become hard to visualize and it is not so clear what "tangent" should mean. But there is at least one multivariable situation where, by exerting some strain on the three-dimensional visualization sectors of our brain, we can geometrically formalize what "tangent" might mean. This is the $m = 2, n = 1$ situation. The graph of such a function is a surface in $\mathbb{R}^3$, and its "tangent" at a point should be a plane.

By analyzing a specific function $f : \mathbb{R}^2 \to \mathbb{R}$, we can get some valuable insights into multivariable derivatives; the analysis will presage many of the concepts we will discuss later in the chapter. Any function would do, but let's focus on the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) = x^2 + y^2$$

for our analysis.

### Describing the graph of $f$

First off, let's try to get a solid understanding of the graph of $f$. The graph $\Gamma$ is a subset of $\mathbb{R}^2 \times \mathbb{R} = \mathbb{R}^3$ defined by

$$\Gamma = \{(x, y, z) : z = f(x, y)\} = \{(x, y, z) : z = x^2 + y^2\}.$$

To visualize $\Gamma$, we will repeatedly slice $\Gamma$ along various planes.

If we consider the slice obtained by setting $x = 0$, we find a parabola $z = y^2$. Similarly, if we consider the slice obtained by setting $x = 1$, we obtain the parabola $z = 1 + y^2$. Fixing $x = 5$, we obtain the parabola $z = 25 + y^2$. In fact, we can see that no matter what value of $x$ we fix, the resulting slice is always a parabola, just translated up from the origin by differing amounts (that amount being $x^2$).

It's also worth considering the "level sets," which are the slices of $\Gamma$ obtained by fixing various values of $z$. When $z = 0$, there is just the single point $x = y = 0$, ie, the origin. Fixing $z = 1$, the slice is given by $1 = x^2 + y^2$, which is a circle of radius 1. Fixing $z = 17$, the slice is $17 = x^2 + y^2$, which is circle of radius $\sqrt{17}$. In fact, all of the level sets are circles.

Putting these slices together, we see that the graph $\Gamma$ is a "big parabolic bowl."

### Tangent plane at $a = (2, 1)$

Now that we understand what the graph of $f$ looks like, let's try to figure out what the "tangent plane" $T$ at a point $a = (2, 1)$ will look like. Of course, it's a plane inside $\mathbb{R}^3$ that passes through the point $(2, 1, f(2, 1)) = (2, 1, 5)$. To specify which plane it actually is, we start by describing some lines that lie on this plane.

Consider the $y = 1$ slice. We know that the graph of $\Gamma$ along this slice is the parabola $z = x^2 + 1$. So, slicing the tangent plane $T$ along $y = 1$ should yield the tangent line to this function of one variable, which we know how to compute from chapter 1. The slope of this tangent line is

$$\frac{d}{dx} f(x, 1)\bigg|_{x=2} = 4.$$

This quantity is called a *partial derivative of* $f$, and will be denoted $(\partial f/\partial x)(a)$. Thus the tangent line is the line parametrized by

$$h \mapsto (2, 1, 5) + h \cdot (1, 0, 4).$$

Similarly, we can consider the $x = 2$ slice. We know that the graph $\Gamma$ along this slice is the parabola $z = 4 + y^2$.The slope of the corresponding tangent line is

$$\frac{\partial f}{\partial y}(a) = \left.\frac{df(2, y)}{dy}\right|_{y=1} = 2.$$

Thus this tangent line is the line parametrized by

$$k \mapsto (2, 1, 5) + k \cdot (0, 1, 2).$$

These two lines must uniquely determine the entire plane $T$. One way of describing $T$ that immediately falls out of the parametrizations we've found of the tangent lines is the plane parametrized by

$$\begin{bmatrix} h \\ k \end{bmatrix} \mapsto \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix} + \begin{bmatrix} h \\ k \\ 4h + 2k \end{bmatrix}.$$

The important part of this expression is the bottom entry on the far right, the $4h + 2k$. To see why, consider the function $\mathbb{R}^2 \to \mathbb{R}$ given by $(h, k) \mapsto 4h + 2k$. This is what we will call the *(total) derivative* of $f$ and denote by $df_a$. Notice that $df_a$ is a linear map $\mathbb{R}^2 \to \mathbb{R}$. Moreover, its graph is a plane passing through the origin that is parallel to $T$. In other words, if we take the graph of $df_a$ and translate it over to the point $(2, 1, 5)$, we obtain exactly the tangent plane $T$. Speaking more loosely, the derivative $df_a : \mathbb{R}^2 \to \mathbb{R}$ records all of the "slopey information" about the tangent plane $T$.

Notice moreover that, if $e_1, e_2$ are the standard basis vectors of $\mathbb{R}^2$ (cf. section 0.4), then

$$\begin{aligned} df_a(e_1) &= 4 = \frac{\partial f}{\partial x}(a) \\ df_a(e_2) &= 2 = \frac{\partial f}{\partial y}(a) \end{aligned}$$

so the standard matrix representation $[df_a]$ (cf. definition 0.5.14) of the linear map $df_a$ is

$$[df_a] = \begin{bmatrix} 4 & 2 \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x}(a) & \frac{\partial f}{\partial y}(a) \end{bmatrix}. \tag{2.1.1}$$

### $df_a$ as an approximation

Consider the function $\epsilon : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$v \mapsto |f(a+v) - f(a) - df_a(v)|.$$

If $v = (h, k)$, we have

$$\epsilon(v) = \epsilon\left(\begin{bmatrix} h \\ k \end{bmatrix}\right) \mapsto |(2+h)^2 + (1+k)^2 - 5 - 4h - 2k| = |h^2 + k^2| = \left|\begin{bmatrix} h \\ k \end{bmatrix}\right|^2 = |v|^2.$$

Notice that $\epsilon$ gets small rapidly as $v \to 0$. More precisely, notice that

$$\lim_{v \to 0} \frac{\epsilon(v)}{|v|} = \lim_{v \to 0} |v| = 0,$$

so $\epsilon = o(|v|)$ as $v \to 0$. Since $\epsilon$ gets small rapidly, we can say that the function $v \mapsto f(a) + df_a(v)$ is a good approximation of the function $v \mapsto f(a+v)$ for small values of $v$. Said differently, letting $x = a + v$, the function $x \mapsto f(a) + df_a(x - a)$ is a good approximation of $f$ near $a$.

### 2.1.A Overview

In what follows, we will turn the above example on its head and *define* the derivative using approximations. In other words, the derivative $df_a$ will be defined to be a linear function which yields a good approximation of $f$ near a point $a$. Then we will prove in theorem 2.3.26 that the partial derivatives can be used to compute the standard matrix representation $df_a$, as we noted in equation (2.1.1) above.

This might seem a bit "backwards," given our analysis of the example above, but there is a good reason for doing this; it turns out that there are some bizarre functions where partial derivatives make sense, but tangent planes do not (cf. exercise 2.3.24 below).

As you may be able to tell, we'll be using linear algebra heavily in this chapter. We'll also need some facts about the operator norm; I encourage you to at least skim through section 0.6.A.

## 2.2 Definition of the derivative

Throughout, $S$ will denote a subset of $\mathbb{R}^m$.

**Definition 2.2.1** (Differentiability at a point)**.** A function $f : S \to \mathbb{R}^n$ is *differentiable at* an interior point $a \in S$ if there exists a linear map $\ell : \mathbb{R}^m \to \mathbb{R}^n$ such that

$$|f(a+h) - f(a) - \ell(h)| = o(|h|) \text{ as } h \to 0.$$

It turns out that there can exist at most one linear map that has the above property.

**Lemma 2.2.2.** *Suppose $f : S \to \mathbb{R}^m$ is differentiable at an interior point $a \in S$. If $\ell$ and $\ell'$ are both linear maps $\mathbb{R}^m \to \mathbb{R}^n$ such that*

$$|f(a+h) - f(a) - \ell(h)| = o(|h|) \textit{ and } |f(a+h) - f(a) - \ell'(h)| = o(|h|) \textit{ as } h \to 0,$$

*then $\ell = \ell'$.*

*Proof.* Let $\phi = \ell' - \ell$. Then $\phi$ is also a linear map $\mathbb{R}^m \to \mathbb{R}^n$. Moreover, observe that

$$\begin{aligned}
|\phi(h)| &= |\ell'(h) - \ell(h)| \\
&= |(f(a+h) - f(a) - \ell(h)) - (f(a+h) - f(a) - \ell'(h)| \\
&\leqslant |f(a+h) - f(a) - \ell(h)| + |f(a+h) - f(a) - \ell'(h)|.
\end{aligned}$$

Since both $|f(a+h) - f(a) - \ell(h)|$ and $|f(a+h) - f(a) - \ell'(h)|$ are $o(|h|)$, exercises 0.1.6 and 0.1.7 imply that $|\phi(h)| = o(|h|)$ also. Then lemma 0.6.5 implies that $\phi = 0$. □

**Exercise 2.2.3.** Look at the exercises from chapter 0 that are invoked in the proof above (namely, exercises 0.1.6, 0.1.7 and 0.6.3) and do any of them that you haven't already done.

Lemma 2.2.2 guarantees that the following definition makes sense.

**Definition 2.2.4** (Derivative at a point)**.** Suppose $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$. Then the *derivative of* $f$ *at* $a$, denoted $df_a$, is the unique linear map $\mathbb{R}^m \to \mathbb{R}^n$ satisfying

$$|f(a+v) - f(a) - df_a(h)| = o(|h|) \text{ as } h \to 0.$$

Sometimes $df_a$ is also called the *total derivative* or the *differential* of $f$ at $a$.

We've now defined the derivative, but it's fairly difficult to use this definition in practice. There are very few examples that can be computed directly from the definition. Here are some.

**Exercise 2.2.5.** Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Prove that $d\ell_a = \ell$ for all $a \in \mathbb{R}^m$.

**Exercise 2.2.6.** Let $\mu : \mathbb{R}^2 \to \mathbb{R}$ be the multiplication map $\mu(x, y) = xy$. Use the definition of the derivative to prove that

$$d\mu_{(a,b)}(h, k) = bh + ak$$

for all $(a, b) \in \mathbb{R}^2$ and $(h, k) \in \mathbb{R}^2$.

**Exercise 2.2.7.** Suppose $v, w \in \mathbb{R}^n$ and $f : \mathbb{R} \to \mathbb{R}^n$ is given by $f(t) = v + tw$. Calculate $df_a$ for any $a \in \mathbb{R}$. What is $df_a(1)$?

We will develop a bit more theory in order to compute derivatives more easily. Meanwhile, we can prove the following directly from the definition.

**Exercise 2.2.8.** If $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$, show that $f$ must be continuous at $a$.

## 2.3 Computing derivatives

### 2.3.A Sum and scalar multiples rule

**Exercise 2.3.1** (Sum rule)**.** Prove that, if $f, g : S \to \mathbb{R}^n$ are both differentiable at an interior point $a \in S$, then $f + g$ is also differentiable at $a$ and

$$d(f + g)_a = df_a + dg_a.$$

**Exercise 2.3.2** (Scalar multiples rule)**.** Prove that, if $c$ is a constant and $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$, then $cf$ is also differentiable at $a$ and

$$d(cf)_a = c \cdot df_a.$$

### 2.3.B Chain rule

The proof of the following is essentially the same as the second proof of the single variable chain rule given in section 1.2.C. Since this proof is basically a repeat, some details are omitted; you are asked to fill in the details in exercise 2.3.5 below.

**Theorem 2.3.3** (Chain rule). *Suppose that* $S$ *and* $T$ *are subsets of* $\mathbb{R}^m$ *and* $\mathbb{R}^n$*, respectively, that* $f : S \to \mathbb{R}^n$ *is differentiable at an interior point* $a$*, that* $f(S) \subseteq T$ *and* $f(a)$ *is an interior point of* $T$*, and that* $g : T \to \mathbb{R}^p$ *is differentiable at* $f(a)$*. Then the composite* $g \circ f : S \to \mathbb{R}^p$ *is also differentiable at* $a$*, and*

$$d(g \circ f)_a = dg_{f(a)} \circ df_a.$$

*Proof.* Define the "error in approximation" functions

$$\begin{aligned} r(h) &= f(a+h) - f(a) - df_a(h) \\ s(k) &= g(f(a)+k) - g(f(a)) - dg_{f(a)}(k) \end{aligned}$$

and then observe that

$$g(f(a+h)) - g(f(a)) - dg_{f(a)}(df_a(h)) = dg_{f(a)}(r(h)) + s(df_a(h) + r(h)). \tag{2.3.4}$$

This means that

$$|g(f(a+h)) - g(f(a)) - dg_{f(a)}(df_a(h))| \leqslant |dg_{f(a)}(r(h))| + |s(df_a(h) + r(h))|.$$

Since $dg_{f(a)}$ is linear and $|r(h)| = o(|h|)$, exercise 2.3.5 guarantees that $|dg_{f(a)}(r(h))| = o(|h|)$ also. Thus, by exercise 0.1.6, it is sufficient to prove that $|s(df_a(h) + r(h))| = o(|h|)$. To do this, define $\eta(k) = |s(k)|/|k|$ and then notice that

$$\begin{aligned} \frac{|s(df_a(h) + r(h))|}{|h|} &= \eta(df_a(h) + r(h)) \cdot \frac{|df_a(h) + r(h)|}{|h|} \\ &\leqslant \eta(df_a(h) + r(h)) \left( \|df_a\| + \frac{|r(h)|}{|h|} \right). \end{aligned}$$

We now take the limit as $h \to 0$. Using the facts that $\eta \circ (df_a + r)$ is continuous at 0 and that $|r(h)| = o(|h|)$, we obtain the result. □

**Exercise 2.3.5.** (a) Prove equation (2.3.4).

(b) Suppose $r : \mathbb{R}^m \to \mathbb{R}^n$ is a function such that $|r(h)| = o(|h|)$ as $h \to 0$. If $\ell : \mathbb{R}^n \to \mathbb{R}^p$ is linear, prove that $|\ell(r(h))| = o(|h|)$ also.

*Possible hint.* Even though the operator norm does not appear anywhere in the statement of this exercise, you might find the concept useful. In particular, you might consider doing exercise 0.6.2 and then using it to prove (b).

(c) Prove that $\eta \circ (df_a + r)$ is continuous at 0.

### 2.3.C Differentiability by components

Recall that we asserted at the beginning of this chapter that the "multi" in the name of this chapter refers to the number of inputs (rather than the number of outputs). This section is where we show that understanding the $n = 1$ case is "enough" to understand general $n$.

**Definition 2.3.6** (Component functions)**.** For any function $f : S \to \mathbb{R}^n$, we define the $j$th *component function* $f_j : S \to \mathbb{R}$ to be the composite $\pi_j \circ f$, where $\pi_j$ is the $j$th projection map (cf. section 0.4).

**Proposition 2.3.7.** *A function $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$ if and only if the component function $f_j : S \to \mathbb{R}$ is differentiable at $a$ for all $j = 1, \dots n$. Moreover,*

$$df_a(h) = \begin{bmatrix} df_{1,a}(h) \\ \vdots \\ df_{n,a}(h) \end{bmatrix}. \tag{2.3.8}$$

*Proof.* If $f$ is differentiable at $a$, then the composite $f_j = \pi_j \circ f$ is also differentiable at $a$ by the chain rule 2.3.3. Moreover, since $\pi_j$ is linear, we know that $d\pi_{j,f(a)} = \pi_j$ by exercise 2.2.5. Thus, by the chain rule,

$$df_{j,a}(h) = (d\pi_{j,f(a)} \circ df_a)(h) = \pi_j(df_a(h))$$

for all $j$, which is precisely equation (2.3.8). For the converse, we turn equation (2.3.8) on its head. Suppose $f_j$ is differentiable at $a$ for all $j$, and let $\ell : \mathbb{R}^m \to \mathbb{R}^n$ be the linear map

$$\ell(h) = \begin{bmatrix} df_{1,a}(h) \\ \vdots \\ df_{n,a}(h) \end{bmatrix}.$$

We now want to show that $|f(a+h) - f(a) - \ell(h)| = o(|h|)$ as $h \to 0$. Observe that the component functions of $h \mapsto f(a+h) - f(a) - \ell(h)$ are given by

$$v \mapsto \pi_j\left(f(a+h) - f(a) - \ell(h)\right) = f_j(a+h) - f_j(a) - df_{j,a}(h)$$

since $\pi_j$ is linear, and we know that $|f_j(a+h) - f_j(a) - df_{j,a}(h)| = o(|h|)$. By exercise 2.3.9 below applied with the function $r(h) = f(a+h) - f(a) - \ell(h)$, we conclude that $|f(a+h) - f(a) - \ell(h)| = o(|h|)$. □

**Exercise 2.3.9.** Show that if $S$ is a neighborhood of $0$ in $\mathbb{R}^m$ and $r : S \to \mathbb{R}^n$ is a function such that $|r_j(h)| = o(|h|)$ as $h \to 0$ for all $j = 1, \ldots, n$, then $|r(h)| = o(|h|)$ also.

*Possible hint.* You may find it useful to do exercise 0.3.10 first.

### 2.3.D Product and quotient rules

We showed in the previous section that it's enough to understand the $n = 1$ case. So in this section and the next, we'll focus our attention on the $n = 1$ case and try to understand this better. Note that, when $n = 1$, we have more operations available to us; we can multiply and divide such functions.

So let's start off by establishing product and quotient rules. As it turns out, we can actually derive these using the multivariable chain rule 2.3.3!

**Proposition 2.3.10** (Product rule)**.** *Suppose $f, g : S \to \mathbb{R}$ are both differentiable at an interior point $a \in S$. Then $fg$ is also differentiable at $a$ and*

$$d(fg)_a = g(a)df_a + f(a)dg_a.$$

*Proof.* Let $f \times g$ denote the function $S \to \mathbb{R}^2$ defined by

$$(f \times g)(x) = (f(x), g(x)).$$

Observe that the first and second component functions of $f \times g$ are $f$ and $g$, respectively; thus it follows from proposition 2.3.7 that

$$d(f \times g)_a(h) = (df_a(h), dg_a(h)).$$

Let $\mu : \mathbb{R}^2 \to \mathbb{R}$ denote the multiplication map $\mu(x, y) = xy$ that we considered in exercise 2.2.6. Then $fg = \mu \circ (f \times g)$, so by the chain rule 2.3.3, we have that

$$\begin{aligned} d(fg)_a(h) &= d\mu_{(f(a),g(a))}(d(f \times g)_a(h)) \\ &= d\mu_{f(a),g(a)}(df_a(h), dg_a(h)) \\ &= g(a)df_a(h) + f(a)dg_a(h), \end{aligned}$$

where we have used the calculation from exercise 2.2.6 for the final step. □

**Exercise 2.3.11.** Suppose $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$ and that $f(a) \neq 0$. Show that $1/f$ is also differentiable at $a$, and

$$d(1/f)_a = -\frac{df_a}{f(a)^2}.$$

*Possible hint.* Consider the function $\iota : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ given by $\iota(x) = 1/x$. This is a single variable function, so we know $d\iota_x$ from chapter 1. Now note that $1/f = \iota \circ f$.

**Exercise 2.3.12** (Quotient rule)**.** Suppose $f, g : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$ and that $g(a) \neq 0$. Show that $f/g$ is also differentiable at $a$, and that

$$d(f/g)_a = \frac{g(a)df_a - f(a)dg_a}{g(a)^2}.$$

*Possible hint.* Combine exercise 2.3.11 and proposition 2.3.10.

### 2.3.E Partial derivatives

**Definition 2.3.13** (Partial derivatives)**.** Suppose $f : S \to \mathbb{R}$ is a function and $a \in S$ is an interior point. The *$i$th partial derivative of $f$ at $a$*, denoted $\partial_i f(a)$, is defined to be the limit

$$\lim_{s \to 0} \frac{f(a + se_i) - f(a)}{s},$$

where $e_i$ denotes the $i$th standard basis vector (cf. section 0.4). If we're using $x$ to denote the input to $f$ and $x_i$ to denote the $i$th component of the input, $\partial_i f(a)$ is sometimes also denoted by one of the following.

$$f_{x_i}(a) \qquad \left.\frac{\partial f}{\partial x_i}\right|_{x=a} \qquad \left.\frac{\partial}{\partial x_i} f(x)\right|_{x=a} \qquad \frac{\partial f}{\partial x_i}(a)$$

You may remember from multivariable calculus that partial derivatives behave a great deal like single variable derivatives. The reason for this is the following important remark, which reinterprets the partial derivative as a single variable derivative.

**Remark 2.3.14.** Consider the function $\xi_i : \mathbb{R} \to \mathbb{R}^m$ given by $\xi_i(s) = a + se_i$. Then 0 is an interior point of $T = \xi_i^{-1}(S) \subseteq \mathbb{R}$, and $f \circ \xi_i$ is a single variable function $T \to \mathbb{R}$. Moreover, we have

$$\partial_i f(a) = (f \circ \xi_i)'(0),$$

just by writing out the definition of both sides.

**Example 2.3.15.** Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ given by $f(x, y) = \sin(xy^2)$. Fix $(a, b) \in \mathbb{R}^2$ and define $\xi : \mathbb{R} \to \mathbb{R}^2$ by $\xi(s) = (a + s, b)$, as in remark 2.3.14. Then

$$(f \circ \xi)(s) = f(a + s, b) = \sin((a + s)b^2).$$

Then

$$\begin{aligned}
\frac{\partial f}{\partial x}(a, b) &= (f \circ \xi)'(0) \\
&= \frac{d}{ds}\left(\sin((a + s)b^2)\right)\Big|_{s=0} \\
&= b^2 \cos((a + s)b^2)\big|_{s=0} \\
&= b^2 \cos(ab^2).
\end{aligned}$$

**Exercise 2.3.16.** With notation as in example 2.3.15, compute $(\partial f/\partial y)(a, b)$.

We can also use remark 2.3.14 to prove some partial derivative analogs of results we proved in chapter 1.

**Exercise 2.3.17** (Product rule for partial derivatives)**.** Suppose $f, g : S \to \mathbb{R}$ are functions and $a \in S$ is an interior point such that $\partial_i f(a)$ and $\partial_i g(a)$ both exist. Prove that $\partial_i(fg)(a)$ also exists and that

$$\partial_i(fg)(a) = g(a)\partial_i(f)(a) + f(a)\partial_i g(a).$$

*Possible hint.* Use remark 2.3.14 and the single variable product rule 1.2.4.

**Exercise 2.3.18** (Quotient rule for partial derivatives)**.** Suppose $f, g : S \to \mathbb{R}$ are functions and $a \in S$ is an interior point such that $g(a) \neq 0$ and $\partial_i f(a)$ and $\partial_i g(a)$ both exist. Show

that $\partial_i(f/g)(a)$ also exists and

$$\partial_i(f/g)(a) = \frac{g(a)\partial_i f(a) - f(a)\partial_i g(a)}{g(a)^2}.$$

*Possible hint.* Use remark 2.3.14 and the single variable quotient rule 1.2.7.

**Exercise 2.3.19** (Interior extremum theorem for partial derivatives)**.** Suppose $f : S \to \mathbb{R}$ is a function and an interior point $a \in S$ is a local extremum of $f$. Show that, if $\partial_i f(a)$ exists, then $\partial_i f(a) = 0$.

*Possible hint.* Use remark 2.3.14 and the single variable interior extremum theorem 1.2.20.

As we noted in chapter 1, the single variable interior extremum theorem 1.2.20 is false (cf. exercise 1.2.21). The same is true for the interior extremum theorem for partial derivatives 2.3.19, except that there are now even more ways for the converse to fail. Here are two examples to illustrate this.

**Example 2.3.20.** Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f(x, y) = x^2 - y^2.$$

Let $\xi(s) = 0 + se_1 = (s, 0)$ and $\eta(s) = 0 + se_2 = (0, s)$. Then $f \circ \xi$ has a minimum at 0, so $(f \circ \xi)'(0) = (\partial f)/(\partial x)(0) = 0$. Similarly, $f \circ \eta$ has a maximum at 0, so $(f \circ \eta)'(0) = (\partial f)/(\partial y)(0) = 0$ also. But clearly these two facts together mean that 0 cannot be a local extremum of $f$.

**Exercise 2.3.21.** Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f(x, y) = (y - x^2)(y - 3x^2).$$

(a) Show that the origin is *not* a local extremum of $f$.

*Possible hint.* Consider the values of $f$ on points of the form $(0, b)$ and $(a, 2a^2)$ near the origin.

(b) Show that $f$ restricted to the $x$-axis does have an absolute minimum at the origin.

(c) Show that $f$ restricted to the $y$-axis also has an absolute minimum at the origin.

(d) Show that $f$ restricted to an arbitrary line through the origin has an absolute minimum at the origin.

Now for the main result of this section: partial derivatives is that they help us compute derivatives. Note that we will generalize this result in theorem 2.3.26 below.

**Proposition 2.3.22.** *If $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$, then the $i$th partial derivative $\partial_i f(a)$ exists for all $i = 1, \dots m$, and the standard matrix representation $[df_a]$ of the linear map $df_a : \mathbb{R}^m \to \mathbb{R}$ is a $1 \times m$ matrix whose $i$th entry is $\partial_i f(a)$.*

$$[df_a] = \begin{bmatrix} \partial_1 f(a) & \partial_2 f(a) & \cdots & \partial_m f(a) \end{bmatrix}$$

*Proof.* Let $r$ be the "error in approximation" function

$$r(h) = f(a+h) - f(a) - df_a(h),$$

so that $|r(h)| = o(|h|)$ as $h \to 0$. Letting $h = se_i$, we see that

$$r(se_i) = f(a+se_i) - f(a) - df_a(se_i) = f(a+se_i) - f(a) - sdf_a(e_i)$$

by linearity of $df_a$, so

$$\frac{r(se_i)}{s} = \frac{f(a+se_i) - f(a) - sdf_a(e_i)}{s} = \frac{f(a+se_i) - f(a)}{s} - df_a(e_i).$$

Then

$$\begin{aligned} \lim_{s\to 0} \left| \frac{f(a+se_i) - f(a)}{s} - df_a(e_i) \right| &= \lim_{s\to 0} \left| \frac{r(se_i)}{s} \right| \\ &= \lim_{s\to 0} \frac{|r(se_i)|}{|s|} \\ &= \lim_{s\to 0} \frac{|r(se_i)|}{|se_i|} \\ &= 0 \end{aligned}$$

where we used the fact that $|r(h)| = o(|h|)$ as $h \to 0$ for the final step. This tells us that

$$\partial_i f(a) = \lim_{s\to 0} \frac{f(a+se_i) - f(a)}{s}$$

exists and equals $df_a(e_i)$. □

**Remark 2.3.23.** If $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$, then the vector

$$[df_a]^T = \begin{bmatrix} \partial_1 f(a) \\ \partial_2 f(a) \\ \vdots \\ \partial_m f(a) \end{bmatrix} \in \mathbb{R}^m$$

is often called the *gradient* of $f$ at $a$, and is denoted $\nabla f(a)$.

Now for the important caveat: unfortunately, the converse to proposition 2.3.22 is *not* true.

**Exercise 2.3.24.** Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined as follows.

$$f(x, y) = \begin{cases} \dfrac{xy}{x^2 + y^2} & \text{if } (x, y) \neq 0 \\ 0 & \text{if } (x, y) = 0 \end{cases}$$

Show that the partial derivatives of $f$ exist at the origin, but that $f$ is not even continuous at the origin (so it cannot be differentiable, by exercise 2.2.8).

*Possible hint.* Approach the origin along the line $x = y$.

Thankfully, this phenomenon is fairly pathological; theorem 2.4.5 will provide a converse to proposition 2.3.22 under fairly mild additional hypotheses.

### 2.3.F Jacobian matrix

**Definition 2.3.25** (Jacobian matrix)**.** Suppose $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$. The *Jacobian matrix* of $f$ at $a$, denoted $f'(a)$, is the $n \times m$ matrix whose $(j, i)$-entry is $\partial_i f_j(a)$. In other words,

$$f'(a) = \begin{bmatrix} \partial_1 f_1(a) & \partial_2 f_1(a) & \cdots & \partial_m f_1(a) \\ \partial_1 f_2(a) & \partial_2 f_2(a) & \cdots & \partial_m f_2(a) \\ \vdots & \vdots & \ddots & \vdots \\ \partial_1 f_n(a) & \partial_2 f_n(a) & \cdots & \partial_m f_n(a) \end{bmatrix}$$

Note that propositions 2.3.7 and 2.3.22 tell us that all of these partial derivatives exist.

**Theorem 2.3.26.** *If* $f : S \to \mathbb{R}^m$ *is differentiable at an interior point* $a \in S$*, then* $[df_a] = f'(a)$.

**Exercise 2.3.27.** Prove theorem 2.3.26.

*Possible hint.* Use propositions 2.3.7 and 2.3.22.

Using theorem 2.3.26, many of the rules for differentiation we've seen acquire forms we may be more familiar with from single variable calculus. For example, since composition of linear maps corresponds to multiplication of the corresponding matrix representations, the chain rule 2.3.3 states that

$$(g \circ f)'(a) = [d(g \circ f)_a] = [dg_{f(a)} \circ df_a] = [dg_{f(a)}][df_a] = g'(f(a))f'(a).$$

Though the equality $(g \circ f)'(a) = g'(f(a))f'(a)$ looks superficially identical to the equality from the single variable chain rule 1.2.9, it's worth noticing that the terms in this multivariable version are *matrices*. In particular, this means that that order of the multiplication $g'(f(a))f'(a)$ cannot be interchanged, whereas this was a non-issue in the single variable case.

Similarly, the matrix version of the multivariable product rule 2.3.10 states that

$$(fg)'(a) = g(a)f'(a) + g(a)f'(a),$$

which again looks superficially identical to the single variable product rule 1.2.4.

### 2.3.G Rank of a differentiable map

**Definition 2.3.28** (Rank of a differentiable map)**.** Suppose $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$. The rank of the linear map $df_a : \mathbb{R}^m \to \mathbb{R}^n$ is also called the *rank of* $f$ *at* $a$, denoted $\operatorname{rank}_a(f)$. Note that $\operatorname{rank}_a(f) \leqslant \min\{m, n\}$.

- If $\operatorname{rank}_a(f) = m$, then we say that $f$ is *immersive at* $a$. Note that this is equivalent to asking that $df_a$ be injective.
- If $\operatorname{rank}_a(f) = n$, then we say that $f$ is *submersive at* $a$. Note that this is equivalent to asking that $df_a$ be surjective.
- If $\operatorname{rank}_a(f) = m = n$, then we say that $f$ is *étale at* $a$. Note that this is equivalent to asking that $df_a$ be an isomorphism.

- If $\operatorname{rank}_a(f) = \min\{m, n\}$, then we say that $a$ is a *regular point* of $f$.
- If $\operatorname{rank}_a(f) < \min\{m, n\}$, then we say that $a$ is a *critical point* of $f$.

There are lots of relationships between these notions, depending on how $m$ compares to $n$. You should convince yourself of the following facts.

- If $m > n$...
  - $f$ cannot be immersive at any point.
  - $a$ is a regular point of $f$ if and only if $f$ is submersive at $a$.
- If $m < n$...
  - $f$ cannot be submersive at any point.
  - $a$ is a regular point of $f$ if and only if $f$ is immersive at $a$.
- If $m = n$...
  - $a$ is a regular point of $f$ if and only if $f$ is immersive at $a$, if and only if $f$ is submersive at $a$, if and only if $f$ is étale at $a$.

*Caution.* A map $f$ being immersive or submersive at an interior point $a \in S$ is always defined as it is above, and critical points are always defined to be the opposite of regular points. But some mathematicians use a different definition of regularity: they define a point to be regular if $\operatorname{rank}_a(f) = n$, so that, by definition, it's *always* true (for all $m, n$) that $f$ is a submersion at $a$ if and only if $a$ is a regular point of $f$. If $m \geqslant n$, there's no difference between these alternative definitions and our definitions. But, if $m < n$, there is a difference: these other mathematicians would say that all points are critical in this case, but that isn't true for us. So, if you see the words "regular point" and "critical point" in another text and it's possible that $m < n$, it's worth looking back to make sure you know what the author means!

**Example 2.3.29.** Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be the map

$$f(r, \theta) = (r\cos\theta, r\sin\theta).$$

See figure 2.3.30. We compute partial derivatives.

$$\frac{\partial f_1}{\partial r} = \cos\theta \quad \frac{\partial f_2}{\partial r} = \sin\theta \quad \frac{\partial f_1}{\partial \theta} = -r\sin\theta \quad \frac{\partial f_2}{\partial \theta} = r\cos\theta.$$

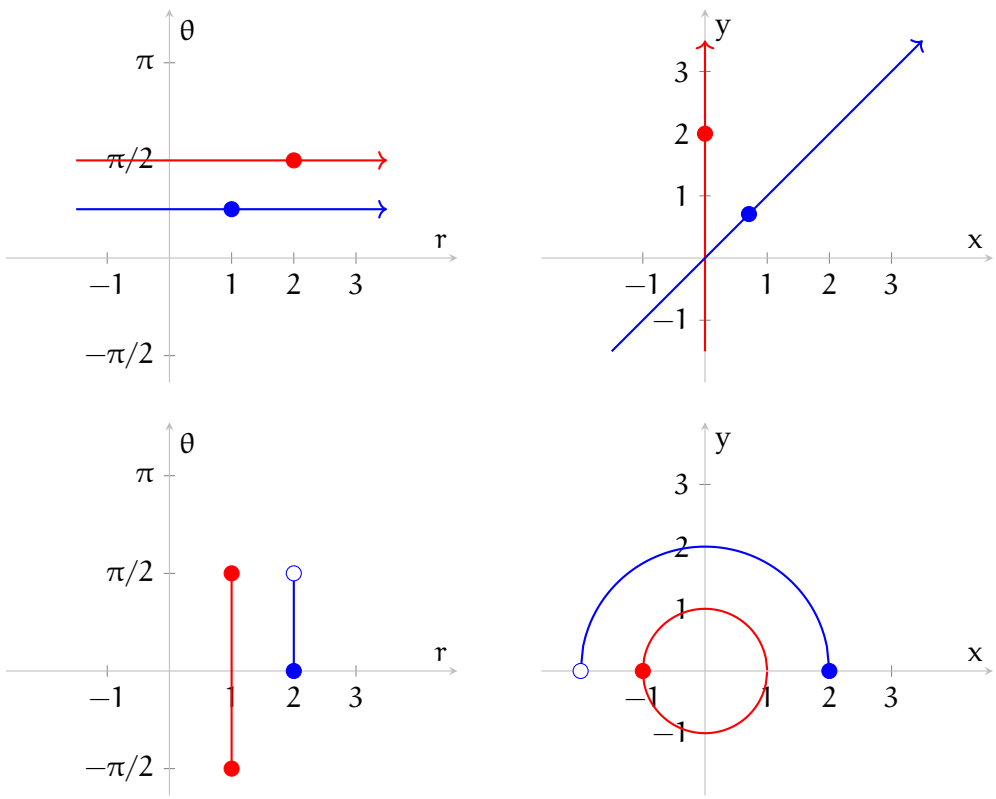


Figure 2.3.30: The function $f(r, \theta) = (r\cos\theta, r\sin\theta)$ of example 2.3.29 transforms the pictures on the right to the pictures on the left.

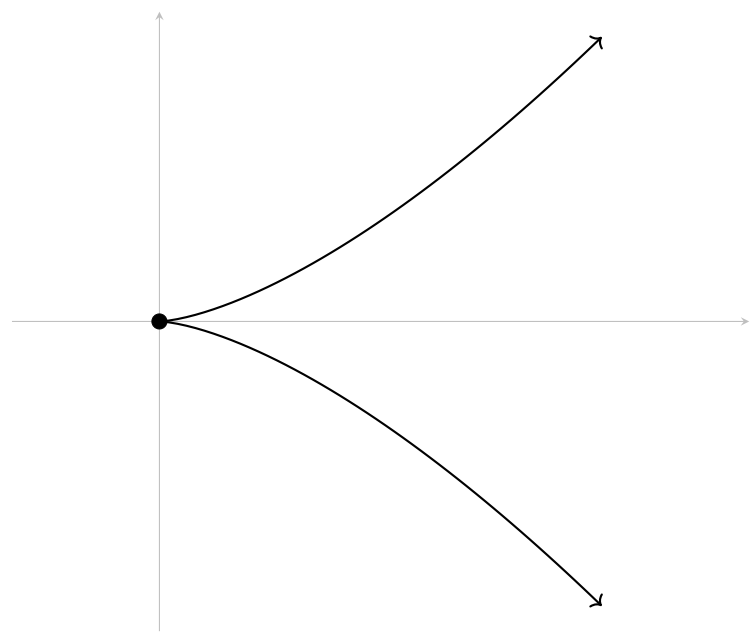

Figure 2.3.32: The image of the function $f(x) = (x^2, x^3)$. The origin is $f(0)$.

Thus

$$f'(r, \theta) = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}.$$

Since this is a square matrix, $df_{(r,\theta)}$ has non-maximal rank if and only if its determinant vanishes. But

$$\det f'(r, \theta) = r\cos^2\theta + r\sin^2\theta = r,$$

so the critical locus (ie, the set of all critical points) of $f$ is the vertical line $r = 0$, and $f$ is étale away from this line.

**Exercise 2.3.31.** Consider the function $f : \mathbb{R} \to \mathbb{R}^2$ defined by

$$f(x) = (x^2, x^3).$$

See figure 2.3.32 for a picture of the image of $f$. Calculate $f'(x)$ for all $x \in \mathbb{R}$. Then show that 0 is a critical point of $f$, and that $f$ is immersive away from 0. Loosely speaking, the fact that 0 is a critical point of $f$ corresponds geometrically to the fact that $f(0)$ is a "pointy" cusp.

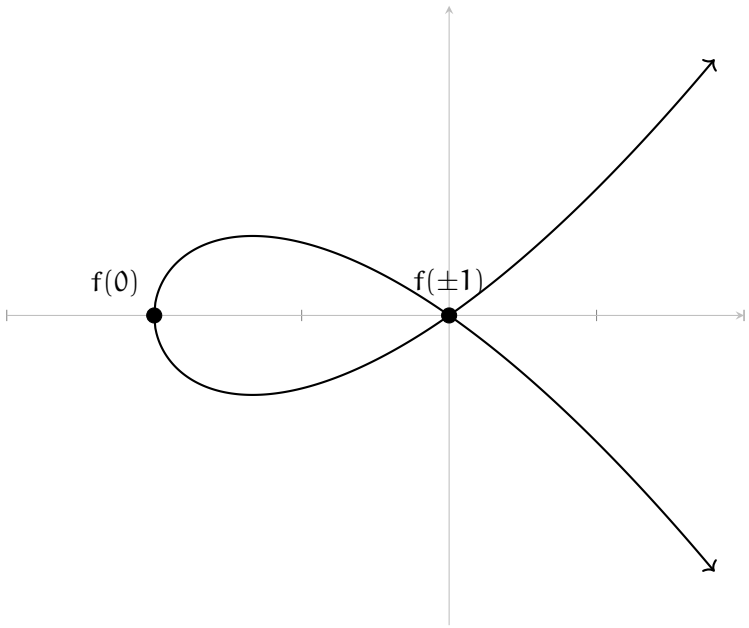


Figure 2.3.34: The image of the function $f(x) = (x^2 - 1, x^3 - x)$. We have $f(0) = (-1, 0)$ and $f(\pm 1) = (0, 0)$.

**Exercise 2.3.33.** Consider the function $f : \mathbb{R} \to \mathbb{R}^2$ defined by

$$f(x) = (x^2 - 1, x^3 - x).$$

See figure 2.3.34 for a picture of the image of $f$. Calculate $f'(x)$ for all $x \in \mathbb{R}$, and show that $f$ is immersive everywhere.

**Exercise 2.3.35.** Show that an interior point $a \in S$ is a critical point of $f : S \to \mathbb{R}$ if and only if $\partial_i f(a) = 0$ for all $i$.

## 2.4 Levels of differentiability

Throughout this section, $U$ denotes an open subset of $\mathbb{R}^m$.

### 2.4.A Differentiable functions

**Definition 2.4.1** (Differentiability)**.** A function $f : U \to \mathbb{R}^n$ is *differentiable* if it is differentiable at every $a \in U$. The *derivative* of $f$, denoted $df$, is the function $U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ given by $a \mapsto df_a$. Sometimes the function $df$ is also called the *total derivative* or the *differential* of $f$.

**Definition 2.4.2.** If $f : U \to \mathbb{R}^n$ is differentiable, the $i$*th partial derivative* of $f$, denoted $\partial_i f$, is the function $a \mapsto \partial_i f(a)$. If we're using $x$ to denote the input to $f$ and $x_i$ to denote the $i$th component of the input, $\partial_i f(a)$ is sometimes also denoted by one of the following.

$$f_{x_i} \qquad \frac{\partial f}{\partial x_i}$$

Here is a multivariable generalization of proposition 1.3.4. In fact, we will use te single variable version to prove this multivariable version.

**Proposition 2.4.3.** *Suppose* $B$ *is an open ball and* $f : B \to \mathbb{R}^n$ *is a differentiable function. Then* $df = 0$ *if and only if* $f$ *is constant.*

*Proof.* Taking $w = 0$ in exercise 2.2.7 shows that, if $f$ is constant, then $df = 0$. We can assume, without loss of generality, that $n = 1$ (cf. exercise 2.4.4). We want to show that, for any $a, b \in U$, we have $f(a) = f(b)$. Consider the function $\gamma : [0, 1] \to B$ given by

$$\gamma(t) = tb + (1 - t)a.$$

In other words, $\gamma$ is the "straight line path" that starts at $a$ and ends at $b$. Notice that $\gamma(t) \in B$ for all $t$ because $a$ and $b$ are both inside $B$ and $B$ is an open ball. Also, $\gamma$ is differentiable (cf. exercise 2.4.4). Thus $f \circ \gamma$ is a differentiable function $[0, 1] \to \mathbb{R}$ and

$$(f \circ \gamma)'(t) = f'(\gamma(t))\gamma'(t) = 0$$

where we used the (matrix version of the) chain rule 2.3.3 at the second step, and the fact that $df = 0$ for the final step. Thus it follows from proposition 1.3.4 that $f \circ \gamma$ is constant, which means that

$$f(a) = f(\gamma(0)) = f(\gamma(1)) = f(b). \qquad \square$$

**Exercise 2.4.4.** (a) Explain why proving that $df = 0$ implies that $f$ is constant for $n = 1$ implies the same fact for general $n$.

(b) Show that the straight line path $\gamma$ defined in the proof above is differentiable.

### 2.4.B Continuous differentiability

#### Continuity of partials

The following theorem is the promised sort-of-converse to proposition 2.3.22. Recall from exercise 2.3.24 above that $\partial_i f$ existing for all $i$ is *not* sufficient to guarantee differentiability. But, insisting that $\partial_i f$ exist *and be continuous* turns out to be enough to guarantee differentiability of $f$. The proof is harder than you might expect.

**Theorem 2.4.5.** *Suppose $f : U \to \mathbb{R}$ is a function such that the partial derivatives $\partial_i f : U \to \mathbb{R}$ exist for all $i = 1, \dots m$. If $a \in U$ is a point such that $\partial_i f$ is continuous at $a$ for all $i$, then $f$ is differentiable at $a$.*

*Proof* $\star$. If $f$ is to be differentiable at $a$, we know from theorem 2.3.26 what $df_a$ needs to be in terms of the partial derivatives; so let's turn this on its head by writing down the linear map defined by the partial derivatives, and then proving that that linear map is in fact $df_a$.

More precisely, we mean the following. Let $\ell$ be the linear map $\mathbb{R}^m \to \mathbb{R}$ such that

$$[\ell] = \begin{bmatrix} \partial_1 f(a) & \cdots & \partial_m f(a) \end{bmatrix}.$$

In other words, $\ell : \mathbb{R}^m \to \mathbb{R}$ is the linear map given by

$$\ell(h_1, \dots, h_m) = h_1 \partial_1 f(a) + \cdots + h_m \partial_m f(a).$$

We will show that

$$|f(a+h) - f(a) - \ell(h)| = o(|h|) \text{ as } h \to 0,$$

which implies that $f$ is differentiable at $a$ and that $df_a = \ell$. In other words, we will show that, for any $\epsilon > 0$, there exists a $\delta > 0$ such that, for all $h \in \mathbb{R}^m$ with $|h| < \delta$, we have

$$|f(a+h) - f(a) - \ell(h)| \leqslant |h|\epsilon. \tag{2.4.6}$$

Let $\delta_0 > 0$ be small enough that the open ball $B(a, \delta_0)$ is contained in $U$. We will soon make it smaller to get inequality (2.4.6) to hold, but for the moment it is sufficient that $B(a, \delta_0) \subseteq U$. Suppose $h = (h_1, \dots, h_m) \in \mathbb{R}^m$ with $|h| < \delta_0$. Then consider the following

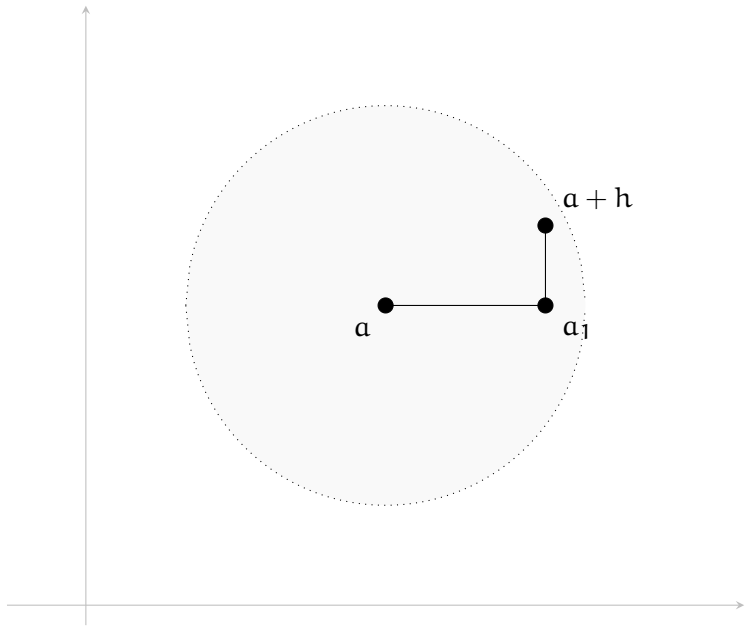


Figure 2.4.7: This is a picture of an open ball of some radius $\delta_0$ around a point $a \in \mathbb{R}^2$. The vector $h \in \mathbb{R}^2$ has $|h| < \delta$, so that $a + h$ is a point inside this ball. Defining $a_1$ as in the proof of theorem 2.4.5, the vector $a_1$ is also inside this ball. Then, defining $\xi_1$ and $\xi_2$ as in the proof, the point $a + \xi_1 e_1$ is somewhere along the horizontal line joining $a$ and $a + h_1$, and the point $a + h_1 + \xi_2 e_2$ is somewhere on the vertical line joining $a + h_1$ and $a + h$.

vectors in $\mathbb{R}^m$.

$$\begin{aligned}
a_0 &= a \\
a_1 &= a + (h_1, 0, 0 \ldots, 0, 0) \\
a_2 &= a + (h_1, h_2, 0, \ldots, 0, 0) \\
&\vdots \\
a_{m-1} &= a + (h_1, h_2, h_3, \ldots, h_{m-1}, 0) \\
a_m &= (h_1, h_2, h_3, \ldots, h_{m-1}, h_m) = a + h
\end{aligned}$$

The vectors $a_0, \ldots, a_m$ define a sort of "zigzag path" from $a$ to $a + h$. See figure 2.4.7. Clearly $|h_i| \leqslant |h|$ for all $i$.

Notice that

$$f(a + h) - f(h) = \sum_{i=1}^{m} \left( f(a_i) - f(a_{i-1}) \right).$$

Let us fix $i$ temporarily. Notice that $a_i = a_{i-1} + h_i e_i$, so

$$f(a_i) - f(a_{i-1}) = f(a_{i-1} + h_i e_i) - f(a_{i-1}).$$

Suppose $h_i \neq 0$. We can then apply the mean value theorem 1.3.2 to the single variable function

$$t \mapsto f(a_{i-1} + te_i)$$

on the closed interval between $0$ and $h_i$. This function is well-defined and differentiable on this interval: its derivative at a value $t$ is $\partial_i f(a_{i-1} + te_i)$, by definition of partial derivatives. The mean value theorem 1.3.2 says that there exists $\xi_i$ between $0$ and $h_i$ such that

$$\partial_i f(a_{i-1} + \xi_i e_i) = \frac{f(a_{i-1} + h_i e_i) - f(a_{i-1})}{h_i} = \frac{f(a_i) - f(a_{i-1})}{h_i},$$

or, in other words,

$$f(a_i) - f(a_{i-1}) = \xi_i \partial_i f(a_{i-1} + \xi_i e_i).$$

If $h_i = 0$, we set $\xi_i = 0$ as well. See figure 2.4.7 again. Unfixing $i$, we find that

$$f(a + h) - f(a) = \sum_{i=1}^{m} (f(a_i) - f(a_{i-1})) = \sum_{i=1}^{m} \xi_i \partial_i f(a_{i-1} + \xi_i e_i).$$

Thus

$$|f(a + h) - f(a) - \ell(h)| \leqslant \sum_{i=1}^{m} |\xi_i| \cdot |\partial_i(a_{i-1} + \xi_i e_i) - \partial_i f(a)|.$$

Since $\partial_i f$ is continuous at $a$, there exists $\delta_i > 0$ such that $|\partial_i(a + k) - \partial_i(a)| < \epsilon/m$ for all $|k| < \delta_i$. Let $\delta = \min\{\delta_0, \delta_1, \ldots, \delta_m\}$. If $|h| < \delta$, then

$$|\partial_i(a_{i-1} + \xi_i e_i) - \partial_i f(a)| < \epsilon/m,$$

which means that

$$|f(a + h) - f(a) - \ell(h)| \leqslant \sum_{i=1}^{m} \frac{|\xi_i|\epsilon}{m} \leqslant |h|_{\max}\epsilon \leqslant |h|\epsilon,$$

where we use the fact that $|\xi_i| \leqslant |h_i|$ for all $i$ for the second step, and exercise 0.3.10 for the final step. This proves that $f$ is differentiable at $a$. $\square$

### Continuous differentiability

**Definition 2.4.8** (Continuously differentiable functions)**.** We say that $f$ is *continuously differentiable* if $\partial_i f_j : U \to \mathbb{R}$ is continuous for all $i = 1, \ldots, m$ and $j = 1, \ldots, n$. Theorem 2.4.5 guarantees that such a function is, in particular, differentiable.

Before proceeding with a discussion of properties of continuously differentiable functions, it is useful to have a non-example in mind. Here is a multivariable version of exercise 1.2.25.

**Exercise 2.4.9.** Let $f : \mathbb{R}^2 \to \mathbb{R}$ denote the following function.

$$f(x, y) = \begin{cases} x^2 \sin(1/x) + y^2 \sin(1/y) & \text{if } x, y \neq 0 \\ x^2 \sin(1/x) & \text{if } x \neq 0 \text{ and } y = 0 \\ y^2 \sin(1/y) & \text{if } x = 0 \text{ and } y \neq 0 \\ 0 & \text{if } x = y = 0 \end{cases}$$

Show that $f$ is differentiable, but not continuously differentiable.

As we saw in the single variable case, continuously differentiable functions have lots of nice properties. For example, recall the calculation we did in example 2.3.29 showing that the critical locus was a line inside $\mathbb{R}^2$. Here is a generalization of this observation. This also generalizes exercise 1.4.3.

**Exercise 2.4.10.** Suppose $f : U \to \mathbb{R}^n$ is continuously differentiable. Show that the "critical locus" of $f$ (ie, the set of all critical points of $f$) is a closed subset of $U$.

Another important property of continuously differentiable functions is the following.

**Theorem 2.4.11** (Inverse function theorem)**.** *Suppose $f : U \to \mathbb{R}^n$ is continuously differentiable, and suppose that $f$ is étale at a point $a \in U$. Then there exists an open neighborhood $V$ of $a$ inside $U$ such that the restriction $f|_V$ of $f$ to $V$ is injective, $f(V)$ is an open subset of $\mathbb{R}^n$, and the inverse function $(f|_V)^{-1} : f(V) \to V$ is continuously differentiable.*

Note that, if $f$ is étale at $a$, then $df_a : \mathbb{R}^m \to \mathbb{R}^n$ is an isomorphism, so we must have $m = n$. The inverse function theorem can be regarded as a multivariable version of exercise 1.2.10, but its proof is much more difficult. We defer the proof to a starred section below. Here are some applications.

**Exercise 2.4.12.** Suppose $f : U \to \mathbb{R}^n$ is continuously differentiably and étale at all points in $U$. Show that $f$ is an open map (in other words, show that, if $V$ is an open subset of $U$, then $f(V)$ is an open subset of $\mathbb{R}^n$).

**Exercise 2.4.13.** You might recall proving at some point in a previous class that $\mathbb{R}^2$ and $\mathbb{R}$ have the same cardinality: in other words, that there exists a bijection $f : \mathbb{R}^2 \to \mathbb{R}$. Show that any such bijection cannot be continuously differentiable.

*Possible hint.* Suppose for a contradition that $f : \mathbb{R}^2 \to \mathbb{R}$ is a continuously differentiable bijection. Let $U$ be an open subset of $\mathbb{R}^2$ such that $\partial_1 f(x) \neq 0$ for all $x \in U$, and then consider the function $g : U \to \mathbb{R}^2$ given by $g(x_1, x_2) = (f(x_1, x_2), x_2)$.

#### Proof of the inverse function theorem $\star$

There are many proofs of the inverse function theorem, but none of them are particularly easy. The proof we give here is the one from [Spi65, chapter 2]. It begins with the following multivariable version of exercise 1.3.5 (and again, we will actually use the single variable version to prove the multivariable version).

**Lemma 2.4.14.** *Suppose* $B$ *is an open ball and* $f : B \to \mathbb{R}^n$ *is a differentiable function. Suppose further that there exists a constant* $M$ *such that* $|\partial_i f_j(x)| \leqslant M$ *for all* $i = 1, \ldots, m$ *and* $j = 1, \ldots, n$. *Then*

$$|f(b) - f(a)| \leqslant m\sqrt{n}M|b - a|$$

*for all* $a, b \in B$.

*Proof.* Let us first show that it is enough to prove the lemma for $n = 1$. Indeed, if it is true for $n = 1$, then it is true for all of the component functions $f_j : B \to \mathbb{R}$, so

$$|f_j(b) - f_j(a)| \leqslant mM|b - a|$$

for all $a, b \in B$, which means that

$$|f(b) - f(a)|_{\max} \leqslant mM|b - a|,$$

where $| - |_{\max}$ denotes the max norm defined in section 0.4. But then

$$|f(b) - f(a)| \leqslant \sqrt{n}|f(b) - f(a)|_{\max} \leqslant m\sqrt{n}M|b - a|,$$

where we have used exercise 0.3.10 for the first step.

So we can assume that $n = 1$. The idea of this proof is very similar to the idea we used in the proof of theorem 2.4.5. Set $h = b - a = (h_1, \ldots, h_m)$, and then define the following list of vectors.

$$\begin{aligned} a_0 &= a \\ a_1 &= a + (h_1, 0, 0 \ldots, 0, 0) \\ a_2 &= a + (h_1, h_2, 0, \ldots, 0, 0) \\ &\vdots \\ a_{m-1} &= (a_1, a_2, a_3, \ldots, a_{m-1}, 0) \\ a_m &= (a_1, a_2, a_3, \ldots, a_{m-1}, a_m) = b \end{aligned}$$

Notice that

$$f(b) - f(a) = \sum_{i=1}^{m} (f(a_i) - f(a_{i-1}))$$

We're now going to restrict $f$ to the zigzag path from $a$ to $b$ made by the points $a_0, \ldots, a_m$, looking at each segment in turn, just as we did in the proof of theorem 2.4.5. The function

$$t \mapsto f(a_{i-1} + te_i)$$

is a single variable function whose derivative at a point $t$ is given by $\partial_i f(a_{i-1} + te_i)$, which we know is bounded in absolute value by $M$. Since this function has the value $f(a_{i-1})$ at $t = 0$ and $f(a_i)$ at $t = s_i$, exercise 1.3.5 gives us the first step of the following chain.

$$|f(a_i) - f(a_{i-1})| \leqslant M|h_i|.$$

But then, using exercise 0.3.10, we have

$$|f(a_i) - f(a_{i-1})| \leqslant M|h_i| \leqslant M|h|_{max} \leqslant M|h| = M|b - a|$$

so we have

$$|f(b) - f(a)| = \left| \sum_{i=1}^{m} f(a_i) - f(a_{i-1}) \right| \leqslant \sum_{i=1}^{m} |f(a_i) - f(a_{i-1})| \leqslant mM|b - a|. \qquad \square$$

*Proof of the inverse function theorem 2.4.11.* Observe that

$$d(df_a^{-1} \circ f)_a = df_a^{-1} \circ df_a = \mathrm{id}$$

using the chain rule 2.3.3 and exercise 2.2.5. By the first two parts of exercise 2.4.17 with $\ell = df_a^{-1}$, we can freely replace $f$ with $df_a^{-1} \circ f$ in order to assume that $df_a = \mathrm{id}$. In other words, we have

$$\partial_i f_j(a) = \delta_{i,j} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise.} \end{cases}$$

We will now construct an open ball $B$ around $a$, and then shrink it so that all points in $B$ have certain desirable properties. First of all, notice that

$$\lim_{v \to 0} \frac{|f(a+v) - f(a) - df_a(v)|}{|v|} = 0,$$

but if $f(a+v) = f(a)$, then

$$\frac{|f(a+v) - f(a) - df_a(v)|}{|v|} = \frac{|v|}{|v|} = 1,$$

which means that there must exist an open ball $B$ around $a$ such that $f(a+v) \neq f(a)$ whenever $a + v \in \bar{B} \setminus \{a\}$.

We now shrink $B$ a bit. Consider the map $U \mapsto \mathbb{R}$ given by $x \mapsto \det(df_x)$. Since $\det(df_x)$ is a polynomial in the partial derivatives of $f$, which are all continuous at $a$ since $f$ is continuously differentiable at $a$, the map $x \mapsto \det(df_x)$ must also be continuous at $a$. But $\det(df_a) \neq 0$, so by shrinking $B$, we can assume that $\det(df_x) \neq 0$ for all $x \in B$. Finally, since $\partial_i f_j$ is continuous at $a$, we can shrink $B$ further in order to assume that

$$|\partial_i f_j(x) - \partial_i f_j(a)| = |\partial_i f_j(x) - \delta_{i,j}| \leqslant \frac{1}{2n^{3/2}}$$

for all $i, j$ and all $x \in B$. In other words, we have constructed an open ball $B$ around $a$ inside $U$ with the following three properties.

(1) $f(x) \neq f(a)$ for all $x \in \bar{B} \setminus \{a\}$.

(2) $df_x$ is invertible for all $x \in B$.

(3) $|\partial_i f_j(x) - \delta_{i,j}| \leqslant 1/2n^{3/2}$ for all $i, j$ and $x \in B$.

The set $V$ we want to construct will be an open subset of $B$. The remainder of the proof is a series of claims.

*Claim A*. $|x_1 - x_2| \leqslant 2|f(x_1) - f(x_2)|$ for all $x_1, x_2 \in B$.

*Proof of claim A*. Consider the function $g(x) = f(x) - x$. Then $\partial_i g_j(x) = \partial_i f_j(x) - \delta_{i,j}$, so

$$|\partial_i g_j(x)| = |\partial_i f_j(x) - \delta_{i,j}| \leqslant \frac{1}{2n^{3/2}}$$

using property (3). By lemma 2.4.14, we see that

$$|g(x_1) - g(x_2)| \leqslant \frac{1}{2}|x_1 - x_2|.$$

But

$$|g(x_1) - g(x_2)| = |(f(x_1) - x_1) - (f(x_2) - x_2)| \geqslant |f(x_1) - f(x_2)| - |x_1 - x_2|$$

by the reverse triangle inequality. Combining these two inequalities proves claim A.

*Claim B*. There exists a positive real number $d$ such that $|f(x) - f(a)| \geqslant d$ for all $x \in \partial B$.

*Proof of claim B*. The function $\partial B \to \mathbb{R}$ given by

$$x \mapsto |f(x) - f(a)|$$

is continuous. Let $d$ denote the infimum of the range of this function. Clearly $d \geqslant 0$, and to prove claim B, we only need to prove that $d > 0$. Suppose for a contradiction that $d = 0$. Since $\partial B$ is compact, by the extreme value theorem there must exist $x \in \partial B$ such that $|f(x) - f(a)| = 0$. This contradicts property (1).

*Claim C*. Let $d$ be the positive real number defined in claim B, and let

$$W = \{y \in \mathbb{R}^n : |y - f(a)| < d/2\}.$$

Then for every $y \in W$, there exists a unique $x \in B$ such that $f(x) = y$.

*Proof of claim C*. Fix $y \in W$ and consider the function $\bar{B} \to \mathbb{R}$ given by

$$g(x) = |y - f(x)|^2 = \sum_{j=1}^{n} (\pi_j(y) - f_j(x))^2.$$

This function is continuous and $\bar{B}$ is compact, so the extreme value theorem guarantees that $g$ achieves its minimum at some $x \in \bar{B}$. Notice that, if $x \in \partial B$, then

$$g(x) = |y - f(x)|^2 \geqslant d^2$$

whereas

$$g(a) = |y - f(a)|^2 < d^2/4,$$

so $g(a) < g(x)$, which contradicts the fact that $x$ is the minimum of $g$. Thus $x \in \bar{B} \setminus \partial B = B$. By the interior extremum theorem 2.3.19, we know that

$$0 = \partial_i g(x) = \sum_{j=1}^{n} 2(\pi_j(y) - f_j(x))\partial_i f_j(x) = \pi_i \left(2f'(x)^T(y - f(x))\right)$$

for all $i$. Here, $f'(x)^T$ denotes the transpose of the Jacobian matrix. This means that $f'(x)^T(y - f(x)) = 0$. Now recall that $df_x$ is an invertible linear map using property (2). Thus $f'(x)$ is an invertible matrix, so $f'(x)^T$ is also an invertible matrix, so we must have $y - f(x) = 0$. In other words, $y = f(x)$. This proves the existence assertion of the claim.

For the uniqueness assertion, if $x'$ is any other point in $B$ such that $f(x') = y$, then we would have

$$0 = 2|y - y| = 2|f(x') - f(x)| \geqslant |x' - x| \geqslant 0$$

using claim A, so $x' = x$. This completes the proof.

*Claim D*. With $W$ as in claim C, define

$$V = f^{-1}(W) = \{x \in B : f(x) \in W\}.$$

Then $V$ is open and $f|_V : V \to W$ is a homeomorphism. In particular, $f(V) = W$ and $W$ is open.

*Proof of claim D.* Recall from claim C that $W$ is the open ball of radius $d/2$ around $f(a)$. Since $f$ is continuous, $f^{-1}(W)$ must also be open. Claim C tells us that $f|_V$ is injective, and claim A says that

$$|f^{-1}(y_1) - f^{-1}(y_2)| \leqslant 2|y_1 - y_2| \tag{2.4.15}$$

for all $y_1, y_2 \in W$. This implies that $(f|_V)^{-1}$ is continuous.

*Claim E.* $(f|_V)^{-1}$ is differentiable.

*Proof of claim E.* For $x \in V$, let $y = f(x) \in f(V)$. We will show that $(f|_V)^{-1}$ is differentiable at $y$ with derivative $df_x^{-1}$. This means that, if we set

$$s(w) = f^{-1}(y + w) - f^{-1}(y) - df_x^{-1}(w),$$

we want to show that $|s(w)| = o(|w|)$ as $w \to 0$. In fact, since $df_x$ is an invertible linear map, it is sufficient to show that $|df_x(s(w))| = o(|w|)$ (cf. exercise 2.3.5 part (b)).

Define $v(w) = f^{-1}(y + w) - f^{-1}(w)$. Since $f|_V$ is injective, we see that $v(w) = 0$ if and only if $w = 0$. Also, equation (2.4.15) tells us that

$$|v(w)| \leqslant 2|w|. \tag{2.4.16}$$

By definition of $s(w)$ and $v(w)$, we have

$$s(w) = v(w) - df_x^{-1}(w)$$

so

$$\begin{aligned} df_x(s(w)) &= df_x(v(w)) - w \\ &= df_x(v(w)) - y + w + y \\ &= df_x(v(w)) - f(x + v(w)) + f(x). \end{aligned}$$

In other words, if we consider the "error in approximation" function

$$r(v) = f(x + v) - f(x) - df_x(v),$$

then

$$df_x(s(w)) = -r(v(w))$$

for all $w$. This means that, for all $w \neq 0$, we have

$$\frac{|df_x(s(w))|}{|w|} = \frac{|r(v(w))|}{|w|} = \frac{|r(v(w))|}{|v(w)|}\frac{|v(w)|}{|w|} \leqslant 2 \cdot \frac{|r(v(w))|}{|v(w)|}.$$

Note that we have used the fact that $v(w)$ is nonzero for nonzero $w$, in order to avoid a division by zero. Equation (2.4.16) tells us that

$$\lim_{w \to 0} v(w) = 0$$

so

$$\lim_{w \to 0} \frac{|r(v(w))|}{|v(w)|} = 0$$

since $|r(v)| = o(|v|)$ as $v \to 0$. By the squeeze theorem, we conclude that

$$\lim_{w \to 0} \frac{|df_x(s(w))|}{|w|} = 0,$$

which is precisely the statement that $|df_x(s(w))| = o(|w|)$ as $w \to 0$.

We've proved that $(f|_V)^{-1}$ is differentiable. It follows from lemma 2.4.18 below that it must in fact be continuously differentiable. □

**Exercise 2.4.17.** (a) Suppose $\ell : \mathbb{R}^n \to \mathbb{R}^n$ is an invertible linear map. Show that $f : U \to \mathbb{R}^n$ is étale at a point $a \in U$ if and only if $\ell \circ f$ is étale at $a$.

(b) Suppose $\ell : \mathbb{R}^n \to \mathbb{R}^n$ is an invertible linear map. Show that the conclusion of the inverse function theorem is true for $f$ if and only if it is true for $\ell \circ f$.

(c) Show that, if $K$ is a compact subset of a metric space $X$ and $a \in X \setminus K$, then

$$\inf\{d(a, x) : x \in K\} > 0.$$

**Lemma 2.4.18.** *Suppose $f : U \to \mathbb{R}^n$ is continuously differentiable, injective, and that $f(U)$ is open in $\mathbb{R}^n$. If $f^{-1} : f(U) \to U$ is differentiable, then it is continuously differentiable.*

*Proof.* Since $f^{-1} \circ f = \mathrm{id}$, we have

$$\mathrm{id} = d(\mathrm{id})_x = d(f^{-1} \circ f)_x = d(f^{-1})_{f(x)} df_x$$

for all $x \in U$, where we used exercise 2.2.5 for the first step and the chain rule theorem 2.3.3 for the last. This implies, in particular, that $df_x$ is an invertible linear map. Said differently, for any $y \in f(U)$, we have

$$d(f^{-1})_y = (df_{f^{-1}(y)})^{-1}.$$

Taking standard matrix representations, we get

$$[d(f^{-1})_y] = [(df_{f^{-1}(y)})^{-1}] = [df_{f^{-1}(y)}]^{-1}.$$

By theorem 2.3.26, the $(j, i)$-entry on the left hand side is $\partial_i(f^{-1})_j(y)$, so this equality says that $\partial_i(f^{-1})_j : f(U) \to \mathbb{R}$ is equal to the following composite.

$$f(U) \xrightarrow{f^{-1}} U \xrightarrow{x \mapsto [df_x]} GL_n \xrightarrow{A \mapsto A^{-1}} GL_n \xrightarrow{(j,i)-\text{entry}} \mathbb{R}$$

Here $GL_n$ denotes the set of invertible $n \times n$ matrices as in section 0.5.B. Each of the functions being composed here is continuous (cf. exercise 2.2.8 and lemma 0.5.18), so we conclude that $\partial_i(f^{-1})_j$ is continuous. □

#### Continuous differentiability via the operator norm ⋆

For this section, we will need to make substantial use of the operator norm. This might make it a bit challenging for those who have not used the operator norm before, but the results contained in this section are also not terribly important. The only reason it is included is because proposition 2.4.21 is aesthetically a rather satisfying statement.

**Lemma 2.4.19.** *Suppose* $f : U \to \mathbb{R}^n$ *is differentiable. Then* $df : U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ *is continuous if and only if* $df_j : \mathcal{L}(\mathbb{R}^m, \mathbb{R})$ *is continuous for all* $j = 1, \ldots, n$.

*Proof.* We will use proposition 2.3.7, which tells us that $f$ is differentiable if and only if $f_j$ is differentiable for all $j = 1, \ldots, n$, and moreover that $df_{j,a} = \pi_j \circ df_a$ for all $a \in U$. In other words, the following diagram commutes.

$$\begin{tikzcd} U \arrow[r, "df"] \arrow[dr, bend right, "df_j"'] & \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \arrow[d, "\pi_j \circ -"] \\ & \mathcal{L}(\mathbb{R}^m, \mathbb{R}) \end{tikzcd}$$

You will verify in exercise 2.4.20 below that the map "postcompose with $\pi_j$" map $\pi_j \circ - :$

$\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \to \mathcal{L}(\mathbb{R}^m, \mathbb{R})$ is continuous. Thus, if $df$ is continuous, then $df_j$ is also continuous at $a$.

Conversely, suppose that $df_j$ is continuous at $a$ for all $j$. Let us show that $df$ is continuous at $a$. Suppose $\epsilon > 0$. Since $df_j$ is continuous, there exists $\delta_j > 0$ such that

$$\|df_{j,x} - df_{j,a}\| < \frac{\epsilon}{\sqrt{n}}$$

for all $x \in U$ such that $|x - a| < \delta_j$. Let $\delta := \min\{\delta_1, \ldots, \delta_n\}$. Then $\|df_x - df_a\| < \epsilon$ for all $x \in U$ such that $|x - a| < \delta$, as you will verify in exercise 2.4.20 below. □

**Exercise 2.4.20.** (a) Check that the "postcompose with $\pi_j$" map $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n) \to \mathcal{L}(\mathbb{R}^m, \mathbb{R})$ (ie, the one given by $\phi \mapsto \pi_j \circ \phi$) is continuous. Alternatively, prove the more general statement given in exercise 0.6.9 and then derive this statement.

(b) With notation as in the latter part of the proof above, verify that $\|df_a - df_b\| < \epsilon$.

**Proposition 2.4.21.** *Suppose $f : U \to \mathbb{R}^n$ is differentiable. Then $f$ is continuously differentiable if and only if $df : U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ is continuous.*

*Proof.* Thanks to lemma 2.4.19, we can assume without loss of generality that $n = 1$. If $f$ is differentiable, then proposition 2.3.22 tells us that $\partial_i f = df(e_i)$. In other words, the following diagram commutes.

$$\begin{tikzcd} U \arrow[r, "df"] \arrow[dr, bend right, "\partial_i f"'] & \mathcal{L}(\mathbb{R}^m, \mathbb{R}) \arrow[d, "\text{“evaluate at } e_i\text{”}"] \\ & \mathbb{R} \end{tikzcd}$$

The "evaluate at $e_i$" map is continuous by exercise 0.6.7. So, if $f$ is continuously differentiable (ie, if $df$ is continuous), then $\partial_i f$ must also be continuous, as it is the composite of two continuous functions.

Conversely, suppose that all of the partial derivatives $\partial_i f$ exist and are continuous. Then theorem 2.4.5 guarantees that $f$ is differentiable. The final step is to show that $df : U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R})$ is continuous; This is left for you to check; see exercise 2.4.22 below. □

**Exercise 2.4.22.** (a) Elaborate on the first sentence of the above proof (in other words, explain why proving theorem 2.4.5 for $n = 1$ is sufficient to prove it for all $n$).

(b) If you haven't already done it, do exercise 0.6.7.

(c) Complete the proof above by showing that $df : U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R})$ is continuous.

**Remark 2.4.23.** This reformulation of continuous differentiability is often convenient in proofs. For example, in the proof of lemma 2.4.18, we calculated that

$$d(f^{-1})_y = (df_{f^{-1}(y)})^{-1}.$$

This says that the $d(f^{-1}) = \text{inversion} \circ df \circ f^{-1}$.

$$\begin{tikzcd} f(U) \arrow[r, "d(f^{-1})"] \arrow[d, "f^{-1}"'] & GL(\mathbb{R}^n) \\ U \arrow[r, "df"] & GL(\mathbb{R}^n) \arrow[u, "\text{inversion}"'] \end{tikzcd}$$

We know that $f^{-1}$ is continuous by exercise 2.2.8, that $df$ is continuous by proposition 2.4.21, and inversion is continuous by exercise 0.6.12. Thus $d(f^{-1})$ is continuous, so proposition 2.4.21 says that $f^{-1}$ is continuously differentiable.

### 2.4.C $C^k$ functions

**Definition 2.4.24** ($C^k$ functions)**.** We say that a function $f : U \to \mathbb{R}^n$ is $C^0$ if it is continuous. Then, inductively, for any positive integer $k$, we say that a function $f : U \to \mathbb{R}^n$ is $C^k$ if the partial derivatives $\partial_i f_j$ all exist and are $C^{k-1}$ functions $U \to \mathbb{R}$.

By definition, $f : U \to \mathbb{R}^n$ is $C^1$ if and only if it is continuously differentiable. It is clear from the definitions that a function $f : U \to \mathbb{R}^n$ is $C^k$ if and only if its component functions $f_j : U \to \mathbb{R}$ are $C^k$. Here are some results showing that reasonable ways of combining $C^k$ functions still yield $C^k$ functions.

**Exercise 2.4.25.** Show that the set of all $C^k$ functions $U \to \mathbb{R}^n$ form a vector space.

**Exercise 2.4.26.** Suppose $U$ is an open subset $\mathbb{R}^m$, that $V$ is an open subset of $\mathbb{R}^n$, and that $f : U \to \mathbb{R}^n$ and $g : V \to \mathbb{R}^p$ are both $C^k$, and that $f(U) \subseteq V$. Show that $g \circ f$ is also $C^k$.

**Exercise 2.4.27.** Suppose $f, g : U \to \mathbb{R}$ are both $C^k$. Show that $fg$ is also $C^k$.

**Exercise 2.4.28.** Suppose $f : U \to \mathbb{R}$ is $C^k$ and $f(x) \neq 0$ for all $x \in U$. Show that $1/f$ is also $C^k$.

*Possible hint.* Note that $1/f$ is $f$ composed with the single variable inversion function.

**Exercise 2.4.29.** Suppose $f : U \to \mathbb{R}^n$ is a $C^k$ injection for some $k \geqslant 1$ and that $f$ is étale at every $a \in U$. Show that $f^{-1} : f(U) \to \mathbb{R}^n$ is also $C^k$.

*Possible hint.* This is a multivariable version of lemma 1.4.10, and can be proved by using a similar induction. Some ingredients you might think about combining are the inverse function theorem 2.4.11 and lemmas 0.5.18 and 2.4.18 (and their proofs).

### 2.4.D Equality of mixed partials ⋆

**Definition 2.4.30** (Mixed partials)**.** Suppose $f : U \to \mathbb{R}$ is $C^2$. The partial derivative $\partial_i f : U \to \mathbb{R}$ for any $i = 1, \ldots, m$ is a continuously differentiable function, so its $j$th partial deriative $\partial_j(\partial_i f)$ is a continuous function $U \to \mathbb{R}$ for any $j = 1, \ldots m$. We denote the function $\partial_j(\partial_i f)$ more succinctly as $\partial_{j,i} f$, or as

$$\frac{\partial^2 f}{\partial x_j x_i}$$

**Theorem 2.4.31** (Equality of mixed partials)**.** *Suppose $f : U \to \mathbb{R}$ is $C^2$. Then, for all $i, j = 1, \ldots, m$, we have $\partial_{j,i} f = \partial_{i,j} f$.*

*Proof.* Observe that

$$\begin{aligned}
\partial_{j,i} f(a) &= \lim_{h_j \to 0} \frac{\partial_i f(a + h_j e_j) - \partial_i f(a)}{h_j} \\
&= \lim_{h_j \to 0} \frac{\lim\limits_{h_i \to 0} \dfrac{f(a + h_i e_i + h_j e_j) - f(a + h_j e_j)}{h_i} - \lim\limits_{h_i \to 0} \dfrac{f(a + h_i e_i) - f(a)}{h_i}}{h_j} \\
&= \lim_{h_j \to 0} \lim_{h_i \to 0} \frac{f(a + h_i e_i + h_j e_j) - f(a + h_i e_i) - f(a + h_j e_j) + f(a)}{h_i h_j}.
\end{aligned}$$

Notice that, if we unwind the definition of $\partial_{i,j} f(a)$ in the same way, we end up taking a double limit of the same expression, but the order of the limits is interchanged. In other words, if we define

$$\sigma(h_i, h_j) = \frac{f(a + h_i e_i + h_j e_j) - f(a + h_i e_i) - f(a + h_j e_j) + f(a)}{h_i h_j},$$

then

$$\partial_{j,i} f(a) = \lim_{h_j \to 0} \lim_{h_i \to 0} \sigma(h_i, h_j) \quad \text{whereas} \quad \partial_{i,j} f(a) = \lim_{h_i \to 0} \lim_{h_j \to 0} \sigma(h_i, h_j).$$

So, to prove the theorem, it is sufficient to prove that

$$\partial_{j,i} f(a) = \lim_{h \to 0} \sigma(h, h).$$

Fix a nonzero real number $h$ and consider the function

$$\alpha(t) = f(a + te_i + he_j) - f(a + te_i).$$

Notice that

$$\sigma(h, h) = \frac{\alpha(h) - \alpha(0)}{h^2}.$$

Observe that $\alpha$ is differentiable, with

$$\alpha'(t) = \partial_i f(a + te_i + he_j) - \partial_i f(a + te_i).$$

Applying the mean value theorem 1.3.2 to $\alpha$ on the closed interval between 0 and $h$, there exists $\xi_i(h)$, strictly between 0 and $h$, such that

$$\sigma(h, h) = \frac{\alpha(h) - \alpha(0)}{h^2} = \frac{\alpha'(\xi_i(h))}{h} = \frac{\partial_i f(a + \xi_i(h)e_i + he_j) - \partial_i f(a + \xi_i e_i)}{h}.$$

Now consider the function

$$\beta(t) = \partial_i f(a + \xi_i(h)e_i + te_j).$$

Then $\beta$ is differentiable with

$$\beta'(t) = \partial_{j,i} f(a + \xi_i e_i + te_j),$$

so, applying the mean value theorem 1.3.2 to $\beta$ on the closed interval between 0 and $h$, we find that there exists $\xi_j(h)$, strictly between 0 and $h$, such that

$$\sigma(h, h) = \frac{\beta(h) - \beta(0)}{h} = \beta'(\xi_j(h)) = \partial_{j,i} f(a + \xi_i(h)e_i + \xi_j(h)e_j).$$

Now notice that, since both $\xi_i(h)$ and $\xi_j(h)$ are between 0 and $h$, we have $\xi_i(h), \xi_j(h) \to 0$ as $h \to 0$. Since $\partial_{j,i} f$ is continuous at $a$, we have

$$\lim_{h \to 0} \sigma(h, h) = \lim_{h \to 0} \partial_{j,i} f(a + \xi_i(h) e_i + \xi_j(h) e_j) = \partial_{j,i} f(a). \qquad \square$$

**Remark 2.4.32.** If $f : U \to \mathbb{R}$ is $C^2$, the *hessian* of $f$ at $a$, denoted $f''(a)$, is the matrix of second partial derivatives.

$$f''(a) = \begin{bmatrix} \partial_{1,1} f(a) & \partial_{1,2} f(a) & \cdots & \partial_{1,m} f(a) \\ \vdots & \vdots & \ddots & \vdots \\ \partial_{m,1} f(a) & \partial_{m,2} f(a) & \cdots & \partial_{m,m} f(a) \end{bmatrix}$$

Theorem 2.4.31 is equivalent to the assertion that $f''(a)$ is a symmetric matrix.

There are many versions of this theorem, all with slightly different hypotheses. For instance, there is the version due to Peano [Rud76, theorem 9.41]. In any case, it's worth remarking that some continuity condition on the mixed partials is crucial; it is not enough that the mixed partials exist.

**Exercise 2.4.33.** Consider the following function $f : \mathbb{R}^2 \to \mathbb{R}$.

$$f(x, y) = \begin{cases} xy \dfrac{x^2 - y^2}{x^2 + y^2} & \text{if } x \neq 0 \text{ or } y \neq 0 \\ 0 & \text{if } x = y = 0 \end{cases}$$

(a) Calculate all of the following.

$$\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \quad \frac{\partial^2 f}{\partial y \partial x} \quad \frac{\partial^2 f}{\partial x \partial y}$$

(b) Show that the mixed partials are discontinuous at the origin.

(c) Show that the mixed partials disagree at the origin.

### 2.4.E Taylor's theorem ⋆

The main difficulty in the multivariable Taylor's theorem is notation. Let us first discuss some notational conventions to make the statement of Taylor's theorem more convenient.

**Definition 2.4.34** (Multi-index)**.** A *multi-index* is an $m$-tuple $\alpha = (\alpha_1, \ldots, \alpha_m)$ where $\alpha_i$ is a non-negative integer. For such a multi-index, we define the following.

$$|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_n$$
$$\alpha! = \alpha_1!\alpha_2! \cdots \alpha_m!$$

If $x = (x_1, \ldots, x_m) \in \mathbb{R}^m$, we define

$$x^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_m^{\alpha_m}.$$

Also, if $f : U \to \mathbb{R}$ is $C^{|\alpha|}$, we define

$$\partial^\alpha f = \partial_1^{\alpha_1} \partial_2^{\alpha_2} \cdots \partial_m^{\alpha_m} f.$$

**Theorem 2.4.35** (Taylor)**.** *Suppose $f : U \to \mathbb{R}$ is $C^k$. Then there exists a unique polynomial $p_k$ of degree $k$ in $m$ variables, called the* degree $k$ Taylor polynomial of $f$ at $a$*, such that $|f(a+h) - p_k(h)| = o(|h|^k)$. Moreover, we have*

$$p_k(v) = \sum_{|\alpha| \leqslant k} \frac{\partial^\alpha f(a)}{\alpha!} h^\alpha. \tag{2.4.36}$$

### 2.4.F Smooth functions

**Definition 2.4.37** (Smoothness)**.** We say that a function $f : U \to \mathbb{R}^n$ is $C^\infty$, or *infinitely differentiable*, or *smooth*, if it is $C^k$ for all $k$.

In section 1.4.D, we formulated the principle that smooth single variable functions can be tailored to almost arbitrary specifications. The same principle remains true of smooth multivariable functions, and we can often bootstrap up from the single variable case. Here is one example.

**Exercise 2.4.38.** Let $\psi : \mathbb{R} \to \mathbb{R}$ denote the bump function from example 1.4.26. Then define $\Psi : \mathbb{R}^m \to \mathbb{R}$ by

$$\Psi(x_1, \ldots, x_m) = \psi(x_1)\psi(x_2) \cdots \psi(x_m).$$

Show that $\Psi$ is a bump function (ie, a smooth function with compact support).

# 3 Manifolds

At this point, we will need to use the language of topological spaces. If you have seen metric spaces but not topological spaces, you should look over section 0.7.A at this point.

## 3.1 Definition of a manifold

Throughout, let $X$ be a topological space.

**Definition 3.1.1** (Chart)**.** A *chart* on $X$ is a pair $(U, x)$ consisting of an open subset $U \subseteq X$ and a function $x : U \to \mathbb{R}^n$ which is a homeomorphism onto an open subset of $\mathbb{R}^n$ for some $n$. We call the functions $x_i := \pi_i \circ x : U \to \mathbb{R}$ *coordinate functions* on $U$ with respect to the chart $(U, x)$, and for any $p \in U$, the list of numbers $x_1(p), \ldots, x_n(p)$ are the *coordinates* of $p$ with respect to the chart $(U, x)$.

**Exercise 3.1.2.** Let $X = \mathbb{R}^2$ and let $U$ denote $\mathbb{R}^2$ minus the non-negative real axis. Let $\text{polar} : U \to \mathbb{R}^2$ be the function which sends a point $p$ to its polar representation $(r(p), \theta(p))$, where $r(p) = |p|$ and $\theta(p)$ is the angle in radians strictly between 0 and $2\pi$ formed between $P$ and the non-negative real axis. Show that this function is a homeomorphism onto its image.

**Definition 3.1.3.** Suppose $(U, x)$ and $(V, y)$ are charts on $X$. The *transition function* from $(U, x)$ to $(V, y)$ is the function $y \circ x^{-1}$.

$$x(U \cap V) \xrightarrow{x^{-1}} U \cap V \xrightarrow{y} y(U \cap V)$$

**Definition 3.1.4** (Compatibility of charts)**.** The two charts $(U, x)$ and $(V, y)$ on $X$ are *smoothly compatible*, or just *compatible*, if both of the transition maps between $(U, x)$ and $(U, y)$ are smooth.

**Remark 3.1.5.** One could also make a definition of $C^k$*-compatibility* for any $k = 0, 1, \ldots, \infty$, but we won't ever use this.

**Exercise 3.1.6.** Let $\mathrm{id} : \mathbb{R}^2 \to \mathbb{R}^2$ denote the identity map and $\mathrm{polar} : U \to \mathbb{R}^2$ the polar representation map from exercise 3.1.2. Show that $(\mathbb{R}^2, \mathrm{id})$ and $(U, \mathrm{polar})$ are compatible charts.

**Exercise 3.1.7.** Show that the relation "is compatible with" is *not* an equivalence relation on the set of all charts on $X$.

*Possible hint.* The function $\mathbb{R} \to \mathbb{R}$ given by $x \mapsto x^3$ is a smooth homeomorphism, but its inverse is not differentiable.

**Exercise 3.1.8.** Suppose $(U, x)$ and $(V, y)$ are compatible charts, where $x : U \to \mathbb{R}^m$ and $y : V \to \mathbb{R}^n$, and that $U \cap V$ is nonempty. Show that $m = n$.

*Possible hint.* Pick a point $p \in U \cap V$, and consider the derivatives of $x \circ y^{-1}$ at $y(p)$ and of $y \circ x^{-1}$ at $x(p)$.

**Definition 3.1.9** (Atlas)**.** An *atlas* on $X$ is a collection $\mathcal{A}$ of charts such that covers $X$. In other words,

$$\bigcup_{(U,x)\in\mathcal{A}} U = X.$$

An atlas is *maximal* if there is no strictly larger atlas containing it.

**Exercise 3.1.10** (Maximal atlases)**.** Prove that every atlas $\mathcal{A}$ is contained in a unique maximal atlas.

*Possible hint.* Let $\mathcal{A}'$ be the set of all charts that are compatible with all of the charts in $\mathcal{A}$. Prove that $\mathcal{A}'$ is an atlas (caution: keep exercise 3.1.7 in mind). Then prove that it is maximal, and that it is the only maximal atlas containing $\mathcal{A}$.

This leads us to the following definition.

**Definition 3.1.11** (Manifold)**.** A *smooth manifold*, or just a *manifold*, is a pair $(X, \mathcal{A}_X)$ where $X$ is a topological space and $\mathcal{A}_X$ is a maximal atlas. Often, write simply "$X$" in place of the pair $(X, \mathcal{A}_X)$, and sometimes $\mathcal{A}_X$ is called the "manifold structure on $X$."

Of course, exercise 3.1.10 tells us that any not-necessarily-maximal atlas on a metric space $X$ defines a manifold structure on $X$ in a canonical way. When constructing examples, we will usually give an explicit non-maximal atlas and then use exercise 3.1.10 to extend it to a maximal atlas. But we require maximality in the definition in part because, when dealing

with abstract manifolds in proofs, we will often find it convenient to assume that the atlas that the manifold comes equipped with is already maximal.

We can also make the following definition.

**Definition 3.1.12** (Dimension)**.** Suppose $X$ is a manifold and $p \in X$ is a point. By exercise 3.1.8, there is a unique integer $n$ such that, for every chart $(U, x) \in \mathcal{A}_X$ with $p \in U$, the codomain of $x$ is $\mathbb{R}^n$. This integer is called the *dimension of* $X$ *at* $p$, and is denoted $\dim_p(X)$. We say that $X$ is $n$-*dimensional* if $\dim_p(X) = n$ for all $p \in X$.

## 3.2 Examples

### 3.2.A First examples

**Example 3.2.1.** The single chart $(\mathbb{R}^n, \mathrm{id})$, where $\mathrm{id} : \mathbb{R}^n \to \mathbb{R}^n$ is the identity map, defines an atlas on $\mathbb{R}^n$. We can extend this to a maximal atlas, called the *euclidean atlas* on $\mathbb{R}^n$. Unless explicitly specified otherwise, we always regard $\mathbb{R}^n$ as a manifold by equipping it with the euclidean atlas. By exercise 3.1.6, the polar representation chart $(U, \mathrm{polar})$ is an element of the euclidean atlas.

**Exercise 3.2.2.** The single chart $(\mathbb{R}, f)$ where $f : \mathbb{R} \to \mathbb{R}$ is the function $f(x) = x^3$ is an atlas on $\mathbb{R}$. Show that the corresponding maximal atlas is *not* the euclidean atlas.

**Exercise 3.2.3** (The circle $S^1$)**.** Let $S^1$ denote the set of points in $\mathbb{R}^2$ along the unit circle. In other words,

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

(a) There are two ways you might think to regard $S^1$ as a topological space. First, you might restrict the euclidean metric on $\mathbb{R}^2$ to $S^1$, so that $S^1$ becomes a metric space, and then you can forget this restricted metric and just remember the resulting topology as in example 0.7.2. Alternatively, you can first forget the metric on $\mathbb{R}^2$ and regard it as a topological space as in example 0.7.2, and then regard $S^1$ as a subspace of $\mathbb{R}^2$ (cf. definition 0.7.13). Show that you end up with the same topology on $S^1$ in both cases.

(b) Let $U = S^1 \setminus \{(0, 1)\}$. For $p \in S^1$, let $x(p)$ denote the $x$-coordinate of the point where the line $y = -1$ intersects the line passing through $(0, 1)$ and $p$. See figure 3.2.4. If $p = (a, b)$, find a formula for $x(p)$ in terms of $a$ and $b$.

(c) Show that $x : U \to \mathbb{R}$ is a homeomorphism. Thus $(U, x)$ is a chart on $S^1$.

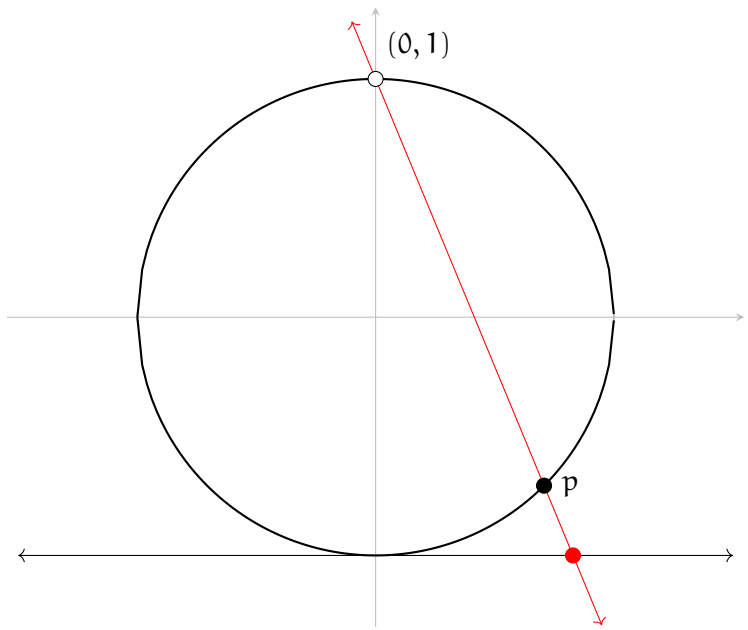


Figure 3.2.4: Given a point $p \in U = S^1 \setminus \{(0,1)\}$, we draw a line connecting $p$ to $(0,1)$, indicated in red. The point where that line intersects the line $y = -1$ (in black) is sometimes called the *projection of* $p$ *from* $(0,1)$ *onto* $\mathbb{R} \times \{-1\}$. The $x$-coordinate of this point is $x(p)$.

*Possible hint.* Use geometry to write down a formula for the inverse function to $x$.

(d) Let $U' = S^1 \setminus \{(0,-1)\}$. For $p \in V$, let $x'(p)$ denote the $x$-coordinate of the point where the line $y = 1$ intersects the line passing through $(0,-1)$ and $p$. Show that $x' : U' \to \mathbb{R}$ is also a homeomorphism.

(e) Show that $(U, x)$ and $(U', x')$ are compatible charts.

Thus $\mathcal{A} = \{(U, x), (U', x')\}$ is an atlas on $S^1$. Extending $\mathcal{A}$ to a maximal atlas, we have defined a manifold structure on $S^1$. It is evidently a 1-dimensional manifold.

**Exercise 3.2.5** (The $n$-sphere $S^n$)**.** Let $S^n$ denote the set of points in $\mathbb{R}^{n+1}$ of norm 1. Generalize exercise 3.2.3 and show that "projection from the north pole" and "projection from the south pole" define an atlas on $S^n$.

**Example 3.2.6** (Matrices)**.** There is an isomorphism of vector spaces $x : \mathcal{M}_{n\times m} \to \mathbb{R}^{mn}$ where, if $A \in \mathcal{M}_{n\times m}$ has entries $a_{i,j}$, then

$$x(A) = \begin{bmatrix} a_{1,1} \\ a_{1,2} \\ \vdots \\ a_{1,m} \\ a_{2,1} \\ \vdots \\ a_{n,m} \end{bmatrix}.$$

If we endow $\mathcal{M}_{n\times m}$ with the canonical topology, then $x$ is automatically a homeomorphism. In other words, the single chart $(\mathcal{M}_{n\times m}, x)$ is an atlas on $\mathcal{M}_{n\times m}$. Extending it to a maximal atlas using exercise 3.1.10, we see that we can regard $\mathcal{M}_{n,m}$ as a manifold. It is $mn$-dimensional.

**Exercise 3.2.7** (Open submanifolds)**.** Let $X$ be a manifold and $U \subseteq X$ an open subset. Show that

$$\mathcal{A}_U = \{(V, x) \in \mathcal{A}_X : V \subseteq U\}$$

is a maximal atlas on $U$. We call $U$, equipped with this maximal atlas, an *open submanifold* of $X$.

**Exercise 3.2.8** (Product manifold)**.** Let $X$ and $Y$ be manifolds. Let $\mathcal{A}$ be the set of charts of the form $(U \times V, x \times y)$, where $(U, x : U \to \mathbb{R}^m) \in \mathcal{A}_X$, $(V, y : V \to \mathbb{R}^n) \in \mathcal{A}_Y$, and $x \times y$ denotes the natural map $U \times V \to \mathbb{R}^m \times \mathbb{R}^n = \mathbb{R}^{m+n}$. Show that $\mathcal{A}$ is an atlas on $X \times Y$. We call $X \times Y$, equipped with the corresponding maximal atlas, the *product manifold*.

### 3.2.B Möbius strip

Let $B$ denote the box $[0, 1] \times (0, 1)$ inside $\mathbb{R}^2$, and define an equivalence relation $\sim$ on $B$ by declaring $(0, b) \sim (1, 1 - b)$ for all $b \in (0, 1)$. We will denote the equivalence class of a point $(a, b) \in B$ as $[a, b]$.

**Exercise 3.2.9.** Check that $\sim$ is an equivalence relation.

The *Möbius strip* is the set of equivalence classes $M := B/\sim$. In other words, we take the box $[0, 1] \times (0, 1)$ and "glue" its left and right edges, with a twist. If you've ever made a Möbius strip out of paper, you should be able to see that we've formalized exactly that same process. We regard $M$ as a topological space using the quotient topology, as in example 0.7.3.

To give $X$ the structure of a manifold, we will now produce an atlas on $X$. It will consist of two charts. The first chart is easy to describe. Let

$$U = \{[a, b] \in M : a \neq 0, 1\}$$

and let $x : U \to \mathbb{R}^2$ be the function $x([a, b]) = (a, b)$.

**Exercise 3.2.10.** Show that $x : U \to \mathbb{R}^2$ is well-defined and that it is a homeomorphism onto its image.

Intuitively, what we've done here is cut along the same edge where we glued the Mobius strip together. What we're left with is $U$, which is just a flat rectangle and therefore embeds into $\mathbb{R}^2$.

The second chart is a bit harder to describe mathematically, but the intuitive idea is the same. Instead of cutting the Möbius strip along the line where we glued, we'll cut along the line where $a = 1/2$. This should still leave us with a flat rectangle, which we can embed into $\mathbb{R}^2$. Formally, let

$$V = \{[a, b] \in M : a \neq 1/2\}$$

and let $y : V \to \mathbb{R}^2$ be the function defined by

$$y([a,b]) = \begin{cases} (a,b) & \text{if } a < 1/2 \\ (a-1, 1-b) & \text{if } a > 1/2 \end{cases}$$

**Exercise 3.2.11.** Show that $y : V \to \mathbb{R}^2$ is well-defined and that it is a homeomorphism onto its image.

**Exercise 3.2.12.** Show that $(U, x)$ and $(V, y)$ are compatible.

Thus $\mathcal{A} = \{(U, x), (V, y)\}$ is an atlas on the Möbius strip $M$. Letting $\mathcal{A}_M$ denote the corresponding maximal atlas, we obtain the structure of a manifold on $M$.

### 3.2.C More examples

**Exercise 3.2.13** (Torus)**.** Let $B$ denote the box $[0,1] \times [0,1]$ inside $\mathbb{R}^2$ and define the equivalence relation $\sim$ by $(a,0) \sim (a,1)$ and $(0,b) \sim (1,b)$ for all $a, b \in [0,1]$. The *torus* $T$ is defined to be the quotient $B/\sim$. Describe an atlas on $T$.

**Exercise 3.2.14** (Projective line)**.** Let $S^1$ denote the circle as in exercise 3.2.3. Define an equivalence relation $\sim$ on $S^1$ which identifies antipodal points; in other words, every point on $S^1$ is declared to be equivalent to the point that is diametrically opposite it. The *projective line* $\mathbb{P}^1$ is defined to be the quotient $S^1/\sim$. Describe an atlas on $\mathbb{P}^1$.

**Exercise 3.2.15** (Projective plane)**.** Let $S^2$ denote the 2-sphere as in exercise 3.2.5. Define an equivalence relation $\sim$ on $S^2$ which identifies antipodal points; in other words, every point on $S^2$ is declared to be equivalent to the point that is diametrically opposite it. The *projective plane* $\mathbb{P}^2$ is defined to be the quotient $S^2/\sim$. Describe an atlas on $\mathbb{P}^2$.

## 3.3 Smooth functions

We will now define what it means for a function between two manifolds to be smooth. Recall that in chapter 2 we saw that the case when the codomain is $\mathbb{R}$ was the most important. The same will be true here.

### 3.3.A Smooth real-valued functions

**Definition 3.3.1** (Smooth real-valued function)**.** Let $X$ be a manifold. A function $f : X \to \mathbb{R}$ is said to be *smooth* if, for any chart $(U, x) \in \mathcal{A}_X$, the function $f \circ x^{-1}$ is smooth.

$$x(U) \xrightarrow{x^{-1}} U \xrightarrow{f} \mathbb{R}$$

There are two disconcerting things about this definition. First, there is the issue that checking smoothness of a particular function seems next to impossible: the maximal atlas $\mathcal{A}_X$ will have *many, many* charts, and the definition makes it seem like we need to check smoothness on each of these charts! Thankfully, we don't need to check on all possible charts. As soon as we check smoothness on a collection of charts that cover $X$, we're done.

**Lemma 3.3.2.** *Let $X$ be a manifold and $f : X \to \mathbb{R}$ a function. Suppose that there exists an atlas $\mathcal{A} \subseteq \mathcal{A}_X$ such that $f \circ x^{-1}$ is smooth for any $(U, x) \in \mathcal{A}$. Then $f$ is smooth.*

*Proof.* Suppose $(V, y) \in \mathcal{A}_X$. For any $(U, x) \in \mathcal{A}$, we have maps as follows.

$$\begin{tikzcd}
y(U \cap V) \arrow[r, "y^{-1}"] \arrow[d, "x \circ y^{-1}"'] & U \cap V \arrow[r, "f"] & \mathbb{R} \\
x(U \cap V) \arrow[ur, bend right, "x^{-1}"'] & &
\end{tikzcd}$$

Since $(V, y)$ and $(U, x)$ are both elements of the maximal atlas $\mathcal{A}_X$, we know that the transition function $x \circ y^{-1}$ is smooth. Since $(U, x) \in \mathcal{A}$, we know by assumption that $f \circ x^{-1}$ is smooth. Thus

$$f \circ y^{-1} = (f \circ x^{-1}) \circ (x \circ y^{-1})$$

is a smooth function $y(U \cap V) \to \mathbb{R}$.

Now note that, since $\mathcal{A}$ is an atlas, its charts cover all of $X$, so

$$\bigcup_{(U,x) \in \mathcal{A}} y(U \cap V) = y(V).$$

Since $f \circ y^{-1}$ is smooth when restricted to each the open subset $y(U \cap V)$ for all $(U, x) \in \mathcal{A}$, we conclude that $f \circ y^{-1}$ must be smooth on all of $y(V)$. $\square$

The second disconcerting thing is the following. Suppose $U \subseteq \mathbb{R}^m$ is an open subset and $f : U \to \mathbb{R}$ is a function. On the one hand, we defined what it means for $f$ to be smooth back

in **??**. On the other hand, we can regard $U$ as an open submanifold of $\mathbb{R}^n$ (cf. exercise 3.2.7) and then definition 3.3.1 gives us another definition of what it means to be smooth. A priori, it could be that these definitions conflict. Thankfully, they do not.

**Exercise 3.3.3.** Suppose $U \subseteq \mathbb{R}^m$ is an open subset and $f : U \to \mathbb{R}$ is a function. Show that $f$ is smooth in the sense of **??** if and only if it is smooth in the sense of definition 3.3.1.

*Possible hint.* You might find lemma 3.3.2 useful.

**Definition 3.3.4.** If $X$ is a manifold, we write $\mathcal{O}(X)$ for the set of smooth functions $X \to \mathbb{R}$.

**Exercise 3.3.5.** Show that $\mathcal{O}(X)$ is a vector space.

**Exercise 3.3.6.** If $f, g \in \mathcal{O}(X)$, show that $fg \in \mathcal{O}(X)$ also.

**Proposition 3.3.7.** *Suppose $K$ is a compact subset of $X$ and $U$ is an open neighborhood of $K$. Then there exists a smooth function $f : X \to [0, 1]$ such that $f(x) = 1$ for all $x \in K$ and $f(x) = 0$ for all $x \notin U$.*

### 3.3.B Partial derivatives of real-valued functions

**Definition 3.3.8** (Partial derivatives)**.** Let $X$ be a manifold and $(U, x) \in \mathcal{A}_X$ a chart. Recall that we then have coordinate functions $x_1, \ldots, x_n : U \to \mathbb{R}$, where $x_i = \pi_i \circ x$. Given a smooth function $f : X \to \mathbb{R}$ and a point $p \in U$, we define the *partial derivative of* $f$ *at* $p$ *with respect to* $x_i$ by

$$\left.\frac{df}{dx_i}\right|_p := \partial_i(f \circ x^{-1})(x(p)).$$

In other words, $df/dx_i = \partial_i(f \circ x^{-1}) \circ x$ as functions $U \to \mathbb{R}$.

### 3.3.C Smooth functions between manifolds

**Definition 3.3.9** (Smoothness)**.** Suppose $X$ and $Y$ are manifolds and $f : X \to Y$ is a function. Then $f$ is *smooth* if, for every chart $(V, y) \in \mathcal{A}_Y$ and every chart $(U, x) \in \mathcal{A}_X$ such that

$f(U) \subseteq V$, the function $y \circ f \circ x^{-1} : x(U) \to y(V)$ is smooth.

$$\begin{tikzcd}
X \arrow[r, "f"] & Y \\
U \arrow[u, hook] \arrow[r, "f"] \arrow[d, "x"'] & V \arrow[u, hook] \arrow[d, "y"] \\
x(U) \arrow[r, "y \circ f \circ x^{-1}"] \arrow[d, hook] & y(V) \arrow[d, hook] \\
\mathbb{R}^m & \mathbb{R}^n
\end{tikzcd}$$

Once again, we don't need to check all possible charts. The precise statement is a bit complicated though; there's a long string of quantifiers.

**Exercise 3.3.10.** Suppose $X$ and $Y$ are manifolds and $f : X \to Y$ is a function. Suppose further that there exists an atlas $\mathcal{A} \subseteq \mathcal{A}_Y$ such that, for every $(V, y) \in \mathcal{A}$, there exists an atlas $\mathcal{A}' \subseteq \mathcal{A}_{f^{-1}(V)}$, such that for every $(U, x) \in \mathcal{A}'$, the function $y \circ f \circ x^{-1} : x(U) \to y(V)$ is smooth. Show that $f$ is smooth.

We should also note that we haven't redefined the word "smooth" in any conflicting ways. More precisely, we have the following two exercises.

**Exercise 3.3.11.** If $U \subseteq \mathbb{R}^m$ is an open subset and $f : U \to \mathbb{R}^n$ is a function, show that $f$ is smooth in the sense of **??** if and only if it is smooth in the sense of definition 3.3.9.

**Exercise 3.3.12.** Show that a function $f : X \to \mathbb{R}$ is smooth in the sense of definition 3.3.1 if and only if it is smooth in the sense of definition 3.3.9.

Finally, we have the following result which tells us that the case when the codomain is $\mathbb{R}$ is the most important case.

**Proposition 3.3.13.** *Suppose $f : X \to Y$ is a function between two manifolds. Then the following are equivalent.*

*(a)* $f$ *is smooth.*

*(b) For every smooth function $g : Y \to \mathbb{R}$, the composite $g \circ f$ is a smooth function $X \to \mathbb{R}$.*

### 3.3.D Diffeomorphisms

**Definition 3.3.14** (Diffeomorphism)**.** Suppose $X$ and $Y$ are manifolds. A function $f : X \to Y$ is a *diffeomorphism* if it is smooth, bijective, and the inverse function $f^{-1} : Y \to X$ is also smooth. If there exists a diffeomorphism $f : X \to Y$, then $X$ and $Y$ are *diffeomorphic*.

**Exercise 3.3.15.** Regard $\mathbb{R}$ as a manifold in two different ways: once with the euclidean atlas, and next with the maximal atlas containing $x \mapsto x^3$. Show that these two manifolds are diffeomorphic.

**Exercise 3.3.16.** If $X$ is a manifold and $(U, x) \in \mathcal{A}_X$, prove that $x : U \to x(U)$ is a diffeomorphism.

**Exercise 3.3.17.** Suppose $f : X \to Y$ is a diffeomorphism. Show that, for any open subset $U \subseteq X$, the image $f(U)$ is open in $Y$.

**Definition 3.3.18** (Local diffeomorphism)**.** Suppose $X$ and $Y$ are manifolds. A function $f : X \to Y$ is a *local diffeomorphism* at a point $p \in X$ if there exists an open neighborhood $U$ of $p$ such that $f|_U$ is a diffeomorphism $U \to f(U)$.

# 4 Tangent Spaces

## 4.1 Tangent space

Throughout this section, we let $X$ denote a manifold and $p \in X$ a point. We also let $n := \dim_p(X)$.

### 4.1.A Curves based at a point

**Definition 4.1.1.** A *curve in* $X$ *based at* $p$ is a smooth function $\gamma : (-\epsilon, \epsilon) \to X$ for some $\epsilon > 0$ such that $\gamma(0) = p$.

Suppose that $(U, x) \in \mathcal{A}_X$ is a chart containing $p$ and that $\gamma : (-\epsilon, \epsilon) \to \mathbb{R}$ is a curve based at $p$. Since $U$ is an open subset of $X$ and $\gamma$ is continuous, the preimage $\gamma^{-1}(U)$ is an open subset of $(-\epsilon, \epsilon)$ containing $0$, so there exists $0 < \epsilon' \leqslant \epsilon$ such that the image of the smaller interval $(-\epsilon', \epsilon)$ is entirely contained inside $U$. We let $x \circ \gamma$ denote the composite function

$$(-\epsilon', \epsilon') \longrightarrow U \longrightarrow \mathbb{R}^n.$$

We can then take its derivative at $0$ as in chapter 2. This derivative $d(x \circ \gamma)_0$ is a linear map $\mathbb{R} \to \mathbb{R}^n$, so its matrix representation

$$(x \circ \gamma)'(0)$$

is a column vector in $\mathbb{R}^n$.

**Definition 4.1.2.** Two curves $\gamma_1, \gamma_2$ in $X$ based at $p$ are *equivalent* if $d(x \circ \gamma_1)_0 = d(x \circ \gamma_2)_0$ for every chart $(U, x) \in \mathcal{A}_X$ containing $p$.

As usual, the first point of order is to check that we don't need to check all possible charts.

**Exercise 4.1.3.** Show that, if $\gamma_1, \gamma_2$ are two curves in $X$ based at $p$ and there exists a chart $(U, x) \in \mathcal{A}_X$ containing $p$ and

$$d(x \circ \gamma_1)_0 = d(x \circ \gamma_2)_0,$$

then $\gamma_1$ is equivalent to $\gamma_2$.

**Definition 4.1.4.** We define the *tangent space of* $X$ *at* $p$, denoted $T_pX$, to be the set of equivalence classes of curves in $X$ based at $p$. Elements of $T_pX$ are called *tangent vectors* at $p$. If $\gamma$ is a curve based at $p$, we will write $v_\gamma$ for the corresponding tangent vector in $T_pX$.

### 4.1.B Vector space structure

We will now put a vector space structure on the tangent space $T_pX$. There are a lot of details involved in the process, so we begin with a high-level overview. If we choose a chart $(U, x) \in \mathcal{A}_X$ containing $p$, there is a bijection $\sigma : T_pX \to \mathbb{R}^n$ (cf. lemma 4.1.5). This means that we can define vector space operations on $T_pX$ by "pulling back along $\sigma$." More precisely, this mean the following: given $v \in T_pX$ and $\lambda \in \mathbb{R}$, we define

$$\lambda v := \sigma^{-1}(\lambda\sigma(v)),$$

and given $v_1, v_2 \in T_pX$, we define

$$v_1 + v_2 := \sigma^{-1}(\sigma(v_1) + \sigma(v_2)).$$

We can then show that these operations define the structure of a vector space on $T_pV$ by showing that all of the vector space axioms are satisfied (cf. lemma 4.1.7). The final step is to show that these operations do not depend on the choice of chart $(U, x)$ (cf. exercise 4.1.11). Thus there is a canonical, coordinate-free, vector space structure on $T_pX$.

**Lemma 4.1.5.** *Suppose* $(U, x) \in \mathcal{A}_X$ *is a chart containing* $p$. *There is a well-defined bijection* $\sigma : T_pX \to \mathbb{R}^n$ *given by* $v_\gamma \mapsto [d(x \circ \gamma)_0] = d(x \circ \gamma)_0(1)$.

*Proof.* The fact that $\sigma$ is well-defined follows immediately from definition 4.1.2. For a vector $w \in \mathbb{R}^n$, define $\gamma_w : (-\epsilon, \epsilon) \to X$ by

$$\gamma_w(t) = x^{-1}(x(p) + tw)$$

where $\epsilon$ is chosen to be small enough that $x(p) + tw \in x(U)$ for all $t \in (-\epsilon, \epsilon)$ (cf. exercise 4.1.6). Then $\gamma_w(0) = x^{-1}(x(p)) = p$, so $\gamma_w$ is a curve in $X$ based at $p$. Notice that

$$(x \circ \gamma_w)(t) = x(x^{-1}(x(p) + tw)) = x(p) + tw$$

so $[d(x \circ \gamma_w)_0] = w$ (cf. exercise 4.1.6). It follows from this that $w \mapsto v_{\gamma_w}$ is inverse to $\sigma$, proving that $\sigma$ is bijective (cf. exercise 4.1.6). □

**Exercise 4.1.6.** (a) Explain why there must exist $\epsilon > 0$ such that $x(p) + tw \in x(U)$ for all $t \in (-\epsilon, \epsilon)$.

(b) Explain why $[d(x \circ \gamma)_0] = w$.

(c) Explain how it follows from (b) that $w \mapsto v_{\gamma_w}$ is inverse to $\sigma$.

**Lemma 4.1.7.** *Suppose* $(U, x) \in \mathcal{A}_X$ *is a chart containing* $p$ *and* $\sigma : T_pX \to \mathbb{R}^n$ *is the bijection of lemma 4.1.5. Then* $T_pX$*, equipped with the addition and scalar multiplication operations obtained by pulling back along* $\sigma$*, is a vector space. The zero element is the equivalence class of the constant curve at* $p$*.*

*Proof.* This follows immediately from the fact that $\mathbb{R}^n$ is a vector space and that $\sigma$ is bijective, but writing out all of the details is fairly tedious. To explain the idea, let's show that vector addition is commutative. For $v_1, v_2 \in T_pX$, we have

$$\begin{aligned} v_1 + v_2 &= \sigma^{-1}(\sigma(v_1) + \sigma(v_2)) \\ &= \sigma^{-1}(\sigma(v_2) + \sigma(v_1)) \\ &= v_2 + v_1 \end{aligned}$$

where we used the definition of addition in $T_pX$ on the first and last steps, and the fact that addition in $\mathbb{R}^n$ is commutative for the middle step.

Let $\gamma$ denote the constant curve $\gamma(t) = p$ for all $t$. Then $x \circ p$ is also a constant function which always takes the value $x(p)$, so its derivative is 0. Thus, for any $v \in T_pX$, we have

$$\begin{aligned} v + v_\gamma &= \sigma^{-1}(\sigma(v) + \sigma(v_\gamma)) \\ &= \sigma^{-1}(\sigma(v) + 0) \\ &= \sigma^{-1}(\sigma(v)) \\ &= v, \end{aligned}$$

proving that $v_\gamma$ is the zero element of $T_pX$.

The rest of the proof is left to you (cf. exercise 4.1.8) $\square$

**Exercise 4.1.8.** Check all of the other axioms that need to be checked to ensure that $T_pX$ is a vector space.[1]

**Exercise 4.1.9.** Suppose $(U, x) \in \mathcal{A}_X$ is a chart containing $p$ and $\sigma : T_pX \to \mathbb{R}^n$ is the bijection of lemma 4.1.5. If $T_pX$ is equipped with the vector space operations obtained by pulling back along $\sigma$, then $\sigma : T_pX \to \mathbb{R}^n$ is a linear map. Therefore, it is an isomorphism of vector spaces, and $\dim T_pX = n$.

**Lemma 4.1.10.** *Suppose* $(U, x), (U', x') \in \mathcal{A}_X$ *are both charts containing* $p$ *and that* $\sigma, \sigma' : T_pX \to \mathbb{R}^n$ *are the bijections from lemma 4.1.5 corresponding to* $(U, x), (U', x')$*, respectively. Then* $\sigma' = d(x' \circ x)_{x(p)} \circ \sigma$*, where* $x' \circ x$ *denotes the transition map (cf. definition 3.1.3).*

$$\begin{tikzcd} T_pX \arrow[r, "\sigma"] \arrow[dr, bend right, "\sigma'"'] & \mathbb{R}^n \arrow[d, "d(x' \circ x^{-1})_{x(p)}"] \\ & \mathbb{R}^n \end{tikzcd}$$

*Proof.* Let $\gamma$ be a curve in $X$ based at $p$. Then

$$x' \circ \gamma = x' \circ (x^{-1} \circ x) \circ \gamma = (x' \circ x) \circ (x \circ \gamma),$$

so the chain rule 2.3.3 says that

$$d(x' \circ \gamma)_0 = d(x' \circ x)_{(x \circ \gamma)(0)} \circ d(x \circ \gamma)_0 = d(x' \circ x)_{x(p)} \circ d(x \circ \gamma)_0.$$

Evaluating at 1, we find that

$$\sigma'(v_\gamma) = d(x' \circ \gamma)_0(1) = d(x' \circ x)_{x(p)}(d(x \circ \gamma)_0(1)) = d(x' \circ x)_{x(p)}(\sigma(v_\gamma)).$$

Since this is true for all $\gamma$, we conclude that $\sigma' = d(x' \circ x)_{x(p)} \circ \sigma$. $\square$

[1] During a class I took with him, George Bergman once said something along the following lines: "If you're sure you can write the details of a proof, you probably don't need to. But if you're not sure you can formalize the details, you need to do it." I encourage you to apply this principle for exercise 4.1.8.

**Exercise 4.1.11.** Suppose $(U, x), (U', x') \in \mathcal{A}_X$ are both charts containing $p$ and that $\sigma, \sigma' : T_pX \to \mathbb{R}^n$ are the bijections from lemma 4.1.5 corresponding to $(U, x), (U', x')$, respectively. Show that the vector space operations on $T_pX$ defined by pulling back along $\sigma$ are the same as those defined by pulling back along $\sigma'$.

*Possible hint.* Use lemma 4.1.10. The key is that $d(x' \circ x^{-1})_{x(p)}$ is an invertible linear map.

This completes our construction of a vector space structure on $T_pX$. Notice that $T_pX$ is an example of a finite dimensional vector space that does not come equipped with any canonical basis! More precisely, this means the following. If we choose a chart $(U, x) \in \mathcal{A}_X$ containing $p$, the bijection $\sigma : T_pX \to \mathbb{R}^n$ of lemma 4.1.5 is an isomorphism of vector spaces (cf. exercise 4.1.9), so $\sigma^{-1}(e_1), \ldots, \sigma^{-1}(e_n)$ is a basis of $T_pX$. But then, if we choose a different chart $(U', x') \in \mathcal{A}_X$ containing $p$ and $\sigma'$ is the corresponding isomorphism $T_pX \to \mathbb{R}^n$, then $\sigma'^{-1}(e_1), \ldots, \sigma'^{-1}(e_n)$ will in general be a different basis of $T_pX$, even though the vector space structures defined by pulling back along $\sigma$ and $\sigma'$ are the same! So, for a general manifold $X$ and point $p$, since there's no "canonical" choice of a chart containing $p$, there is also no "canonical" choice of a basis on $T_pX$. I strongly encourage you to work through the following exercise to make sense of this.

**Exercise 4.1.12.** Consider the unit circle $S^1$, regarded manifold using the charts $(U, x)$ and $(U', x')$ defined in exercise 3.2.3. Any point $p \in S^1$ can be written as $(\cos\theta, \sin\theta)$ for some $\theta \in \mathbb{R}$.

(a) Observe that $\gamma : \mathbb{R} \to S^1$ given by

$$\gamma(t) = (\cos(\theta + t), \sin(\theta + t))$$

is a curve in $S^1$ based at $p$. Explain why $v_\gamma$ is a basis for $T_pS^1$.

(b) Suppose $p \in U$ and let $\sigma : T_pS^1 \to \mathbb{R}$ be the isomorphism of lemma 4.1.5 corresponding to the chart $(U, x)$. Find $\lambda \in \mathbb{R}$ such that $\sigma(\lambda v_\gamma) = 1$.

(c) Suppose $p \in U'$ and let $\sigma' : T_pS^1 \to \mathbb{R}$ be the isomorphism of lemma 4.1.5 corresponding to the chart $(U', x')$. Find $\lambda' \in \mathbb{R}$ such that $\sigma'(\lambda' v_\gamma) = 1$.

So, if $p \in U \cap U'$, we see that there are three possible bases we might choose on $T_pS^1$ ($v_\gamma$, or $\lambda v_\gamma$, or $\lambda' v_\gamma$), none of these is any more "canonical" than another. This is what is meant when we say that $T_pS^1$ has no canonical choice of basis.

**Remark 4.1.13.** Suppose $U$ is an open subset of $X$ containing $p$ regarded as an open submanifold (cf. exercise 3.2.7). A curve $\gamma$ in $U$ based at $p$ is also automatically a curve in $X$, which means there is a natural "inclusion" map $T_pU \to T_pX$ given by $v_\gamma \mapsto v_\gamma$. But we can "shrink" curves in $X$ to curves in $U$ without changing their equivalence class, so $T_pU \to T_pX$ is actually bijective. Moreover, if we choose a chart in $\mathcal{A}_U$ containing $p$, we can use this chart to define the vector space structure on both $T_pU$ and $T_pX$, so $T_pU \to T_pX$ is actually an isomorphism of vector spaces. Since this isomorphism is tautological, we sometimes write $T_pU = T_pX$.

**Remark 4.1.14.** There is one situation where we do have a canonical choice of a chart: $\mathbb{R}^n$. Namely, the identity map $\mathrm{id} : \mathbb{R}^n \to \mathbb{R}^n$ defines a chart. If $\gamma$ is a curve in $\mathbb{R}^n$ based at a point $p$, then $d(\mathrm{id} \circ \gamma)_0 = d\gamma_0$, so the isomorphism $\sigma : T_p\mathbb{R}^n \to \mathbb{R}^n$ is given by $\sigma(v_\gamma) = [d\gamma_0]$. We will call this the "canonical isomorphism" $T_p\mathbb{R}^n \to \mathbb{R}^n$, which we sometimes also write as $T_p\mathbb{R}^n = \mathbb{R}^n$. But this is fairly abusive and it's worth remembering that $T_p\mathbb{R}^n$ is not literally equal to $\mathbb{R}^n$. The former is equivalence classes of curves, the latter is column vectors, and the relationship between the two is given by taking the derivative at $0$.

### 4.1.C Derivations $\star$

There is another perspective on tangent vectors which is often useful, though it is decidedly less geometric at first glance. It begins with the following definition.

**Definition 4.1.15.** Let $X$ be a manifold and $p \in X$ a point. A *derivation* on $X$ based at $p$ is a linear function $\partial : \mathcal{O}(X) \to \mathbb{R}$ such that

$$\partial(fg) = \partial(f)g(p) + f(p)\partial(g).$$

We let $\mathrm{Der}_p(X)$ denote the set of all derivations on $X$ based at $p$.

The following is not very hard to prove.

**Exercise 4.1.16.** Show that $\mathrm{Der}_p(X)$ is a vector space.

It turns out that $T_pX$ and $\mathrm{Der}_p(X)$ are isomorphic, as we will soon prove.

**Remark 4.1.17.** Sometimes, people define the tangent space to be $\mathrm{Der}_p(X)$. It's not very geometric, but it at least makes the vector space operations on the tangent space fairly obvious (cf. exercise 4.1.16); compare this with our geometric definition of the tangent

space in terms of curves, where we had to do a lot of work to regard the tangent space as a vector space (cf. section 4.1.B). That said, even if one defines the tangent space to be $\mathrm{Der}_p(X)$, it's not clear how to produce examples of derivations, or to prove that $\mathrm{Der}_p(X)$ is finite dimensional. One way or another, one needs to show that derivations are somehow linked to curves. This is what we do next.

**Definition 4.1.18.** Suppose $\gamma$ is a curve in $X$ based at $p$. If $f : X \to \mathbb{R}$ is a smooth function, then $f \circ \gamma$ is a single variable function $(-\epsilon, \epsilon) \to \mathbb{R}$. We define $\partial_\gamma : \mathcal{O}(X) \to \mathbb{R}$ by

$$\partial_\gamma(f) = (f \circ \gamma)'(0).$$

**Lemma 4.1.19.** *If $\gamma$ is a curve in $X$ based at $p$, the function $\partial_\gamma$ is a derivation.*

*Proof.* Suppose $f, g \in \mathcal{O}(X)$ and $\lambda \in \mathbb{R}$. To see that $\partial_\gamma$ is linear, we use the sum and scalar multiple rules from exercises 1.2.2 and 1.2.3.

$$\begin{aligned}
\partial_\gamma(f + \lambda g) &= ((f + \lambda g) \circ \gamma)'(0) \\
&= ((f \circ \gamma) + \lambda(g \circ \gamma))'(0) \\
&= (f \circ \gamma)'(0) + \lambda(g \circ \gamma)'(0) \\
&= \partial_\gamma(f) + \lambda \partial_\gamma(g)
\end{aligned}$$

To see that it is a derivation, we use the single variable product rule 1.2.4.

$$\begin{aligned}
\partial_\gamma(fg) &= (fg \circ \gamma)'(0) \\
&= ((f \circ \gamma)(g \circ \gamma))'(0) \\
&= (f \circ \gamma)'(0)(g \circ \gamma)(0) + (f \circ \gamma)(0)(g \circ \gamma)'(0) \\
&= \partial_\gamma(f)g(p) + f(p)\partial_\gamma(g)
\end{aligned}$$

This completes the proof. □

**Lemma 4.1.20.** *If $\gamma_1$ and $\gamma_2$ are equivalent curves in $X$ based at $p$, then $\partial_{\gamma_1} = \partial_{\gamma_2}$.*

*Proof.* Choose a chart $(U, x) \in \mathcal{A}_X$ containing $p$ and let $\gamma$ be a curve in $X$ based at $p$. Then

$$\begin{aligned}\partial_\gamma(f) &= (f \circ \gamma)'(0) \\ &= d(f \circ \gamma)_0(1) \\ &= d(f \circ x^{-1} \circ x \circ \gamma)_0(1) \\ &= d(f \circ x^{-1})_{x(p)}\left(d(x \circ \gamma)_0(1)\right)\end{aligned}$$

where we used the chain rule 2.3.3 at the last step. Since $\partial_\gamma(f)$ only depends on $d(x \circ \gamma)_0$, we see that equivalent curves must give rise to the same derivation. □

**Theorem 4.1.21.** *The function* $T_pX \to \mathrm{Der}_p(X)$ *given by* $v_\gamma \mapsto \partial_\gamma$ *is an isomorphism.*

**Definition 4.1.22.** If $v \in T_pX$, we let $\partial_v$ denote the corresponding element in $\mathrm{Der}_p(X)$.

## 4.2 Pushforward

### 4.2.A Definition of the pushforward

Let $f : X \to Y$ be a smooth map of manifolds and $p \in X$ a point. Given a curve $\gamma$ in $X$ based at $p$, observe that $f \circ \gamma$ is a curve in $Y$ based at $f(p)$.

**Lemma 4.2.1.** *Let* $f : X \to Y$ *be a smooth map of manifolds and* $p \in X$ *a point. If* $\gamma_1$ *and* $\gamma_2$ *are equivalent curves in* $X$ *based at* $p$*, then* $f \circ \gamma_1$ *and* $f \circ \gamma_2$ *are equivalent curves in* $Y$ *based at* $f(p)$*.*

This allows us to make the following definition. We will shortly see that this is the "ultimate" generalization of derivatives.

**Definition 4.2.2.** Let $f : X \to Y$ be a smooth map of manifolds and $p \in X$ a point. If $\gamma$ is a curve in $X$ based at $p$, we define its *pushforward* $f_*(v_\gamma) \in T_{f(p)}Y$ to be the equivalence class of the curve $f \circ \gamma$. This defines the *pushforward map* $f_* : T_pX \to T_{f(p)}Y$.

**Proposition 4.2.3.** *Let* $f : X \to Y$ *be a smooth map of manifolds and* $p \in X$ *a point. The pushforward map* $f_* : T_pX \to T_{f(p)}Y$ *is linear.*

The following result states that the pushforward generalizes the derivative at a point.

**Proposition 4.2.4.** *Let $U$ is an open subset of $\mathbb{R}^m$, $p \in U$ is a point, and $f : U \to \mathbb{R}^n$ is a smooth map. Then, using the identifications of remarks 4.1.13 and 4.1.14, we have $f_* = df_p$.*

$$\begin{tikzcd}
T_pU \arrow[r, equal] \arrow[d, "f_*"'] & T_p\mathbb{R}^m \arrow[r, equal] & \mathbb{R}^m \arrow[d, "df_p"] \\
T_{f(p)}\mathbb{R}^n \arrow[rr, equal] & & \mathbb{R}^n
\end{tikzcd}$$

*Proof.* Let $\sigma$ denote the isomorphisms $T_p\mathbb{R}^m \to \mathbb{R}^m$ and $T_p\mathbb{R}^n \to \mathbb{R}^n$ of remark 4.1.14. Suppose $\gamma$ is a curve in $U$ based at $p$. Then

$$(df_p \circ \sigma)(v_\gamma) = df_p(d\gamma_0(1)) = (df_p \circ d\gamma_0)(1) = d(f \circ \gamma)_0(1),$$

where we used the chain rule 2.3.3 for the final step. On the other hand, we also have

$$(\sigma \circ f_*)(v_\gamma) = \sigma(v_{f\circ\gamma}) = d(f \circ \gamma)_0. \qquad \square$$

**Remark 4.2.5.** Sometimes, the map $f_* : T_pX \to T_pY$ is denoted $df_p$, similar to the notation we were using in chapter 2. This creates no conflicts, thanks to proposition 4.2.4.

In light of proposition 4.2.4, the following is a generalization of the chain rule.

**Proposition 4.2.6.** *Suppose $f : X \to Y$ and $g : Y \to Z$ are smooth maps of manifolds, and $p \in X$ is a point. Then $g_* \circ f_* = (g \circ f)_*$.*

$$\begin{tikzcd}
T_pX \arrow[r, "f_*"] \arrow[dr, bend right, "(g\circ f)_*"'] & T_{f(p)}Y \arrow[d, "g_*"] \\
& T_{g(f(p))}Z
\end{tikzcd}$$

*Proof.* If $\gamma$ is a curve in $X$ based at $p$, then

$$g_*(f_*(v_\gamma)) = g_*(v_{f\circ\gamma}) = v_{g\circ f\circ\gamma} = (g \circ f)_*(v_\gamma). \qquad \square$$

This proof looks absurdly easy, and that's because it is. You might be wondering why the proof of this generalization of the chain rule looks so easy, while the proof of the chain rule in chapter 2 was significantly more work. It's because the pushforward is so abstractly defined that it's basically useless and uncomputable until we know that it actually

generalizes the derivative from chapter 2 and therefore can be computed using techniques from chapter 2. This was the content of proposition 4.2.4, and notice that we used the chain rule 2.3.3 to prove proposition 4.2.4.

### 4.2.B Rank of a smooth map

In light of proposition 4.2.4, the following generalizes definition 2.3.28.

**Definition 4.2.7.** Suppose $f : X \to Y$ is a smooth map between manifolds and $p \in X$ is a point. The rank of the linear map $f_* : T_pX \to T_{f(p)}Y$ is also called the *rank of* $f$ *at* $p$, denoted $\text{rank}_p(f)$. Note that $\text{rank}_p(f) \leqslant \min\{\dim_p(X), \dim_{f(p)}(Y)\}$.

- If $\text{rank}_p(f) = \dim_p(X)$, then we say that $f$ is *immsersive at* $p$. This is equivalent to requiring that $f_*$ is an injective map $T_pX \to T_pY$.
- If $\text{rank}_p(f) = \dim_{f(p)}(Y)$, then we say that $f$ is *submersive at* $p$. This is equivalent to requiring that $f_*$ is a surjective map $T_pX \to T_pY$.
- If $\text{rank}_p(f) = \dim_p(X) = \dim_{f(p)}(Y)$, then we say that $f$ is *étale at* $p$. This is equivalent to requiring that $f_*$ is an isomorphism $T_pX \to T_pY$.
- If $\text{rank}_p(f) = \min\{\dim_p(X), \dim_{f(p)}(Y)\}$, then we say that $p$ is a *regular point* of $f$.
- If $\text{rank}_p(f) < \min\{\dim_p(X), \dim_{f(p)}(Y)\}$, then we say that $p$ is a *critical point* of $f$.

**Definition 4.2.8.** Suppose $f : X \to Y$ is a smooth map between manifolds.

- If $f$ is immersive at every $p \in X$, then $f$ is an *immersion*.
- If $f$ is submersive at every $p \in X$, then $f$ is a *submersion*.
- if $f$ is étale at every $p \in X$, then $f$ is *étale*.

# Bibliography


[Axl15] Sheldon Axler. *Linear algebra done right*. Undergraduate Texts in Mathematics. Springer, third edition, 2015. doi:10.1007/978-3-319-11080-6.

[GO03] Bernard R. Gelbaum and John M. H. Olmsted. *Counterexamples in analysis*. Dover Publications, Inc., 2003. Corrected reprint of the second (1965) edition.

[GVL13] Gene H. Golub and Charles F. Van Loan. *Matrix computations*. Johns Hopkins Studies in the Mathematical Sciences. Johns Hopkins University Press, fourth edition, 2013.

[Hef19] Jim Hefferon. *Linear algebra*. Third edition, 2019. URL: http://joshua.smcvt.edu/linearalgebra/.

[PM91] Murray H. Protter and Charles B. Morrey, Jr. *A first course in real analysis*. Undergraduate Texts in Mathematics. Springer-Verlag, second edition, 1991. doi:10.1007/978-1-4419-8744-0.

[Pom07] Dimitrie D. Pompeiu. Sur les fonctions dérivées. *Mathematische Annalen*, 63(3):326–332, 1907. doi:10.1007/BF01449201.

[Pug15] Charles C. Pugh. *Real mathematical analysis*. Undergraduate Texts in Mathematics. Springer, second edition, 2015. doi:10.1007/978-3-319-17771-7.

[Rud76] Walter Rudin. *Principles of mathematical analysis*. McGraw-Hill Book C, third edition, 1976. International Series in Pure and Applied Mathematics.

[Spi65] Michael Spivak. *Calculus on manifolds*. W. A. Benjamin, Inc., 1965.

[Spi05] Michael Spivak. *A Comprehensive Introduction to Differential Geometry*, volume I. Publish or Perish, Inc, third edition, 2005.

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